
Low-cost Full Fine-tuning: Learning What to Update for LLMs

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Abstract

While large language models (LLMs) have strong abilities, they generally rely on fine-tuning to supplement downstream task-specific knowledge. Due to the prohibitive memory overhead of full fine-tuning (FT), existing parameter-efficient fine-tuning techniques, *e.g.*, LoRA and Adapters, update parameters only in low-rank or restricted subspaces. However, they fail to approximate FT—the performative fine-tuner—and risk performance degradation in tough tasks. Therefore, we naturally raise a *Low-cost Full Fine-tuning* question: Can we approach standard full fine-tuning in theory, yet with much lower costs in practice? Our key insight is that performing selective updates at each step can, theoretically, recover FT asymptotically, while being cost-effective and ignoring no parameter direction. This motivates a new general fine-tuning paradigm (called *Think-Touch*): we first predict potentials of parameter groups (*think*) and then update only the selected (*touch*) in one step. Theoretically, we show that under a very weak sufficient condition—divergence of the cumulative coverage of the expected gradient norm—any selection strategy can converge in the full-parameter space to a stationary point at which the FT admits no further first-order improvement. Besides, we further derive the general convergence rate for our paradigm and identify a post-hoc greedy strategy that is rate-optimal. Unfortunately, this strategy cannot be directly ap-

plied in practice due to its reliance on full and accurate gradient information. Thus, we propose a bandit-based method to online approximate this ideal strategy in the long run with a rigorous regret guarantee. Extensive experimental results on various tasks demonstrate the potential of our paradigm, including much lower space overheads against FT and better performance than LoRAs.

1. Introduction

Despite strong general capabilities, Large language models (LLMs) still require fine-tuning (Devlin et al., 2019; Brown et al., 2020). This need for efficient adaptation becomes more pronounced as LLMs and deep neural models are applied to specialized domains—from AI-for-science readiness and recommendation to chemical and molecular reasoning, career mobility and job-skill modeling, and self-interpretable learning—where downstream knowledge and constraints are often domain-specific (Qin et al., 2025; Xin et al., 2025; Zhang et al., 2025b;a; Cui et al., 2026; Ji et al., 2025; Sun et al., 2025).

To reduce the high cost of *full-parameter fine-tuning* (FT) (Rajbhandari et al., 2020), *parameter-efficient fine-tuning* (PEFT) methods, *e.g.*, Adapters and LoRAs (Houlsby et al., 2019; Hu et al., 2022), restrict their parameter updates in low-rank or fixed subspaces, and thus may ignore critical directions and scarify their expressivities, risking performance degradation in difficult tasks (Han et al., 2024b).

Therefore, we raise a **Low-cost Full Fine-tuning** problem: *Can we approach standard full fine-tuning in theory, yet with much lower costs in practice?*

Our key insight is that once a few parameter groups (*e.g.*, rows, modules, layers) are allowed to be updated in a step, we can still emulate FT abilities in the long run theoretically, ignoring no parameter direction, while remaining cost-effective. It is rooted in an observation that at most of training steps in FT, a few parameters obtain most of full gradient energy (Sec. 6.4), and thus we need only to spend budget where it really matters.

This insight motivates a new general fine-tuning paradigm, *i.e.*, *Think-Touch* (Fig. 1), wherein in each training step we

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first *think*—predict the current potentials for the downstream task based on historical steps’ knowledge and select a few parameter groups; and then *touch*—update only the selected groups. The superiority of our paradigm is two-fold. First, it significantly reduces memory by avoiding maintaining gradients or optimizer states of untouched groups. Second, due to its ability to freely select groups per step, it can strategically explore directions that may be missed by standard PEFTs over time. To further release its potential and avoid possible degeneration, we provide some theoretical analysis and insights to essentially understand its behaviors.

We provide a theoretical guarantee (Corollary 4.8) that *Think-Touch* can approximate a stationary point admitted by FT in the full-parameter space under a very weak sufficient condition that the cumulative coverage of expected gradient energy (Definition 4.4) diverges. This implies tolerances on infinite numbers of failures (*i.e.*, suboptimal selections) and provides feasibility for many strategies, including uniform and time-varying stochastic strategies. Such a broad feasible strategy space highlights the generality and flexibility of our paradigm. Then, we derive its convergence rate (Theorem 4.10), which theoretically shows that long-term neglect of critical groups would slow down and even degrade convergence.

Motivated by our theoretical analysis, we can maximize its rate and naturally yield an *ideal* strategy resembling a post-hoc greedy rule—selecting the groups with the highest current-step gradient norms. Unfortunately, this ideal strategy cannot be directly applied in our low-cost setting, due to its reliance on full and accurate gradient information. Therefore, to accurately approximate it, we cast group selection as an online decision problem and propose a bandit-based framework to learn groups’ potentials from incomplete history. We theoretically guarantee that our online selector tracks the ideal strategy in hindsight while keeping its convergence rate over time via regret analysis.

Our main contributions are summarized as follows:

(1) New paradigm. We propose a *Low-cost Full Fine-tuning* problem and a general *Think-Touch* paradigm, with a very weak sufficient condition showing its general FT-emulating ability as well as a convergence-rate bound that provides insights on strategy comparisons.

(2) New fine-tuning framework. We propose a bandit-based framework with a regret guarantee showing it can online approximate the impractical ideal strategy maximizing our derived rate, under our low-cost setting.

(3) Empirical results. Extensive experiments show that we save substantial space against standard FT while outperforming common PEFT baselines such as LoRA.

Code Availability. The implementation is available at

<https://anonymous.4open.science/r/ICML2026-LOWCOST-SFT-BDE4/>.

2. Preliminaries

Large language models (LLMs) are auto-regressive Transformers p_θ with billions of parameters θ , large-scale pre-trained for general-purpose initializations θ_0 , where given a sequence $x_{1:n}$, $p_\theta(x_{1:n}) = \prod_{t=1}^n p_\theta(x_t | x_{<t})$.

Supervised fine-tuning on a dataset $\mathcal{D} = \{(u_i, v_i)\}_{i=1}^N$ seeks a solution θ^* minimizing the negative log-likelihood $\min_\theta \mathcal{L}(\theta) := -\frac{1}{N} \sum_{(u,v) \in \mathcal{D}} \sum_{t \in \mathcal{I}(u,v)} \log p_\theta(x_t | x_{<t})$ from an initialization θ_0 , where sequence x concatenates the input query u and the desired output v , with $\mathcal{I}(u, v)$ denoting the output positions. *Full fine-tuning* (FT) updates all parameters θ per step, extremely expensive in memory.

LoRAs freeze the pretrained weights and low-rankly reparameterize total updates of some matrices $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$:

$$W' = W + \Delta W, \quad \Delta W = \frac{\alpha}{r} BA, \quad \text{rank}(\Delta W) \leq r,$$

with a scaling factor α , an rank $r \ll \min(d_{\text{out}}, d_{\text{in}})$, and trainable $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$ (Hu et al., 2022). Despite the reduced memory, the low-rank structure fails to express full-rank updates in general and thus intrinsically restricts the expressivity—only a *subset* of full parameter space $\mathbb{R}^{|\theta|}$ (Def. 3.1)—which is discussed in Sec. 6.4.

Adapters freeze the pretrained weights and insert small bottleneck modules into each layer (Houlsby et al., 2019):

$$\text{Adapter}(h) = h + U \sigma(Dh),$$

with learnable down-/up-projections $D \in \mathbb{R}^{m \times d}$ and $U \in \mathbb{R}^{d \times m}$, a hidden state $h \in \mathbb{R}^d$, and nonlinearity $\sigma(\cdot)$. Similar to LoRAs, the bottleneck dimension $m \ll d$ reduces the cost, yet risking performance degradation against FT.

3. Problem Formulation: Low-Cost Full Fine-Tuning (LC-FT)

This work focuses on the *Cost-Expressivity dilemma*, *i.e.*, how to achieve the theoretical expressivity of FT, yet practically with a reduced cost similar to LoRAs and Adapters.

Specifically, we formalize a parametrized problem as below:

Definition 3.1 (Low-Cost Fine Tuning, LC-FT $(M, \mathcal{S})$). Given two parameters—per-step cost budget M (*e.g.*, amount of parameters) and final expressivity parameter $\mathcal{S} \subseteq \mathbb{R}^{|\theta|}$ —finding such a Cost-Aware Updating Operator $\mathcal{U}(\theta; M)$ that by iteratively applying $\theta_{t+1} \leftarrow \mathcal{U}(\theta_t; M)$ under a per-step budget M , one can obtain a solution $\theta_{+\infty} = \theta^{*[\mathcal{S}]}$ minimizing the task-specific loss function $\mathcal{L}(\cdot)$ within the subset $\mathcal{S}$ of the full-parameter space $\mathbb{R}^{|\theta|}$, *i.e.*, $\theta^{*[\mathcal{S}]} \in \arg \min_{\theta \in \mathcal{S}} \mathcal{L}(\theta)$.

By this formulation, we transform the vanilla problem into an operator designing challenge, which interprets LoRA and its variants as solutions to the problems with $\mathcal{S} \subset \mathbb{R}^{|\theta|}$. However, we focus on the tougher problem with $\mathcal{S} = \mathbb{R}^{|\theta|}$ —*FT-expressivity*—with exactly the same one-step budget M :

Definition 3.2 (Low-Cost Full Fine Tuning, LC-FT (M)). We define the problem of *Low-Cost Full Fine Tuning* as a special version of *Low-Cost Fine Tuning* with $\mathcal{S} = \mathbb{R}^{|\theta|}$.

4. Think-Touch: The First LC-FT Paradigm

Traditionally, there has been a contradiction between cost and expressivity due to the logic chain: low costs $\rightarrow$ few parameters $\rightarrow$ low expressivity. To break this dilemma, existing works such as LoRAs—*mainly focusing on how to break the latter half of the logic chain*—improve expressivity but with less learnable parameters. **Yet, reducing the number of trainable parameters would unavoidably sacrifice the full expressivity**, which clearly interprets why LoRAs fail in approximating FT and have only expressivity $\mathcal{S} \subsetneq \mathbb{R}^{|\theta|}$. Different from prior works, we solve the problem by *breaking the former half of the above logic chain, not the latter half*: low costs $\rightarrow$ full parameters $\rightarrow$ full expressivity.

4.1. Intuitive Insights

Conventionally, breaking the former half and realizing low costs $\rightarrow$ full parameters is still fundamentally impossible in one training step, since the calculation and storage of the full-gradient are well-known to be expensive. However, note that we need to realize this goal *only in the long run rather than in just one step*. We empirically observe that *at almost all steps in FT, only a few parameters obtain most of the full-gradient energy* (Sec. 6.4).

Therefore, after dividing the full parameters into many parameter groups (e.g., rows, columns, modules, blocks, matrices, layers), our key insight can be summarized that—**even if one or a few parameter groups are allowed to be updated in a step, we may still theoretically emulate FT and recover its full-expressivity long-time fine-tuning abilities**.

4.2. The Think-Touch Paradigm

This insight motivates us to propose a new fine-tuning paradigm, i.e., *Think-Touch*, which is formulated as follows:

Definition 4.1 (Think-Touch Fine-Tuning Paradigm). Given a per-step budget parameter M , Think-Touch designs the following per-step cost-aware updating operator $\mathcal{U}(\theta; M)$:

- (i) *Think*: $\{g_k\}_{k \in [1, M]} \sim \mathbb{P}_t(g)$, $\mathbb{P}_t \leftarrow \text{Strategy}(\mathbb{K}_{t-1})$,
- (ii) *Touch*: $\{\theta_t^{(g_k)}, \mathcal{K}_t^{(g_k)}\}_k \leftarrow \text{Update}\left(\left\{\theta_{t-1}^{(g_k)}\right\}_k; \mathcal{L}\right)$,

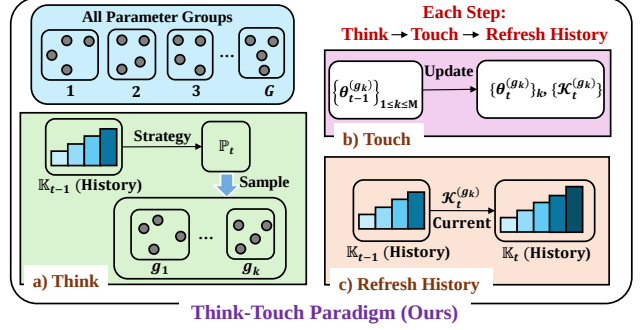


Figure 1. Illustration of the proposed *Think-Touch* paradigm. It consists of three steps: (1) *Think*: predict the potentials for groups from historical energy observations and sample a small touch set; (2) *Touch*: apply gradient updates only to the *touched* groups while keeping the others frozen; and (3) *Refresh History*: update history via current-step knowledge for the next round.

- (iii) *Refresh History* $\mathbb{K}_{t-1}$ into $\mathbb{K}_t$ with $\left\{\mathcal{K}_t^{(g_k)}\right\}_{k \in [1, M]}$,

where $\theta_t^{(g_k)}$ denotes the parameters of group g_k at current step. At the step t of minimizing the objective $\mathcal{L}$, **Think** utilizes a *Strategy* ($\cdot$) operator to learn a strategy $\mathbb{P}_t(g)$ over all groups g based on cumulative available historical knowledge $\mathbb{K}_{t-1}$, and then samples at most M parameter groups $\{g_k\}$ as target groups, while **Touch** utilizes *Update* ($\cdot$) such as gradient descent to calculate the knowledge of target groups $\mathcal{K}_t^{(g_k)}$ and then update the groups correspondingly.

Fig. 1 also high-levelly illustrates our paradigm. The main difference between this paradigm and conventional counterparts is the *Think* substep, which allows us to divide long-term intensive fine-tuning into the composition of many short-term selective updating and have potential to further reduce redundancy by learning a good strategy.

Specifically, our paradigm have tree-fold intrinsic superiority. First, it significantly saves memory, since for untouched groups, it directly avoids their gradient calculation and storage. Second, it may explore those critical directions missed by prior works such as LoRAs via a strategy. Third, compared with some post-hoc heuristics, it is more feasible, since it learns a strategy $\mathbb{P}_t$ only from history $\mathbb{K}_{t-1}$ —assuming no current-step knowledge $\mathcal{K}_t$ available.

4.3. Theoretical Analysis

In this subsection, we provide some theoretical analysis and insights to essentially understand why and how the proposed paradigm can emulate FT, which can further release its potential (converging faster) and avoid possible degeneration.

To provide a basic theoretical guarantee for our *FT-approximability*, we first introduce several key concepts:

Definition 4.2 (Group gradient energy). Let θ be partitioned into G disjoint parameter groups $\{\theta^{(g)}\}_{g=1}^G$. For each group

g , define the group gradient energy at θ as follows:

$$w_g(\theta) := \|\nabla_g \mathcal{L}(\theta)\|^2, \quad (1)$$

where $\nabla_g \mathcal{L}(\theta)$ denotes the partial gradient of $\mathcal{L}$ w.r.t. $\theta^{(g)}$.

Definition 4.3 (Coverage Coefficient of gradient energy, α_t). At step t (assuming $M = 1$), let $w_g(\theta_t) \geq 0$ denote the group gradient energy. Define Coverage Coefficient as:

$$\alpha_t := \frac{\mathbb{E}_{g \sim \mathbb{P}_t(\cdot|\theta_t)}[w_g(\theta_t)]}{\frac{1}{G} \sum_{j=1}^G w_j(\theta_t)} = \frac{\sum_{g=1}^G \mathbb{P}_t(g|\theta_t) w_g(\theta_t)}{\frac{1}{G} \sum_{j=1}^G w_j(\theta_t)}. \quad (2)$$

Definition 4.4 (Cumulative Coverage of Gradient Energy). Given the coverage coefficients $\{\alpha_t\}_{t \geq 0}$, define the cumulative coverage up to time T as follows, with $\text{CC}(0) := 0$:

$$\text{CC}(T) := \sum_{t=0}^{T-1} \alpha_t. \quad (3)$$

Then, we show that the cumulative coverage decides the convergence by providing the next high-probability bound.

Assumption 4.5. We assume that: (i) (*Group smoothness*) for each group $g \in \{1, \dots, G\}$, the partial gradient $\nabla_g \mathcal{L}$ is L_g -Lipschitz; (ii) (*PL condition*) there exists $\mu > 0$ such that for all θ , $\frac{1}{2} \|\nabla \mathcal{L}(\theta)\|_2^2 \geq \mu(\mathcal{L}(\theta) - \mathcal{L}^*)$, where $\mathcal{L}^* := \inf_{\theta} \mathcal{L}(\theta)$; (iii) (*Selective update*, $M = 1$) at step t , only a group $g_t \sim \mathbb{P}_t(\cdot|\theta_t)$ is sampled and updated by $\theta_{t+1}^{(g_t)} = \theta_t^{(g_t)} - \frac{1}{L_{g_t}} \nabla_{g_t} \mathcal{L}(\theta_t)$.

Remark 4.6 (Scope of assumptions). The PL condition in Assumption 4.5 is used as an analytical surrogate for locally well-behaved over-parameterized optimization landscapes, a convention supported by convergence analyses of over-parameterized neural networks (Allen-Zhu et al., 2019; Du et al., 2019). For the bounded-gradient-energy condition used in our high-probability bandit analysis, standard gradient clipping in LLM fine-tuning provides a practical mechanism to control such quantities. The case $M = 1$ is adopted only for notational clarity; when $M > 1$, the same coverage and contraction arguments apply after replacing the single selected group energy by the summed energy over the selected set, which is also the regime used in our experiments.

Theorem 4.7 (A uniform high-probability bound via coverage). Let α_t be the coverage coefficient at step t defined in Eq. 2. Under Assumption 4.5, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$:

$$\forall t \geq 0, \mathcal{L}(\theta_t) - \mathcal{L}^* \leq \frac{\mathcal{L}(\theta_0) - \mathcal{L}^*}{\delta} \exp\left(-c_U \cdot \text{CC}(t)\right),$$

where $L_{\max} := \max_{g \in \{1, \dots, G\}} L_g$ and $c_U := \frac{\mu}{G L_{\max}}$.

The bound naturally gives a concise yet insightful condition.

Corollary 4.8 (A weak sufficient condition for convergence). Under the above definitions and assumptions, almost surely:

$$\text{CC}(T) \rightarrow +\infty \Rightarrow \mathcal{L}(\theta_T) - \mathcal{L}^* \rightarrow 0. \quad (4)$$

This Corollary shows that even under a very weak sufficient condition—the **cumulative coverage diverges**—our paradigm is guaranteed to recover a high-quality solution corresponding to a stationary point in the full-parameter space at which the FT admits no further first-order improvement and thus provably achieve full-expressivity $\mathcal{S} = \mathbb{R}^{|\theta|}$.

Remark 4.9. Intuitively, this result allows infinite numbers of failures (i.e., mismatching gradient norms with near-zero coverage α_t) during training and empowers nearly all non-degenerate strategies, e.g., uniform and some time-varying stochastic policies. **Such a broad feasible strategy space highlights the generality and flexibility of our paradigm.**

Next, to provide further insights on what learned strategies can be faster, we derive a lower bound on convergence rates.

Theorem 4.10 (Expected exponential (linear) convergence). With the above definitions and assumptions, we define:

$$c(\alpha) = c(\{\alpha_t\}) := \inf_{t \geq 0} \inf_{\theta: \sum_{j=1}^G w_j(\theta) > 0} \alpha_t(\theta). \quad (5)$$

If we further assume that $c(\alpha) > 0$, then for all $t \geq 0$:

$$\mathbb{E}[\mathcal{L}(\theta_t) - \mathcal{L}^*] \leq \exp\left(-c_U \cdot c(\alpha) \cdot t\right) (\mathcal{L}(\theta_0) - \mathcal{L}^*). \quad (6)$$

Remark 4.11. The rate of a Uniform strategy can be characterized via the factor c_U (with $c(\alpha) = 1$). Therefore, $c(\alpha)$ becomes a significant indicator measuring the advantage of a general strategy over the Uniform baseline, which theoretically shows that under-investing critical groups would slow down and even degrade convergence, explaining why some prior constrained paradigms can diverge from FT behavior.

5. A Bandit-based Online Selector

In this section, we propose a bandit-based technical framework to learn a strategy that can maximize the convergence speed solely from historical knowledge. To simplify the description and our theoretical analysis, we assume $M = 1$ herein, i.e., selecting one group at a time, and the extension to $M > 1$ is natural and technically straightforward.

In our implementation, we use row-level parameter groups because rows align with the memory layout of Transformer weight matrices and support efficient tensor slicing without breaking vectorized operations. Coarser layer-level groups tend to update many redundant parameters, while element-wise groups create a prohibitive number of bandit arms; practitioners can adjust the granularity according to the desired memory–flexibility trade-off.

Recalling the derived rate bound in Sec. 4.3, we observe that the optimization error g_T contracts at a rate governed by the factor $c(\alpha) = \inf \alpha_t$, which implies that we should accelerate the overall convergence by intuitively maximizing the energy ratio α_t at each step t . This insight motivates our oracle strategy selection and reward function design.

Oracle Strategy. Let $w_g(\theta_t) \geq 0$ denote the population energy of group g at step t —a group-gradient norm¹ in Def. 4.2—and denote the total energy $W_t := \sum_{j=1}^G w_j(\theta_t)$. We naturally choose the following post-hoc greedy oracle:

$$g_t^* \in \arg \max_{g \in [G]} w_g(\theta_t), \quad (7)$$

since it maximizes the energy ratio at each step in hindsight, i.e., $\text{Onehot}(g_t^*) \in \arg \max_{\mathbb{P}_t(\cdot | \theta_t)} \alpha_t$, or equivalently:

$$\alpha_t = \frac{\mathbb{E}_{g \sim \mathbb{P}_t(\cdot | \theta_t)} [w_g(\theta_t)]}{\frac{1}{G} \cdot W_t} \leq \frac{w_{g_t^*}(\theta_t)}{\frac{1}{G} \cdot W_t} = \max\{\alpha_t | \theta_t\}, \quad (8)$$

where the equality holds when $\mathbb{P}_t(\cdot | \theta_t) = \text{Onehot}(g_t^*)$.

Intuitively, this oracle allocates budgets to those groups with maximal potential at each step regarding a specific downstream task, and ideally achieves at least $\sup_{\alpha} c(\alpha) \geq 1$ times² of theoretical acceleration against the Uniform strategy. However, this oracle—*assuming all group energies at the current step are available*—is inherently non-implementable in our low-cost regime, whereas none can be known before practical decision making, which necessitates a framework that can online approximate it. That is also the core difference between our paradigm and existing post-hoc Coordinate Descent methods such as (Nesterov, 2012).

Reward Design. To explicitly connect the bandit objective (i.e., regret) with cumulative coverage $\text{CC}(T)$, we prioritize the most natural design of our reward function as below:

$$r_t^{(\varepsilon)}(g) := \frac{w_g(\theta_t)}{W_t + \varepsilon} \in [0, 1], \quad (9)$$

which is a *normalized energy* that theoretically connects α_t :

$$\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) | \mathcal{F}_{t-1} \right] = \frac{1}{G} \alpha_t^{(\varepsilon)}, \quad (10)$$

where $\mathcal{F}_{t-1}$ is a σ -algebra containing all randomness and historical decisions, and to ensure stability, we explicitly lower-bound the denominators of α_t and r_t by injecting an ε factor, which alters no theoretical properties (Sec. C.1).

How they connect to our paradigm? Challenge & Idea
Maximizing the cumulative rewards (minimizing the bandit

¹It is scaled by a group Lipschitz constant in our theoretical derivations, and we omit it in our practical implementation.

²It is a greedy local maximum rather than a global maximum.

Algorithm 1 Bandit-based online group selector

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1: Input: regularizer  $\varepsilon > 0$ , discount schedule  $\{\lambda_t\}_{t \geq 1}$ ,
   exploration schedule  $\{\gamma_t\}_{t \geq 1}$ , confidence  $\delta \in (0, 1)$ ,
   horizon  $T$ .
2: Initialize:  $N_1(g) \leftarrow 0$ ,  $S_1(g) \leftarrow 0$  for all  $g \in [G]$ .
3: for  $t = 1, 2, \dots, T$  do
4:    $\forall g \in [G]$ ,  $\hat{w}_g(\theta_t) \leftarrow \max\{S_t(g)/N_t(g), 0\}$ .
5:    $\hat{W}_t \leftarrow \sum_{j=1}^G \hat{w}_j(\theta_t)$ .
6:    $\forall g \in [G]$ ,  $\hat{r}_t^{(\varepsilon)}(g) \leftarrow \hat{w}_g(\theta_t)/(\hat{W}_t + \varepsilon)$ .
7:    $\forall g \in [G]$ ,  $\text{UCB}_t(g) \leftarrow \hat{r}_t^{(\varepsilon)}(g) + \eta_t(g)$ .  $\triangleright$  UCB
   indexes: rewards  $\hat{r}_t^{(\varepsilon)}$  and confidence radius  $\eta_t$  (in Eq. 30)
8:    $\hat{g}_t \in \arg \max_{g \in [G]} \text{UCB}_t(g)$ .
9:   Sample  $g_t \sim \pi_t = (1 - \gamma_t)\delta_{\hat{g}_t} + \gamma_t \text{Unif}([G])$ .
10:  Observe and estimate mini-batch energy  $\tilde{w}_{g_t}(\theta_t)$ .
11:  for  $g = 1, 2, \dots, G$  do
12:     $N_{t+1}(g) \leftarrow \lambda_t N_t(g) + \mathbf{1}\{g_t = g\}$ .
13:     $S_{t+1}(g) \leftarrow \lambda_t S_t(g) + \mathbf{1}\{g_t = g\} \tilde{w}_{g_t}(\theta_t)$ .
14:  end for
15: end for
    
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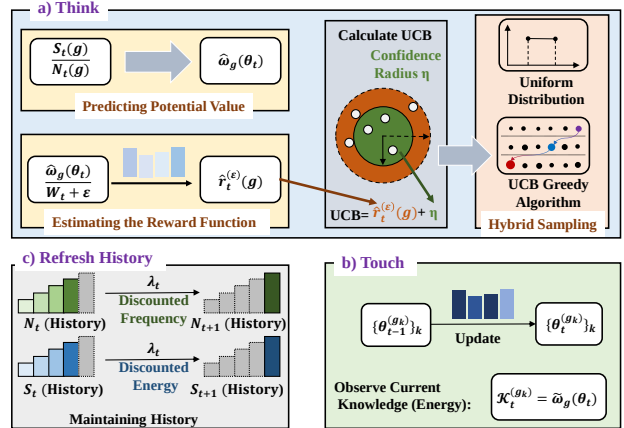


Figure 2. Illustration of our technical framework—how our bandit-based online selector implements the *Think-Touch* paradigm.

regret) is equivalent to maximizing $\text{CC}(T)$ that dominates convergence. However, we CANNOT directly observe real rewards with only sparse and noisy feedback, introducing an irreconcilable conflict between theory and practical algorithms. Our idea is to select groups based on history by tracking their short-term potentials via Exponential Moving Average (EMA) in *Think*, and collect partial current potentials after optimizer updating in *Touch*. In this implementation, our rewards can be interpreted as the normalized potential among all groups and thus measure the advantage of the selected groups over all the other counterparts.

5.1. Our Holistic Technical Framework

Algorithm 1 with its visual illustration (Fig. 2) summarizes the comprehensive pipeline of our selector, where we first

maintain EMA-style statistics for each group to extract historical knowledge and estimate plug-in rewards, then compute the confidence radius and UCB indexes, and finally sample M arms (*i.e.*, groups) optimistically based on these indexes with decreasing Uniform exploration.

We will detail in the next paragraphs some critical designs that solve the core challenge and facilitate the essence of our theoretical analysis on the oracle comparison. More details can be found in Sec. E.3 and Algorithm 2 of Appendix.

(1) Unbiased observation estimation $\tilde{w}_{g_t}$. Each step, we observe only a mini-batch gradient energy $w_{g_t}(\theta_t; \mathcal{B}_t)$ —not an unbiased estimation of the full-batch energy $w_{g_t}(\theta_t)$. Dividing $\mathcal{B}_t$ into two parts, *i.e.*, $\mathcal{B}_t = \mathcal{B}_{t,1} \cup \mathcal{B}_{t,2}$, we let $\tilde{w}_{g_t}(\theta_t) := \langle \nabla_{g_t} \mathcal{L}(\theta_t; \mathcal{B}_{t,1}), \nabla_{g_t} \mathcal{L}(\theta_t; \mathcal{B}_{t,2}) \rangle$. Thus, the fact $\mathbb{E}[\tilde{w}_{g_t}(\theta_t)] = w_{g_t}(\theta_t)$ shows the unbiasedness of $\tilde{w}_{g_t}$.

(2) Discounting schedule $\{\lambda_t\}$. Due to reward drifting, we should focus more on recent history via discounting; yet, strong discounting would degrade EMA counts $N_t(g)$, overly expand confidence radius, and thus slow down convergence. To better trade them off—*simultaneously tracking reward drifting and maintaining statistical stability*—we introduce a gradually increasing discounting schedule $\{\lambda_t\}$ to carefully control the effective discounted window: long enough to keep $N_t(g) \gg 0$ and shrink the confidence radius while short enough to track drift. We provide a simple anytime design in Eq. (80) with logarithmic window growth.

(3) Plug-in reward estimation with guarantees. Regarding bandit learning, it is the rewards that drive our selector to approximate the oracle. Yet, real rewards are NOT practically observable or computable even with an unbiased $\tilde{w}_{g_t}$, since $r_t^{(\epsilon)}(g_t) = w_{g_t}(\theta_t)/(W_t + \epsilon)$ depends on the unknown W_t . Directly summing $\tilde{w}_{g_t}$ to estimate W_t is still not feasible due to sparse feedback and many zero values. The insight to fill the empty numerators is to learn from recent historical knowledge. Therefore, we introduce EMA-style trackers (Lines 12, 13) and unbiasedly estimate both numerators and denominators (Lines 4, 5, omitting $\max\{\cdot, 0\}$ projection). Then, our plug-in reward estimator can be formalized: $\hat{r}_t^{(\epsilon)}(g) \leftarrow \hat{w}_g(\theta_t)/(\hat{W}_t + \epsilon)$ (Line 6 in Algo. 1).

Despite possible bias, it is asymptotically unbiased for real rewards under a sufficiently smooth drift. More rigorously, we prove a uniform high-probability deviation bound for $|\hat{r}_t^{(\epsilon)}(g) - r_t^{(\epsilon)}(g)|$ (Sec. E.11)—decomposed into shrinking confidence radius and drifting terms—and thus successfully connect bandit regret analysis to our LLM fine-tuning task.

(4) Hybrid exploration. We mix greedy selection with uniform exploration via a schedule $\{\gamma_t\}$ and point out a weak sufficient condition— $p_t(g) \geq \gamma_t/G$, $\forall g \in [G]$ and $\sum_{t \leq T} \gamma_t = o(T)$ —that guarantees that each group gradually receives enough feedback to control both confidence

radius and drift accumulation, while keeping total exploration asymptotically negligible in the exponent. One simple example is our log-time decreasing construction in Eq. (82).

5.2. Theoretical Guarantees of Our Online Selector:

“ *Regret $\Rightarrow$ Coverage $\Rightarrow$ Rate* ”

By the above designs, we make our framework not only practically feasible but also theoretically analyzable. In this subsection, we prove that our selector tracks the oracle in dynamic regret and exactly recovers the linear oracle-rate factor up to vanishing sublinear terms—including a high-probability regret bound, cumulative coverage control, and convergence rate recovery. We defer the complete statement, all proofs, and detailed constructions to Sec D-I of Appn.

(1) High-probability bound of dynamic regret.

Definition 5.1 (Dynamic regret against Oracle).

$$\text{Reg}^{(\epsilon)}(T) := \sum_{t=1}^T (r_t^{(\epsilon)}(g_t^*) - r_t^{(\epsilon)}(g_t)).$$

The following bound makes explicit what governs regret.

Theorem 5.2 (High-probability dynamic regret bound). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$, we have:*

$$\begin{aligned} \text{Reg}^{(\epsilon)}(T) \leq & \underbrace{\frac{1}{1 - \gamma_{\max}} \sqrt{2T \log \frac{2}{\delta}}}_{(I) \text{ concentration term}} + \underbrace{\sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t}}_{(II) \text{ exploration}} \\ & + \underbrace{2 \sum_{t=1}^T \eta_t(g_t)}_{(III) \text{ estimation error}} + \underbrace{2 \sum_{t=1}^T \text{Drift}_t}_{(IV) \text{ drift term}}, \end{aligned} \quad (11)$$

where $\gamma_{\max} := \max_{1 \leq t \leq T} \gamma_t$. With mild assumptions and the constructions in Sec. E, the regret becomes sublinear.

Remark 5.3 (What are the four ingredients). (I) *Concentration term* controls martingale deviations between random rewards and their conditional means. (II) *Exploration term* is the price of $\{\gamma_t\}$ -scheduled exploration, facilitating well-conditioned EMA statistics. (III) *Estimation error* captures the gap between the plug-in ratio reward $\hat{r}_t(g)$ and the true ratio $r_t(g)$ caused by denominator estimation. (IV) *Drift term* captures non-stationarity—reward drift—governed by the total variation budget of the reward sequence $\{r_t^{(\epsilon)}\}$, *i.e.*, $V_T^{(\epsilon)} := \sum_{t=2}^T \max_{g \in [G]} |r_t^{(\epsilon)}(g) - r_{t-1}^{(\epsilon)}(g)|$.

Remark 5.4 (How to make the regret $o(T)$). A simple yet sufficient system ensuring $o(T)$ regret is constructed in Sec. E.7, requiring only $V_T^{(\epsilon)} = o(T/\text{polylog } T)$ —a very mild drifting condition. Thereby, the online selector successfully tracks the oracle over time $\text{Reg}^{(\epsilon)}(T)/T \rightarrow 0$.

(2) Regret lower-bounds cumulative coverage. Recall in Sec. 4.7, we show that cumulative coverage $\text{CC}(T)$ governs

the convergence. The next result provides an intuition that we can emulate the Oracle’s $CC(T)$ by controlling regrets.

Theorem 5.5 (From dynamic regret to cumulative coverage). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$, we have:*

$$CC(T) \geq CC^*(T) - G \cdot \mathcal{F}(T, \delta), \quad (12)$$

$$\mathcal{F}(T, \delta) := \text{Reg}^{(\varepsilon)}(T) + \sqrt{2T \log \frac{1}{\delta}}, \quad (13)$$

where $CC^*(T)$ denotes the cumulative coverage obtained by the post-hoc greedy oracle g_t^* within the first T steps. In particular, whenever $\text{Reg}^{(\varepsilon)}(T) = o(T)$, we have $CC^*(T) - CC(T) \leq o(T)$, which shows that **the average missed energy ratio tends to vanish over time**.

(3) Final rate guarantee and oracle-rate recovery. Based on the above results, we further compare our overall convergence rate with the oracle’s by providing the next theorem.

Theorem 5.6 (Rate comparison: Oracle v.s. Online). *For any $\delta \in (0, 1)$, with probability at least $1 - \delta$, we have:*

$$\Delta_{T,orc} \leq \Delta_0 \cdot \exp\left(-c_U \cdot CC^*(T) + \ln \frac{2}{\delta}\right),$$

$$\Delta_T \leq \Delta_0 \cdot \exp\left(-c_U \cdot CC^*(T) + c_U G \mathcal{F}(T, \delta) + \ln \frac{2}{\delta}\right),$$

where $\Delta_T := \mathcal{L}_T - \mathcal{L}^*$ and $\Delta_{T,orc} := \mathcal{L}_{T,orc} - \mathcal{L}^*$ is the loss gap of our selector and the oracle strategy. Whenever $\mathcal{F}(T, \delta) = o(T)$ —the gap vanishing over time—the online selector will recover exactly the same rate as the oracle’s.

Corollary 5.7 (Linear oracle-rate recovery). *Particularly, our selected oracle has $CC^*(T) = c(\alpha^*) \cdot T$ with the advantage factor $c(\alpha^*) \geq 1$, and thus the online selector will recover the factor exactly, i.e., $c(\alpha) = c(\alpha^*)$:*

$$\Delta_T \leq \Delta_0 \cdot \exp\left(-c_U \cdot c(\alpha^*) \cdot T + \ln \frac{2}{\delta}\right), \quad (14)$$

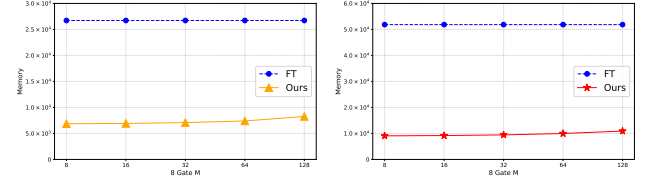
$$= \Delta_0 \cdot \exp\left(-T \cdot (c_U \cdot c(\alpha^*) - o(1))\right), \quad (15)$$

which shows that even under the “one-group-per-step” budget, our selector provably recovers the oracle linear-rate convergence constant, only if it has a sublinear regret.

Remark 5.8. From a decision-making perspective, the selector’s decisions coincide with the oracle’s on all but a vanishing fraction of rounds, which intuitively explains why the selector converges faster by learning from only history.

6. Experiments

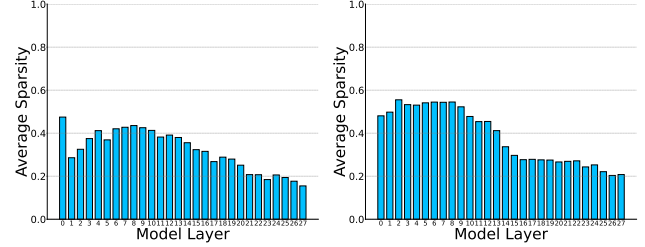
Tasks and datasets. We run experiments on a single commonsense reasoning benchmark comprising eight sub-tasks, each with a predefined train/test split. Following Dora (Liu et al., 2024), we merge the training sets of all eight sub-tasks into a single training corpus, and evaluate on the test set of each sub-task separately.



(a) Qwen2.5-1.5B-Instruct

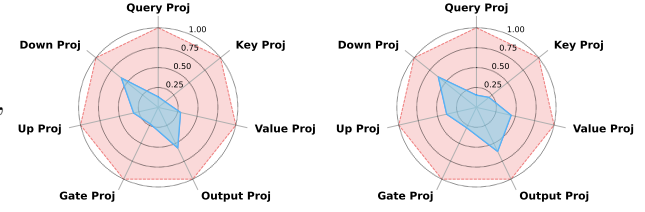
(b) Llama-3.2-3B-Instruct

Figure 3. Comparisons of peak GPU memory (GB) against full fine-tuning (FT), with batchsize = 1. Our method avoids gradient/optimizer states for untouched rows.



(a) Qwen2.5-1.5B-Instruct
(By Layer ID)

(b) Llama-3.2-3B-Instruct
(By Layer ID)



(c) Qwen2.5-1.5B-Instruct
(By Module Types)

(d) Llama-3.2-3B-Instruct
(By Module Types)

Figure 4. **Gradient-energy concentration in dense FT on the mixed commonsense training set.** (1) **Top:** Energy Sparsity Index (ESI; Sec. 6.4) summarized as mean $\pm$ std among different layers. (2) **Bottom:** Normalized energy composition across parameter types (outer polygon is total energy = 1; inner polygon shows the average ratio of FT energy for each module type, e.g. Linear projections and attention matrices W_Q, W_K, W_V , etc.). These plots show that FT gradients are both (i) concentrated in a small subset of groups and (ii) structured by module type, motivating selective low-cost updates that prioritize the most energetic groups.

Backbones, baselines, and implementation. We consider two backbone models: Qwen2.5-1.5B-Instruct (Qwen et al., 2025) and Llama-3.2-3B-Instruct (Grattafiori et al., 2024). In our implementation, we partition each trainable matrix into fine-grained rows and treat each row as a parameter group. We compare against full fine-tuning (FT) and representative PEFT baselines, including LoRA (rank 32 and 64) (Hu et al., 2022), DoRA (Liu et al., 2024), and RandLoRA (Albert et al., 2025). Our method is a bandit-based online selector (Sec. 5) with per-step budgets $M \in \{32, 64, 128\}$, denoted OURS ($M = 32/64/128$).

Additional camera-ready studies requested during review, including larger-model math/code results, FISH Mask comparisons, wall-clock runtime, and empirical coverage-speed

evidence, are reported in Appendix C.

6.1. Main Results

As shown in Tab. 1, on eight commonsense reasoning tasks, OURS consistently achieves the best LC-FT performance among low-cost methods across most sub-tasks. Under comparable trainable-parameter budgets, OURS improves over standard LoRA baselines ($r=32/64$) by 1.5% on average (and up to 4.8% on the best sub-task), suggesting that online, feedback-driven selection of parameter groups yields more effective updates than a fixed low-rank subspace. Compared with DoRA and RandLoRA, OURS also shows a clear advantage of 0.4%–1.0% on average, indicating that adaptivity matters beyond the specific PEFT parameterization.

Moreover, as the per-step budget increases from $M = 32$ to $M = 128$, OURS achieves a monotonic gain of 3.1% (averaged over these tasks) and narrows the gap to full fine-tuning (FT), with the remaining FT–OURS gap reduced to 1.0% at $M = 128$. These results support our motivation: even with a small per-step update budget, an online selector can allocate budgets to the most beneficial parameter groups over time and approach the effectiveness of FT.

The additional results in Appendix C further show that this trend persists on a larger Llama-3-8B-Instruct backbone for math and coding tasks and against a full-rank sparse-update baseline.

6.2. Peak GPU Memory

As expected, OURS substantially reduces peak GPU memory compared with FT (Fig. 3), and the reduction is largest at the smallest budget M . This trend is expected because OURS updates only a limited number of parameter groups per step, which reduces the amount of optimizer state and activation/gradient storage that must be kept resident. ***Crucially, the memory reduction does not sacrifice effectiveness:*** despite using far fewer updated groups per step, OURS remains competitive with FT and ***consistently outperforms strong PEFT baselines*** such as LoRA ($r=32/64$), DoRA, and RandLoRA (Tab. 1). These results highlight a practical advantage of LC-FT: *we can trade a small per-step budget for a large memory saving*, while the online selector allocates that budget to the most useful groups over time, preserving much of FT’s progress. Overall, this aligns with the goal of LC-FT: *retain much of FT’s capability while reducing per-step cost*.

6.3. Convergence Speed

We conducted an experiment on convergence. Under the same update budget, OURS converges faster than UNIFORM on loss curves (Fig. 5), supporting the core message that LC-FT is a *strategy learning* problem rather than a static sparsification rule. This is consistent with our theory: the

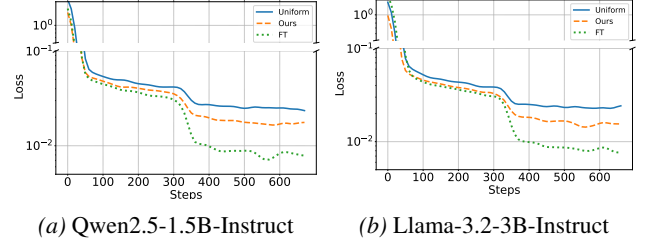


Figure 5. Training loss versus step on commonsense reasoning. We compare FT, UNIFORM, and OURS, showing that OURS reaches a lower loss in fewer steps and tracks FT optimization trajectory more closely than UNIFORM baseline, indicating faster convergence under the same per-step budget. Our bandit-based selector may need some steps to *explore* in early training.

selector is designed to approximate the (infeasible) oracle preference using only past information (Sec. 5), so better oracle tracking translates into a larger effective contraction in the long run (Theorem 5.6).

Appendix C also reports wall-clock measurements: on the 1.5B backbone, OURS uses 0.1632s per step, compared with 1.2803s for FT, so a step-wise lag relative to FT can still yield a favorable computation–accuracy trade-off in practice.

6.4. Motivation: Gradient-Energy Sparsity in FT

Energy sparsity index (in $[0, 1]$). To quantify the empirical motivation for *Think–Touch*, we measure the degree of concentration of the group’s energies during FT. At each step, sort energies in ascending order $w_{(1)} \leq \dots \leq w_{(G)}$, min–max normalize $\tilde{w}_{(i)} := w_{(i)} / \sum_{j=1}^G w_{(j)} \in [0, 1]$, and compute the cumulative sum sequence $s_t := \sum_{i=1}^t \tilde{w}_{(i)}$. Let $x_i := (i - 1)/(G - 1)$, $y_i := \tilde{w}_{(i)}$ and view $\{(x_i, y_i)\}_{1 \leq i \leq G}$ as a discrete curve over $x \in [0, 1]$, and then we define the trapezoidal area as:

$$\text{AUC} := \sum_{i=1}^{G-1} \frac{s_i + s_{i+1}}{2} \cdot \frac{1}{G-1}.$$

Since the diagonal baseline from $(0, 0)$ to $(1, 1)$ has area $1/2$, we define the *Energy Sparsity Index* as follows:

$$\text{ESI} := \text{AUC}/(1/2) = 2 \cdot \text{AUC} \in [0, 1],$$

where smaller values indicate stronger concentration, *i.e.*, few groups dominate the energy.

Discussion. We empirically find that FT exhibits consistently low average ESI (mean over steps; see the radar plot in Fig. 4) across most layers and most module types, implying that a small fraction of groups carries most of the gradient energy at many steps. This provides direct empirical support for LC-FT: if energy is sharply skewed, then *touching only a small number of well-chosen rows per step* can preserve most of FT’s progress, and our online selector

Low-cost Full Fine-tuning

Backbone	Method	#Params (%)	ARC-c	ARC-e	BoolQ	HellaSwag	OBQA	PIQA	SIQA	WinoGrande	Avg.
Qwen2.5-1.5B-Instruct	FT	100.00	78.14	90.29	68.50	91.91	87.20	84.35	78.32	81.18	82.49
	LoRA ($r = 32$)	2.34	75.82	88.65	65.90	90.30	82.81	82.70	75.20	76.25	79.70
	LoRA ($r = 64$)	4.57	76.81	88.98	66.95	90.75	83.79	82.27	75.10	76.64	80.16
	DoRA ($r = 64$)	4.60	77.71	90.34	67.80	89.39	83.60	83.32	76.02	73.07	80.16
	RandLoRA ($r = 64$)	0.35	78.73	90.00	67.77	89.50	84.20	84.02	77.04	75.35	80.83
	OURS ($M = 32$)	1.03	73.44	87.87	65.99	86.97	77.15	80.66	74.90	69.84	77.10
	OURS ($M = 64$)	2.01	77.30	88.82	66.86	90.81	84.57	83.30	75.86	75.86	80.42
	OURS ($M = 128$)	3.85	79.58	90.21	67.68	91.14	85.00	83.26	76.63	75.59	81.14
Llama-3.2-3B-Instruct	FT	100.00	79.86	91.12	72.69	94.11	87.00	87.00	79.89	84.29	84.50
	LoRA ($r = 32$)	1.49	76.28	89.94	67.43	91.87	78.40	83.95	75.86	80.74	80.56
	LoRA ($r = 64$)	2.94	77.13	90.15	68.75	92.08	81.40	84.06	80.04	81.06	81.83
	DoRA ($r = 64$)	2.96	77.65	90.24	70.43	93.25	84.20	84.11	78.92	83.11	82.74
	RandLoRA ($r = 64$)	0.68	78.16	90.49	71.74	93.77	84.00	84.93	80.04	83.66	83.35
	OURS ($M = 32$)	0.73	77.73	90.45	70.40	91.30	82.40	79.53	79.72	82.05	81.70
	OURS ($M = 64$)	1.44	77.65	90.32	70.49	92.69	83.20	85.42	79.89	83.74	82.93
	OURS ($M = 128$)	2.80	79.01	90.32	71.28	93.49	86.00	86.13	79.68	84.69	83.83

Table 1. Performance comparison on eight commonsense reasoning tasks with two LLM backbones. We report accuracy (%) for each task and the average across all tasks. Note that our framework with budget $2M$ has nearly the same para count as LoRA with $r = M$.

is precisely the mechanism that turns this observation into a principled training rule rather than a heuristic.

7. Conclusion

We investigate *Low-cost Full Fine-tuning* (LC-FT): achieving full expressivity of dense full fine-tuning with a limited per-step update budget. We propose the *Think-Touch* paradigm, which selects parameter groups based on past knowledge (*think*) and updates only the selected (*touch*). Our theory shows that convergence is governed by *cumulative coverage* of group gradient energy, yielding a clean rate characterization and motivating an ideal but infeasible post-hoc oracle strategy. To approximate this oracle under partial, noisy, and drifting feedback, we estimate the unobservable rewards and design an online bandit-based selector that balances exploitation, discounting, and exploration. Theoretically, by deriving a regret bound that controls cumulative coverages, we show that the online selector recovers the oracle’s linear-rate constant asymptotically up to vanishing sublinear corrections. Empirically, our framework saves substantial memory against full fine-tuning and performs better than popular parameter-efficient baselines, highlighting *Think-Touch* as a principled and scalable training paradigm. Future work targets richer group structures, contextual signals, and larger-scale training systems, etc. Besides, our work can also be extended to reinforcement learning-based Large language model alignment and some continual pre-training tasks under per-step low-cost constraints, which would be explored in the future.

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Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Related Work

In this section, we review prior work on memory-efficient training and parameter-efficient adaptation, followed by bandit-theoretic foundations that relate to our regret-controlled parameter-group selection for fine-tuning.

Parameter-efficient fine-tuning (PEFT). PEFT methods reduce training and storage costs by updating only a small set of parameters, often by constraining updates to a fixed low-dimensional or modular structure. Adapters insert small trainable modules into each layer and support transfer learning and modular reuse (Pfeiffer et al., 2020; 2021). LoRA-style methods use low-rank updates, and quantization-aware PEFT methods (e.g., QLoRA) further reduce memory by training adapters on top of a quantized backbone (Dettmers et al., 2023). Other directions include tuning only biases (BitFit) (Zaken et al., 2022), activation rescaling (IA³) (Liu et al., 2022), and structured adapter parameterizations such as Compacter (Karimi Mahabadi et al., 2021). UniPELT further combines multiple PEFT modules and learns how to activate them (Mao et al., 2022). While these methods are highly efficient and effective in many transfer settings, their expressivity is fundamentally limited by the chosen parameterization, making them less expressive than full fine-tuning. Our approach explicitly targets full expressivity as the fundamental goal: instead of fixing a restricted update form, we adaptively select parameter groups to update over time and use regret guarantees to control the cumulative deviation from full fine-tuning behavior under constrained per-step updates.

Efficient fine-tuning and training systems. A large part of the cost of fine-tuning comes from memory for activations, gradients, and optimizer states. Classic system-domain techniques reduce activation memory via checkpointing and recomputation (Chen et al., 2016), and reduce compute/memory bandwidth by mixed-precision training (Micikevicius et al., 2018). Beyond system-level optimizations, optimizer-side changes can also reduce memory while still updating the full parameter set, such as low-rank projections of gradients (Zhao et al., 2024). These approaches aim to improve their flexibility but are still less expressive than full fine-tuning, due to, e.g., the low-rank limitation. Moreover, they primarily change the execution pipeline (e.g., recomputation schedules) or the optimizer representation (e.g., projected gradients). In contrast, our work treats *which parameter groups to update* as a sequential decision problem under partial and noisy feedback, and provides an explicit regret guarantee for long-horizon group allocation rather than a system/optimizer redesign. Another line of work reduces cost via sparsity in parameters or computation. The lottery ticket hypothesis suggests that suitably chosen sparse subnetworks can match dense performance (Frankle & Carbin, 2019). In transfer settings, pruning-based fine-tuning methods such as movement pruning explore sparse adaptation while leveraging pretrained initialization (Sanh et al., 2020). Diff pruning learns sparse “diff” parameters relative to a frozen backbone, aiming to improving performance while modifying only a small fraction of weights (Guo et al., 2021). Dynamic sparse training methods (e.g., RigL) adjust sparse connectivity during training under a fixed budget (Evci et al., 2020). Other approaches reduce memory by maintaining only a subset of weights during optimization (e.g., DropBack) (Golub et al., 2018). Selective Backprop focuses on reducing computation by skipping gradient updates for certain data points (Jiang et al., 2019), which is orthogonal to parameter selection. Although these methods can substantially reduce training cost, they rely on enforcing or learning sparsity patterns that constrain the update space at each step, which also risks requiring an unignorable cost, e.g., a holistic training to obtain a high-quality lottery ticket. Our fundamental goal is not to seek a sparse subnetwork. Our method instead preserves the potential expressivity of full fine-tuning across time: by sequentially selecting which parameter groups to update and optimizing this selection via regret minimization, we aim to recover dense-update behavior in aggregate without committing

to a static or heuristic sparsity structure.

Classic Foundations of Bandit Theory. This paragraph summarizes bandit principles most directly related to our regret analysis. Information-theoretic lower bounds characterize the optimal asymptotic regret in stochastic bandits (Lai & Robbins, 1985), and UCB-style exploration achieves logarithmic regret under bounded stochastic rewards (Auer et al., 2002a). Adversarial bandits formalize robustness to nonstochastic feedback, with EXP3 achieving $\tilde{O}(\sqrt{KT})$ regret (Auer et al., 2002b). Thompson sampling provides a randomized exploration strategy with provable efficiency in stochastic settings (Thompson, 1933; Agrawal & Goyal, 2012). KL-UCB refines index design via KL-based confidence bounds for tighter instance-dependent regret (Garivier & Cappé, 2011). Structured bandit models such as stochastic linear bandits exploit feature structure to reduce regret dependence on the number of arms (Dani et al., 2008), with OFUL providing high-probability guarantees (Abbasi-Yadkori et al., 2011). Beyond cumulative regret, best-arm identification focuses on identification objectives under sampling constraints (Audibert & Bubeck, 2010; Kaufmann et al., 2016). Extensions to non-stationary and corrupted environments measure difficulty via variation or corruption budgets (Garivier & Moulines, 2008; Besbes et al., 2014; Gupta et al., 2019). Standard references synthesize these results and techniques (Bubeck & Cesa-Bianchi, 2012b; Lattimore & Szepesvári, 2020; Cesa-Bianchi & Lugosi, 2006). While these works study bandit feedback as a primary abstraction, our contribution leverages regret as a tool to control long-horizon deviation from full fine-tuning when feedback arises indirectly from partial and noisy gradient observations.

Recent Advances in Bandit Theory. Recent work refines minimax guarantees and robustness properties, strengthening regret as a principled objective. Sharp minimax characterizations for bandits with expert advice and contextual linear bandits with bounded context sets are established in (Ito, 2024). Conditions under which linear bandits admit polylogarithmic minimax regret are identified in (Lumbreras & Tomamichel, 2024). Best-of-both-worlds algorithms that adapt between stochastic and adversarial linear contextual regimes are developed in (Kuroki et al., 2024). Near-optimal regret for adversarial sleeping bandits under availability constraints is studied in (Nguyen & Mehta, 2024). Robustness under privacy, corruption, and heavy-tailed noise is analyzed in (Zhou & ZHANG, 2024; Hu & Chen, 2025). More precise finite-time regret characterizations and inference implications for UCB-style algorithms are provided in (Han et al., 2024a). These advances motivate using regret as a unifying design principle; however, rather than proposing a new bandit model, our work instantiates regret minimization to govern parameter-group updates in fine-tuning, with the explicit goal of recovering full-update expressivity over long horizons under memory constraints.

B. More Discussions

B.1. Why LoRA/Adapters Generally Fail to Match FT

Both LoRA and adapters are designed to reduce cost by restricting the update space. Therefore, without additional assumptions on the full fine-tuning solution (such as low-rank assumptions), they cannot *theoretically guarantee* reaching the full fine-tuning optimum.

LoRA: low-rank update is a hard constraint. For each matrix that needs to be adapted or fine-tuned, LoRA restricts ΔW to be rank- $\leq r$ (Hu et al., 2022). In contrast, full fine-tuning can realize an arbitrary update ΔW_{FT} (potentially full rank). If the layer-wise full fine-tuning update ΔW_{FT} has rank larger than r , then no choice of (A, B) can represent it exactly. More formally and quantitatively, let $\Delta W_{\text{FT}} = U\Sigma V^\top$ be the SVD. Among all rank- $\leq r$ matrices, the best approximation error satisfies the classical Eckart–Young characterization:

$$\min_{\text{rank}(\Delta W) \leq r} \|\Delta W_{\text{FT}} - \Delta W\|_F^2 = \sum_{i>r} \sigma_i(\Delta W_{\text{FT}})^2,$$

where $\{\sigma_i\}_i$ are decreasing or non-increasing. Hence, whenever the singular values beyond rank r carry non-negligible mass (*i.e.*, a sum much larger than zero), there is an irreducible “missed-subspace” error. Recent analyses show that low-rank adaptation and full fine-tuning can lead to systematically different spectral structures and generalization behaviors, even when in-distribution task performance is similar (Shuttleworth et al., 2024), (Biderman et al., 2024). Related work further motivates closing the capacity gap between LoRA and full fine-tuning, *e.g.*, by studying why LoRA differs from full fine-tuning and proposing variants that are better similar to full fine-tuning expressivity with separate magnitude and angle modeling (Liu et al., 2024), or by explicitly targeting *full-rank* update capability while keeping few trainable parameters (Albert et al., 2025).

From a practical standpoint, the missed subspace can matter in tasks that require many *independent* functional changes across representation directions. For instance, if a downstream task requires adjusting a linear map along k nearly orthogonal directions (e.g., correcting k distinct features that are not redundant), the resulting ΔW_{FT} can have effective rank on the order of k ; choosing $r \ll k$ forces LoRA to discard these directions, leading to a non-vanishing approximation gap. This perspective is consistent with theoretical investigations of LoRA’s expressive power, which characterize rank thresholds and approximation error when the rank is insufficient (Zeng & Lee, 2024).

Adapters: bottlenecked residual updates restrict the reachable transformations. Adapters restrict updates by only allowing changes through a low-dimensional bottleneck (Houlsby et al., 2019). To see the limitation, consider the (local) linear regime where σ can be approximated by the identity around typical activations. Then the adapter contribution becomes approximately

$$U \sigma(Dh) \approx U D h,$$

which is a linear map of rank at most m (since $UD \in \mathbb{R}^{d \times d}$ factors through $\mathbb{R}^m$). Thus, at the representation level, the adapter can only introduce perturbations that lie in a low-dimensional subspace controlled by m . Full fine-tuning, in contrast, can modify the entire layer transformation in a dense manner. Unless the bottleneck dimension m is taken comparable to d (which defeats the parameter-efficiency goal), adapters cannot guarantee matching the full fine-tuning solution.

This limitation is conceptually different from the observation that some tasks may admit low-dimensional reparameterizations (Aghajanyan et al., 2021), since while low intrinsic dimension suggests *existence* of compact adaptations for certain settings, fixed-structure PEFT methods still commit to a particular restricted family (low-rank updates or bottleneck residuals), which can be mismatched to tasks where the full fine-tuning update distributes across many directions.

B.2. More Discussions on Relationship to Prior Works

This subsection clarifies what our work is relative to several nearby lines of research. Across all comparisons, the most critical point is that we do *not* propose yet another post-hoc, gradient-informed sparse-update heuristic or a fixed PEFT parameterization. Instead, we formalize *low-cost full fine-tuning* as a *budgeted emulation* problem: per step, we can only touch a small number of structured parameter groups, and the choice must be made *before* observing dense or full gradients—using only partial, noisy, and slowly drifting feedback—while still aiming to recover dense-FT expressivity or local minimum (a stationary point in the full-parameter space) over a long horizon with regret-style guarantees. We then (i) separate our setting from classical coordinate descent / Gauss–Southwell rules (which assume access to global per-block scores), (ii) distinguish it from heuristic subset-of-weights training such as DropBack (which relies on current-step gradients and learns a sparse difference from initialization), and (iii) explain why random-basis PEFT (e.g., RandLoRA (Albert et al., 2025)) and DoRA (Liu et al., 2024) address *the low-rank bottleneck in fixed-parameterization* with still restricted expressivity (using a few trainable parameters), whereas we target the more fundamental bottleneck: turning “few parameters *per step*” into effectively “many (full) parameters *over time*” without being restricted to any low-dimensional subspace and thus we achieve full-expressivity like standard full fine-tuning (FT) over the long run.

B.2.1. DISCUSSION ON RELATIONSHIP TO COORDINATE DESCENT AND GAUSS–SOUTHWELL RULES

Although our post-hoc oracle resembles Gauss–Southwell-type greedy block rules in form (e.g., selecting the parameter group with the largest block-gradient norm), our formulation is fundamentally different from classical coordinate descent. Coordinate descent is typically proposed as an alternative optimizer: updating a single coordinate/block is motivated by computational or structural convenience, and greedy rules assume that coordinate scores (e.g., full gradients or per-block utilities) are accessible at each iteration. In contrast, we treat standard dense full fine-tuning (FT) as the reference training regime and study a budgeted emulation problem: ***how to approach a dense FT-admitted minimizer in the full-parameter space under a strict per-step resource constraint***. At the LLM scale, obtaining global block-gradient norms for all groups generally requires near-dense backpropagation and materializing the full backward dependency graph, which collapses to dense FT in cost and defeats the low-cost goal. Therefore, **in our proposed paradigm, the key challenge is not to design another greedy coordinate optimizer, but to learn an online selector from partial, noisy, and gradually drifting feedback that can approximate the post-hoc oracle in the long run**, together with regret guarantees.

From another perspective, our paradigm and technical framework can offer a flexible, general interface for importing a broad family of coordinate-descent rules and their variants into LLM fine-tuning. Under a weak gradual time-variation assumption, such rules—often defined as idealized oracle selectors—can be approximated online from partial and noisy feedback

with formal regret guarantees, potentially bridging classical optimization heuristics and practical, resource-constrained fine-tuning at scale.

B.2.2. RELATION TO COORDINATE DESCENT WITH BANDIT SAMPLING.

Salehi et al. (Salehi et al., 2018) also connect coordinate descent with bandit sampling under partial feedback, but their goal is to accelerate classical coordinate descent for structured objectives, whereas our work formulates LC-FT as a budgeted emulation of dense full fine-tuning for LLMs. Their reward is derived from immediate objective decrease in the coordinate-descent setting; in contrast, our reward is normalized group-gradient energy, which is an unbiased proxy for the coverage coefficient that directly controls our loss-contraction analysis. Our selector also handles multi-group selection, uncertainty-aware exploration, history discounting, and reward drift, which are central to partial, noisy, and non-stationary LLM fine-tuning.

B.2.3. DISCUSSION ON RELATIONSHIP TO HEURISTIC SPARSE TRAINING TECHNIQUES (e.g., DROPBACK) AND OUR KEY DIFFERENCES.

Summary. This subsection notes that both our method and heuristic subset-of-weights training (e.g., DropBack (Golub et al., 2018)) reduce training cost by updating only a subset of parameters. However, DropBack selects weights using current-step gradient signals and enforces a sparse difference-from-initialization parameterization, whereas our goal is to approximate dense full fine-tuning under a strict per-step budget without access to full gradients. We therefore study a before-hoc budget allocation problem with partial/noisy and drifting feedback, and provide a general analysis for convergence to full fine-tuning at the level of parameter groups.

Key point 1: DropBack is post-hoc and gradient-informed (it uses current-step gradients), whereas ours is before-hoc and does not assume access to full gradients.

Key point 2: DropBack learns a sparse difference ΔW from initialization, whereas ours targets low-cost approximation to dense full fine-tuning without restricting the update to a fixed sparse subspace.

Our work is related to weight subset training under a controllable weight budget (e.g., DropBack (Golub et al., 2018)) because both employ *sparse updates* to reduce training cost. Instead of the full memory and compute footprint of dense backpropagation and optimizer states, one updates only a restricted subset of parameters, and this subset may change over time. Despite the similar intuition, the settings are fundamentally different. In DropBack (Golub et al., 2018), subset selection is *gradient-informed* and uses *current-step gradients* to decide which *individual* weights are tracked. Moreover, DropBack (Golub et al., 2018) keeps non-tracked weights at their initialization, which effectively learns a sparse difference $\Delta W := W - W^{(0)}$.

In contrast, our goal is low-cost full fine-tuning rather than sparse- ΔW adaptation. Specifically, we aim to approximate the dense full fine-tuning objective under a strict per-step budget, without restricting the solution to a fixed sparse-difference subspace.

More importantly, our paradigm is *before-hoc*: we must decide which *structured parameter groups* (layers/modules/blocks/rows) to update *without* access to full current gradients, relying only on history and partial/noisy feedback. This is necessary because computing, storing, or ranking fine-grained gradients at large language models is too expensive or even strictly forbidden.

Consequently, different from post-hoc heuristics: our central problem is reliable *budget allocation* under limited observability and non-stationarity, while still recovering the optimization ability and expressivity of dense full fine-tuning. Our contribution therefore goes beyond a new selection rule: we provide a general analysis of this paradigm, identify weak sufficient conditions for convergence to full fine-tuning, and characterize factors that control the convergence speed. Finally, using group-level actions is also a practical design choice: it aligns with the modular structure of Transformers and with selective backward and distributed training, making the framework applicable to modern large-model fine-tuning pipelines. Overall, we target a general, scalable, and theoretically grounded low-cost fine-tuning framework, while recognizing DropBack (Golub et al., 2018) as a meaningful related work in the shared topic of sparse updating.

B.2.4. DISCUSSION ON RELATION TO RANDLoRA-STYLE RANDOM-BASIS PEFT AND OUR KEY INSIGHT.

Summary. This subsection starts from a common explanation of why PEFT methods can lag behind full fine-tuning: low memory often means few trainable parameters, which is often linked to low-rank updates and limited expressivity. It then explains that RandLoRA (Albert et al., 2025) breaks the link between “few parameters” and “low rank”, but argues that the deeper limitation is still the small number of trainable parameters, since it restricts the update space. Finally, it motivates our key idea: under a strict per-step budget, changing the updated parameter groups over time can accumulate into effectively dense changes and can converge to the full fine-tuning objective under weak conditions.

Key point 1: *Decoupling rank from parameter count (as in RandLoRA) helps, but it does not remove the core limit from using only a small number of trainable parameters—unavoidably sacrificing expressivity.*

Key point 2: *With a strict per-step budget, short-term sparsity can still yield long-term expressivity by updating different parameter groups over time, enabling convergence toward full fine-tuning.*

We position our work using a common “folk chain”:

$$\begin{aligned} \text{low memory} &\implies \text{few trainable parameters} \implies \text{low-rank / low-dimensional updates} \\ &\implies \text{limited expressivity} \implies \text{lower performance (hard tasks)}, \end{aligned}$$

where *low memory* often leads to ***few trainable parameters***, which is often linked to ***low-rank updates***, then to limited expressivity and lower performance on hard tasks. Classical LoRA is a standard example: it saves memory by training few parameters, but each matrix receives a rank- r update, which removes directions outside the rank- r subspace. Thus, if the full fine-tuning update ΔW_{FT} has important components beyond rank r , LoRA cannot recover them, and the approximation error can be large. From this view, RandLoRA (Albert et al., 2025) challenges the step “few parameters $\Rightarrow$ low rank”: with fixed random bases and learned coefficients, it can produce high-rank or even full-rank effective updates while keeping trainable parameters small. In short, RandLoRA (Albert et al., 2025) ***decouples rank from parameter count***.

Our view is that this decoupling is useful, but it does not remove the main reason for the full fine-tuning gap. Even with a full-rank update, optimizing only p parameters restricts learning to an update set whose intrinsic dimension is at most p , so it cannot represent an arbitrary dense full fine-tuning update unless p scales with the ambient degrees of freedom. **Therefore, “few parameters $\Rightarrow$ low rank” is not the only issue: one can have few parameters and full rank but still face a restricted update space with reduced freedom and reduced expressivities.** This is central to our motivation: we aim to *approach dense full fine-tuning itself*, rather than producing a high-rank update inside a fixed low-dimensional parameterization.

Concretely, we argue that the real bottleneck is the first step, “low memory $\Rightarrow$ few parameters”, which is often treated as a hard constraint because many updated parameters seem to require storing or communicating their gradients and optimizer states. ***Our key insight is that this implication can be relaxed over a long horizon: few parameters per step does not imply few parameters in total. By selecting different parameter groups over time under a strict per-step budget, the training trajectory can accumulate into rich, effectively dense changes, and under weak conditions it can converge to a full fine-tuning objective minima*** (a solution admitted by FT—at which it cannot improve further with first-order information). Equivalently, we accept per-step sparsity, but we show that long-term expressivity can still be high through time-varying budgeted updates. This yields a more general and practical framework: it targets the main training-time constraints directly, operates at scalable group granularity, and provides a principled analysis of which strategies approach full fine-tuning and what controls the convergence rate. From this angle, RandLoRA (Albert et al., 2025) clarifies why *rank* matters, while our work studies how to overcome the low-memory barrier without a fixed low-dimensional parameterization, using ***time-varying, before-hoc budget allocation*** to recover full fine-tuning expressivity in the long run.

B.2.5. RELATION TO DoRA: WEIGHT-DECOMPOSED LOW-RANK ADAPTATION

Summary. DoRA (Liu et al., 2024) refines LoRA by decomposing weights into magnitude and direction, and then applying low-rank adaptation in a form that better matches typical full fine-tuning changes. However, DoRA is still a fixed PEFT parameterization that updates a pre-defined low-dimensional set of parameters, so ***it does not address our problem of low-cost full fine-tuning under a strict per-step budget*** with before-hoc decisions and partial/noisy feedback. Our framework instead uses time-varying group selection, where short-term sparsity can accumulate into effectively dense updates, with theory giving weak conditions for convergence toward full fine-tuning.

Key point 1: DoRA improves the form of low-rank adaptation, but it remains a fixed low-dimensional parameterization. **Key point 2:** Our setting is a strict per-step budget with before-hoc group selection, where changing groups over time can approach dense full fine-tuning, i.e., obtaining a minimizer of full-parameter space admitted by FT with full expressivity.

DoRA (Liu et al., 2024) is an important refinement within the LoRA family. Classical LoRA uses an additive low-rank update, while DoRA decomposes weights into *magnitude* and *direction* and applies low-rank adaptation in this re-parameterized form. Empirically, this often improves stability and accuracy over vanilla LoRA, and it also highlights that the update geometry (scale and direction) matters, not only the rank.

At the same time, DoRA remains a *fixed-form PEFT parameterization*: it always updates a pre-specified, low-dimensional set of parameters (DoRA/LoRA parameters and small scale variables). Therefore, it still optimizes within a restricted update family, and it does not directly match our central setting: *low-cost full fine-tuning under a strict per-step budget*, where the update subspace can change over time and the choice of parameter groups must be made before observing the full current gradient, often with partial/noisy feedback. In other words, DoRA improves how low-rank adaptation is performed, while our work studies when and where to allocate a limited update budget over a long horizon so that training can approach dense full fine-tuning.

This difference matters for both expressivity and practicality. *DoRA can reduce the gap to full fine-tuning in many cases, but it still has a basic limitation of fixed PEFT families: without extra assumptions, there is no general guarantee that the full fine-tuning optimum lies in the DoRA parameterization, so matching full fine-tuning in principle is not ensured. Our framework takes a complementary route: we do not encode a dense update into a single fixed low-dimensional manifold; instead, we use the fact that short-term sparsity does not imply long-term loss of freedom.* By selecting different parameter groups across steps, a budgeted process can accumulate into rich, effectively dense updates, and our theory gives weak sufficient conditions for such strategies to converge to (and efficiently approach) the full fine-tuning objective. From this viewpoint, DoRA is an orthogonal improvement inside the PEFT toolbox, while our contribution is a more general framework for large-model fine-tuning under system constraints.

B.3. Discussion on The Reward-Computability challenge

In this subsection, we detail a core challenge existing in our technical framework—the online bandit selector—and how we address it.

In addition to the time-varying, partial, and noisy observation challenges, we have another central difficulty: Real rewards are NOT observable or even computable after receiving feedback, *introducing an irreconcilable conflict between theory and practical algorithms*. $r_t^{(\varepsilon)}(g)$ is a ratio whose denominator is both unobserved and drifting with training: **at each step we can form a mini-batch estimate of the selected numerator $w_t(g_t)$, yet we cannot observe $W_t = \sum_g w_t(g)$, and naively substituting a noisy denominator induces ratio bias** that breaks standard regret analyses. Moreover, because the underlying energies drift as the model evolves, even an unbiased numerator estimator does not yield a stationary reward process, so any analysis that treats rewards as *iid.* or even martingale-like at the level of the ratio is fundamentally misaligned with the actual learning dynamics. **Our selector resolves this by maintaining the discounted EMA-style estimates for all groups, constructing a plug-in reward $\hat{r}_t^{(\varepsilon)}$, and then analyzing it through a uniform deviation decomposition** that cleanly separates statistical uncertainty from non-stationary drift:

$$|\hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq \eta_t(g) + \text{Drift}_t, \quad (16)$$

holding simultaneously for all $t \leq T$ and $g \in [G]$ with high probability, where $\eta_t(g)$ is an explicit confidence radius driven by mini-batch noise under discounting and Drift_t aggregates non-stationarity through the variation budget $V_T^{(\varepsilon)}$ (see Sec. G.5 for the full construction and constants). Because (16) does not rely on any effective unbiased ratio estimation at any step, *it is precisely the ingredient that lets an optimistic index method provably approximate the oracle in a drifting, partially observed, ratio-reward environment, and it is exactly why the resulting convergence rate can inherit the oracle exponent up to a sublinear correction*.

C. Additional Camera-ready Empirical Results and Discussions

This section adds the empirical and discussion items requested during review: larger-model generalization beyond common-sense reasoning, comparison with a full-rank sparse-update baseline, wall-clock runtime, the empirical relation between

coverage and convergence speed, and the interpretation of the gap to FT under a strict per-step budget.

Larger backbone on math and coding tasks. Table 2 evaluates Llama-3-8B-Instruct on GSM8K, MATH, and HumanEval. OURS remains much closer to FT than standard PEFT baselines and outperforms LoRA/DoRA across all three tasks, supporting the generality of the Think–Touch paradigm beyond the commonsense benchmark.

Method	GSM8K	MATH	HumanEval
FT	75.28	31.0	64.63
LoRA	68.16	28.8	51.83
DoRA	65.13	29.4	50.00
OURS	74.07	30.6	60.98

Table 2. Additional results on Llama-3-8B-Instruct for math and coding tasks.

Comparison with FISH Mask. FISH Mask (Sung et al., 2021) pre-computes a fixed sparse mask using Fisher information, whereas Think–Touch keeps a strict per-step update budget and learns from partial feedback online. Table 3 shows that OURS attains higher average accuracy with fewer updated parameters on both backbones.

Backbone	Method	#Params (%)	ARC-c	ARC-e	BoolQ	HellaSwag	OBQA	PIQA	SIQA	WinoGrande	Avg.
1.5B	FISH Mask	15.5	77.39	89.65	62.17	85.42	80.00	81.99	74.00	69.14	77.47
1.5B	OURS ($M = 128$)	3.85	79.58	90.21	67.68	91.14	85.00	83.26	76.63	75.59	81.14
3B	FISH Mask	12.7	77.22	90.19	62.17	91.07	79.60	84.17	78.10	78.45	80.12
3B	OURS ($M = 128$)	2.80	79.01	90.32	71.28	93.49	86.00	86.13	79.68	84.69	83.83

Table 3. Comparison with FISH Mask on commonsense reasoning tasks.

Wall-clock runtime. Table 4 reports average runtime per optimization step on the 1.5B backbone. Although FT remains the accuracy upper bound, OURS is substantially faster per step than FT and is also faster than the reported LoRA/DoRA/RandLoRA baselines in this setup.

Method	Runtime per step (s)
FT	1.2803
LoRA ($r = 64$)	0.1775
DoRA ($r = 64$)	0.4481
RandLoRA	0.2746
OURS ($M = 128$)	0.1632

Table 4. Average runtime per step on the 1.5B backbone for commonsense reasoning.

Coverage coefficient and convergence speed. Table 5 empirically supports the theoretical role of cumulative coverage: larger normalized coverage corresponds to fewer steps needed to reach the same convergence criterion.

$CC(T)/(TG)$	Steps to convergence
0.4207	2898
0.5660	2045
0.7375	1278

Table 5. Empirical relation between normalized cumulative coverage and convergence speed.

Budget, expressivity, and the gap to FT. The residual gap between OURS and exact FT should be interpreted as a temporal computation–accuracy trade-off induced by the per-step budget, not as a fundamental expressivity limitation. Our theory targets the full-parameter search space over time; increasing M or training longer can further reduce the gap, and when $M = G$ the update recovers dense FT exactly. At the same time, row-level grouping offers a practical compromise: it captures structured gradient sparsity while avoiding both redundant layer-level updates and the prohibitive number of element-wise bandit arms.

C.1. Why adding the regularizer ε does not change the theory

This subsection explains why introducing the stabilizing ε in the normalization $r_t^{(\varepsilon)}(g) = \frac{w_g(\theta_t)}{W_t + \varepsilon}$ does not alter the form of the main convergence and rate guarantees in Theorem 4.7 and Theorem 4.10. The key points are: (i) the ε -version is a monotone rescaling (i.e., smaller but order-preserving) of the unregularized objective at each round; (ii) all inequalities used by the two theorems remain valid after a one-line rescaling of the right-hand side; and (iii) the induced bandit reward is changed only by a common per-round positive factor, so the oracle comparator and regret ordering are preserved.

(1) Relation between ε -regularized and unregularized normalizations. Recall the unregularized coverage coefficient α_t defined in (2), and let

$$\bar{w}_t := \frac{1}{G} \sum_{g=1}^G w_g(\theta_t), \quad u_t := \sum_{g=1}^G \mathbb{P}_t(g) w_g(\theta_t).$$

With ε -regularization, we have:

$$\alpha_t^{(\varepsilon)} = \frac{\bar{w}_t}{\bar{w}_t + \varepsilon} \alpha_t, \quad 0 \leq \alpha_t^{(\varepsilon)} \leq \alpha_t. \quad (17)$$

(2) Optimization-side implication: Theorem 4.7 and Theorem 4.10 keep the same form. Define the ε -regularized cumulative coverage

$$\text{CC}^{(\varepsilon)}(T) := \sum_{t=0}^{T-1} \alpha_t^{(\varepsilon)}.$$

By (17), we have $\text{CC}^{(\varepsilon)}(T) \leq \text{CC}(T)$ for all T . Both Theorem 4.7 and Theorem 4.10 upper bound their respective optimization errors by a monotone decreasing function of $\text{CC}(T)$ —in particular, an exponential term of the form $\exp(-c_U \text{CC}(T))$. Since $\text{CC}^{(\varepsilon)}(T) \leq \text{CC}(T)$ implies

$$\exp(-c_U \text{CC}(T)) \leq \exp(-c_U \text{CC}^{(\varepsilon)}(T)),$$

the same statements remain valid after replacing $\text{CC}(T)$ by $\text{CC}^{(\varepsilon)}(T)$: the bounds keep exactly the same structure with the same constants, while the ε -regularized normalization is numerically stable when $\sum_g w_g(\theta_t)$ is small.

Summary. For readability and stability, we introduce ε explicitly in our method/technical framework section rather than at the very beginning, but all theoretical arguments can be interpreted as applying to the stabilized objective throughout.

D. Think-Touch Convergence Analysis via Cumulative Coverage

This section provides a theoretical analysis of the convergence behavior of the proposed Think-Touch mechanism from a long-horizon perspective. By introducing the notion of cumulative coverage, we characterize how sequential parameter-group selections aggregate over time and establish conditions under which the resulting updates recover the expressivity of full fine-tuning. The main result shows that, despite updating only a subset of parameter groups at each step, the cumulative effect of Think-Touch ensures asymptotic coverage of the full parameter space, thereby justifying our regret-based design as a principled approach to low-cost full fine-tuning.

D.1. Timeline, Filtration, and Measurability

We make the timeline explicit to avoid measurability issues. Let $\{\mathcal{F}_t\}_{t \geq 0}$ be a filtration such that θ_t is $\mathcal{F}_t$ -measurable and the group index g_t is sampled after $\mathcal{F}_t$ is revealed:

$$\mathbb{P}(g_t = g \mid \mathcal{F}_t) = \mathbb{P}_t(g \mid \theta_t),$$

where $\mathbb{P}_t(\cdot \mid \theta_t)$ is $\mathcal{F}_t$ -measurable (possibly adaptive and nonstationary). The update then produces θ_{t+1} , and $\mathcal{F}_{t+1}$ contains (g_t, θ_{t+1}) and all other randomness at step t . In particular, any quantity depending only on $(\theta_t, \mathbb{P}_t(\cdot \mid \theta_t))$ is $\mathcal{F}_t$ -measurable.

D.2. Standing Assumptions and Definitions

In this section, we introduce the key assumptions and definitions that underpin the theoretical framework of our work. These assumptions are essential for understanding the behavior of the group update mechanism and the convergence analysis

we perform in later sections. Specifically, we define the smoothness conditions for the group updates and introduce the Polyak–Łojasiewicz (PL) condition, which is fundamental to our analysis. The selective update rule and the weighted gradient energy are also discussed to set the stage for our detailed convergence proofs.

Assumption D.1 (Group smoothness). For each group $g \in \{1, \dots, G\}$, there exists $L_g > 0$ such that for all θ and all group increments $u^{(g)}$,

$$\mathcal{L}(\dots, \theta^{(g)} + u^{(g)}, \dots) \leq \mathcal{L}(\theta) + \langle \nabla_g \mathcal{L}(\theta), u^{(g)} \rangle + \frac{L_g}{2} \|u^{(g)}\|^2.$$

Let $L_{\max} := \max_g L_g$.

Assumption D.2 (Polyak–Łojasiewicz (PL)). There exists $\mu > 0$ such that for all θ ,

$$\frac{1}{2} \|\nabla \mathcal{L}(\theta)\|^2 \geq \mu (\mathcal{L}(\theta) - \mathcal{L}^*), \quad \mathcal{L}^* := \inf_{\theta} \mathcal{L}(\theta).$$

Selective update (single-group). At step t , after sampling g_t , the algorithm updates

$$\theta_{t+1}^{(g_t)} = \theta_t^{(g_t)} - \frac{1}{L_{g_t}} \nabla_{g_t} \mathcal{L}(\theta_t), \quad \theta_{t+1}^{(g)} = \theta_t^{(g)} \quad (g \neq g_t).$$

Weighted group gradient energy and coverage coefficient. Define

$$w_g(\theta) := \frac{\|\nabla_g \mathcal{L}(\theta)\|^2}{L_g} \geq 0, \quad W(\theta) := \sum_{j=1}^G w_j(\theta).$$

When $W(\theta) > 0$, define

$$\alpha_t(\theta) := \frac{\mathbb{E}_{g \sim \mathbb{P}_t(\cdot|\theta)}[w_g(\theta)]}{\frac{1}{G} \sum_{j=1}^G w_j(\theta)} = \frac{G \mathbb{E}_{g \sim \mathbb{P}_t(\cdot|\theta)}[w_g(\theta)]}{W(\theta)}.$$

We write $\alpha_t := \alpha_t(\theta_t)$, which is $\mathcal{F}_t$ -measurable by the timeline above.

D.3. One-step decrease and conditional contraction

This subsection establishes a one-step descent bound for a single-group update and then derives a conditional contraction inequality in terms of the coverage coefficient α_t , which will be used in the subsequent convergence analysis.

Lemma D.3 (One-step group decrease). Under Assumption D.1, for any θ and any group g , the group step $\theta^{(g)} \leftarrow \theta^{(g)} - \frac{1}{L_g} \nabla_g \mathcal{L}(\theta)$ satisfies

$$\mathcal{L}(\theta^+) \leq \mathcal{L}(\theta) - \frac{1}{2L_g} \|\nabla_g \mathcal{L}(\theta)\|^2 = \mathcal{L}(\theta) - \frac{1}{2} w_g(\theta).$$

Proof. Let $u^{(g)} := -\frac{1}{L_g} \nabla_g \mathcal{L}(\theta)$. By Assumption D.1,

$$\mathcal{L}(\theta^+) \leq \mathcal{L}(\theta) + \langle \nabla_g \mathcal{L}(\theta), u^{(g)} \rangle + \frac{L_g}{2} \|u^{(g)}\|^2 = \mathcal{L}(\theta) + \left\langle \nabla_g \mathcal{L}(\theta), -\frac{1}{L_g} \nabla_g \mathcal{L}(\theta) \right\rangle + \frac{L_g}{2} \left\| -\frac{1}{L_g} \nabla_g \mathcal{L}(\theta) \right\|^2.$$

Moreover,

$$\left\langle \nabla_g \mathcal{L}(\theta), -\frac{1}{L_g} \nabla_g \mathcal{L}(\theta) \right\rangle = -\frac{1}{L_g} \|\nabla_g \mathcal{L}(\theta)\|^2, \quad \left\| -\frac{1}{L_g} \nabla_g \mathcal{L}(\theta) \right\|^2 = \frac{1}{L_g^2} \|\nabla_g \mathcal{L}(\theta)\|^2,$$

hence

$$\mathcal{L}(\theta^+) \leq \mathcal{L}(\theta) - \frac{1}{L_g} \|\nabla_g \mathcal{L}(\theta)\|^2 + \frac{L_g}{2} \cdot \frac{1}{L_g^2} \|\nabla_g \mathcal{L}(\theta)\|^2 = \mathcal{L}(\theta) - \left(\frac{1}{L_g} - \frac{1}{2L_g} \right) \|\nabla_g \mathcal{L}(\theta)\|^2.$$

Therefore,

$$\mathcal{L}(\theta^+) \leq \mathcal{L}(\theta) - \frac{1}{2L_g} \|\nabla_g \mathcal{L}(\theta)\|^2 = \mathcal{L}(\theta) - \frac{1}{2} \cdot \frac{\|\nabla_g \mathcal{L}(\theta)\|^2}{L_g} = \mathcal{L}(\theta) - \frac{1}{2} w_g(\theta),$$

which is exactly the claimed bound and concludes the proof. $\square$

Lemma D.4 (Conditional contraction in terms of α_t). *Let $\Delta_t := \mathcal{L}(\theta_t) - \mathcal{L}^* \geq 0$ and $c_U := \mu/(GL_{\max})$. Under Assumptions D.1–D.2 and the selective update rule, for all $t \geq 0$,*

$$\mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t] \leq (1 - c_U \alpha_t) \Delta_t \leq e^{-c_U \alpha_t} \Delta_t.$$

Proof. Proof sketch. Step 1 uses Lemma D.3 to express Δ_{t+1} as Δ_t minus a single-group energy term. Step 2 takes $\mathcal{F}_t$ -conditional expectation and rewrites $\mathbb{E}[w_{g_t}(\theta_t) \mid \mathcal{F}_t]$ in terms of α_t and $W(\theta_t)$. Step 3 lower bounds $W(\theta_t)$ by a constant multiple of Δ_t (via $L_{\max}$ and the PL condition) and substitutes back to obtain the factor $(1 - c_U \alpha_t)$.

Step 1 (one-step descent $\Rightarrow$ conditional decrease). By Lemma D.3, conditioned on (θ_t, g_t) ,

$$\mathcal{L}(\theta_{t+1}) \leq \mathcal{L}(\theta_t) - \frac{1}{2} w_{g_t}(\theta_t) \Rightarrow \Delta_{t+1} = \mathcal{L}(\theta_{t+1}) - \mathcal{L}^* \leq \mathcal{L}(\theta_t) - \mathcal{L}^* - \frac{1}{2} w_{g_t}(\theta_t) = \Delta_t - \frac{1}{2} w_{g_t}(\theta_t).$$

Taking $\mathcal{F}_t$ -conditional expectation and using that Δ_t is $\mathcal{F}_t$ -measurable gives

$$\mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t] \leq \Delta_t - \frac{1}{2} \mathbb{E}[w_{g_t}(\theta_t) \mid \mathcal{F}_t].$$

Step 2 (rewrite the averaged decrease via α_t). If $W(\theta_t) = 0$, then $w_j(\theta_t) = 0$ for all j , hence $\nabla \mathcal{L}(\theta_t) = 0$ and Assumption D.2 implies $\Delta_t = 0$, so the claim holds. Assume $W(\theta_t) > 0$. By the definition of α_t and $W(\theta_t) = \sum_{j=1}^G w_j(\theta_t)$,

$$\alpha_t = \frac{\mathbb{E}[w_{g_t}(\theta_t) \mid \mathcal{F}_t]}{W(\theta_t)/G} \Rightarrow \mathbb{E}[w_{g_t}(\theta_t) \mid \mathcal{F}_t] = \frac{\alpha_t}{G} W(\theta_t),$$

and thus

$$\mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t] \leq \Delta_t - \frac{1}{2} \cdot \frac{\alpha_t}{G} W(\theta_t).$$

Step 3 (lower bound $W(\theta_t)$ and conclude). Since $L_j \leq L_{\max}$ for all j ,

$$W(\theta_t) = \sum_{j=1}^G \frac{\|\nabla_j \mathcal{L}(\theta_t)\|^2}{L_j} \geq \sum_{j=1}^G \frac{\|\nabla_j \mathcal{L}(\theta_t)\|^2}{L_{\max}} = \frac{1}{L_{\max}} \|\nabla \mathcal{L}(\theta_t)\|^2.$$

By Assumption D.2, $\frac{1}{2} \|\nabla \mathcal{L}(\theta_t)\|^2 \geq \mu(\mathcal{L}(\theta_t) - \mathcal{L}^*) = \mu \Delta_t$, hence $\|\nabla \mathcal{L}(\theta_t)\|^2 \geq 2\mu \Delta_t$ and therefore $W(\theta_t) \geq \frac{2\mu}{L_{\max}} \Delta_t$. Plugging into the bound from Step 2,

$$\mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t] \leq \Delta_t - \frac{1}{2} \cdot \frac{\alpha_t}{G} \cdot \frac{2\mu}{L_{\max}} \Delta_t = \left(1 - \frac{\mu}{GL_{\max}} \alpha_t\right) \Delta_t = (1 - c_U \alpha_t) \Delta_t.$$

Finally, $1 - x \leq e^{-x}$ implies $(1 - c_U \alpha_t) \Delta_t \leq e^{-c_U \alpha_t} \Delta_t$ for $\Delta_t \geq 0$, which matches the stated bound and concludes the proof. $\square$

D.4. A nonnegative supermartingale and a uniform high-probability bound

This subsection converts the one-step conditional contraction into a time-uniform high-probability guarantee. The key idea is to construct a nonnegative supermartingale by exponentially reweighting the suboptimality gap, and then apply Ville's inequality to control its running maximum.

Theorem D.5 (Supermartingale and Ville's inequality). *Let $\Delta_t := \mathcal{L}(\theta_t) - \mathcal{L}^*$ and define $c_U := \mu/(GL_{\max})$. Define*

$$M_t := \Delta_t \exp \left(c_U \sum_{s=0}^{t-1} \alpha_s \right), \quad t \geq 0,$$

with the convention that the empty sum equals 0, so $M_0 = \Delta_0$. Then $\{M_t\}_{t \geq 0}$ is a nonnegative supermartingale with respect to $\{\mathcal{F}_t\}$: $\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] \leq M_t$ and $M_t \geq 0$. Moreover, for any $\delta \in (0, 1)$,

$$\mathbb{P} \left(\sup_{t \geq 0} M_t \geq \frac{\Delta_0}{\delta} \right) \leq \delta,$$

equivalently, with probability at least $1 - \delta$,

$$\forall t \geq 0 : \quad \Delta_t \leq \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right).$$

Proof. Proof sketch. Step 1 shows that $M_t = \Delta_t \exp(c_U \sum_{s < t} \alpha_s)$ has non-increasing conditional expectation by combining the conditional contraction with measurability of the exponential weight. Step 2 applies Ville's inequality (via optional stopping for nonnegative supermartingales) to bound $\mathbb{P}(\sup_t M_t \geq \lambda)$ for any threshold λ . Step 3 substitutes $\lambda = \Delta_0/\delta$ and rewrites the event $\{\sup_t M_t < \Delta_0/\delta\}$ as the desired uniform bound on Δ_t .

Step 1 (supermartingale). From Lemma D.4, $\mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t] \leq e^{-c_U \alpha_t} \Delta_t$. Since α_t is $\mathcal{F}_t$ -measurable, the weight $\exp(c_U \sum_{s=0}^t \alpha_s)$ is $\mathcal{F}_t$ -measurable. Therefore,

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] = \mathbb{E} \left[\Delta_{t+1} \exp \left(c_U \sum_{s=0}^t \alpha_s \right) \middle| \mathcal{F}_t \right] = \exp \left(c_U \sum_{s=0}^t \alpha_s \right) \mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t] \leq \exp \left(c_U \sum_{s=0}^t \alpha_s \right) e^{-c_U \alpha_t} \Delta_t.$$

Using $\sum_{s=0}^t \alpha_s = \sum_{s=0}^{t-1} \alpha_s + \alpha_t$, we have

$$\exp \left(c_U \sum_{s=0}^t \alpha_s \right) e^{-c_U \alpha_t} = \exp \left(c_U \sum_{s=0}^{t-1} \alpha_s + c_U \alpha_t \right) e^{-c_U \alpha_t} = \exp \left(c_U \sum_{s=0}^{t-1} \alpha_s \right),$$

hence

$$\mathbb{E}[M_{t+1} \mid \mathcal{F}_t] \leq \Delta_t \exp \left(c_U \sum_{s=0}^{t-1} \alpha_s \right) = M_t.$$

Also, $M_t \geq 0$ since $\Delta_t \geq 0$ and the exponential factor is positive.

Step 2 (Ville's inequality). Fix $\lambda > 0$ and define the stopping time $\tau := \inf\{t \geq 0 : M_t \geq \lambda\}$, and the bounded stopping time $\tau \wedge n$. By optional stopping for nonnegative supermartingales, $\mathbb{E}[M_{\tau \wedge n}] \leq \mathbb{E}[M_0] = \Delta_0$. Moreover,

$$M_{\tau \wedge n} = M_\tau \mathbf{1}_{\{\tau \leq n\}} + M_n \mathbf{1}_{\{\tau > n\}} \geq M_\tau \mathbf{1}_{\{\tau \leq n\}} \geq \lambda \mathbf{1}_{\{\tau \leq n\}},$$

so taking expectations yields

$$\mathbb{E}[M_{\tau \wedge n}] \geq \lambda \mathbb{P}(\tau \leq n).$$

Combining the two displays gives $\mathbb{P}(\tau \leq n) \leq \Delta_0/\lambda$. Letting $n \rightarrow \infty$ and using $\{\tau \leq n\} \uparrow \{\tau < \infty\} = \{\sup_{t \geq 0} M_t \geq \lambda\}$, we obtain

$$\mathbb{P} \left(\sup_{t \geq 0} M_t \geq \lambda \right) \leq \frac{\Delta_0}{\lambda}.$$

Step 3 (uniform bound). Set $\lambda = \Delta_0/\delta$ to get $\mathbb{P}(\sup_{t \geq 0} M_t \geq \Delta_0/\delta) \leq \delta$. On the complementary event $\{\sup_{t \geq 0} M_t < \Delta_0/\delta\}$, for every $t \geq 0$,

$$M_t = \Delta_t \exp \left(c_U \sum_{s=0}^{t-1} \alpha_s \right) < \frac{\Delta_0}{\delta} \quad \Rightarrow \quad \Delta_t < \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right),$$

which is exactly the stated uniform high-probability bound and concludes the proof. $\square$

D.5. Weak sufficient condition

This subsection presents a minimal condition on the cumulative coverage $\sum_{t \geq 0} \alpha_t$ that guarantees almost-sure convergence of the suboptimality gap, and records the corresponding time-uniform tail bound inherited from the supermartingale argument.

Corollary D.6 (Weak sufficient condition: $\sum_t \alpha_t = +\infty$ implies convergence). *If $\sum_{t=0}^{\infty} \alpha_t = +\infty$ almost surely, then $\Delta_t \rightarrow 0$ almost surely. Moreover, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$\forall t \geq 0 : \quad \Delta_t \leq \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right).$$

Proof. Proof sketch. Step 1 applies the time-uniform bound from the supermartingale argument to control Δ_t by an exponentially decaying envelope in $\sum_{s \leq t} \alpha_s$. Step 2 uses the assumption $\sum_{s=0}^{\infty} \alpha_s = +\infty$ to show that this envelope converges to 0. Step 3 removes the fixed- δ conditioning by intersecting over a vanishing sequence $\delta \downarrow 0$ and using that the corresponding high-probability events occur eventually almost surely.

Step 1 (uniform bound). The stated high-probability inequality is exactly Theorem D.5, i.e., for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\forall t \geq 0 : \quad \Delta_t \leq \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right).$$

Step 2 (convergence on a high-probability event). Fix $\delta \in (0, 1)$ and work on the event

$$E_\delta := \left\{ \forall t \geq 0 : \Delta_t \leq \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right) \right\}, \quad \mathbb{P}(E_\delta) \geq 1 - \delta.$$

On $E_\delta \cap \{\sum_{s=0}^{\infty} \alpha_s = +\infty\}$, we have $\sum_{s=0}^{t-1} \alpha_s \rightarrow +\infty$, hence

$$-c_U \sum_{s=0}^{t-1} \alpha_s \rightarrow -\infty \quad \Rightarrow \quad \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right) \rightarrow 0,$$

and therefore the right-hand side in the bound on E_δ converges to 0, implying $\Delta_t \rightarrow 0$ on $E_\delta \cap \{\sum_s \alpha_s = +\infty\}$.

Step 3 (upgrade to almost sure convergence). Let $\delta_n := 2^{-n}$ and define $E_n := E_{\delta_n}$, so $\mathbb{P}(E_n) \geq 1 - 2^{-n}$. Then $\mathbb{P}(E_n^c) \leq 2^{-n}$ and $\sum_{n=1}^{\infty} \mathbb{P}(E_n^c) < \infty$, hence by the Borel–Cantelli lemma,

$$\mathbb{P}(E_n^c \text{ infinitely often}) = 0 \quad \Rightarrow \quad \mathbb{P}(\exists N : \forall n \geq N, E_n \text{ occurs}) = 1.$$

Thus, with probability one, there exists a (random) index N such that E_N holds. On the same probability-one event, the assumption gives $\sum_{s=0}^{\infty} \alpha_s = +\infty$. Applying Step 2 with $\delta = \delta_N$ shows $\Delta_t \rightarrow 0$ almost surely, which concludes the proof. $\square$

D.6. Window sufficient condition and linear-rate convergence

This subsection gives stronger sufficient conditions that yield explicit linear (exponential) decay rates. We first assume a uniform lower bound on the *windowed* coverage over blocks of length H , which implies exponential decay along the subsequence $\{\theta_{kH}\}_{k \geq 0}$. We then specialize to a uniform per-iterate lower bound $\inf_t \alpha_t > 0$, which yields exponential decay for all iterates.

Assumption D.7 (Window coverage lower bound). There exist an integer $H \geq 1$ and a constant $\underline{\alpha} > 0$ such that almost surely, for all $k \geq 0$,

$$\sum_{t=kH}^{kH+H-1} \alpha_t \geq H \underline{\alpha}.$$

Corollary D.8 (Window condition implies linear (exponential) convergence on a subsequence). *Under Assumption D.7, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, for all $k \geq 0$,*

$$\Delta_{kH} \leq \frac{\Delta_0}{\delta} \exp(-c_U kH \underline{\alpha}).$$

Proof. Proof sketch. We apply the time-uniform bound from Theorem D.5 at the subsequence $t = kH$, and then lower bound the cumulative coverage $\sum_{s < kH} \alpha_s$ by summing the window lower bounds from Assumption D.7.

On the $(1 - \delta)$ -probability event from Theorem D.5, for all $t \geq 0$,

$$\Delta_t \leq \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right).$$

Setting $t = kH$ yields

$$\Delta_{kH} \leq \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{kH-1} \alpha_s \right).$$

Decompose the cumulative sum as

$$\sum_{s=0}^{kH-1} \alpha_s = \sum_{i=0}^{k-1} \sum_{t=iH}^{iH+H-1} \alpha_t.$$

By Assumption D.7, almost surely for each $i \geq 0$,

$$\sum_{t=iH}^{iH+H-1} \alpha_t \geq H \underline{\alpha},$$

hence

$$\sum_{s=0}^{kH-1} \alpha_s = \sum_{i=0}^{k-1} \sum_{t=iH}^{iH+H-1} \alpha_t \geq \sum_{i=0}^{k-1} H \underline{\alpha} = kH \underline{\alpha}.$$

Substituting this bound into the display for Δ_{kH} gives

$$\Delta_{kH} \leq \frac{\Delta_0}{\delta} \exp(-c_U kH \underline{\alpha}),$$

which is exactly the stated inequality and concludes the proof. $\square$

Corollary D.9 (Uniform condition implies high-probability linear (exponential) convergence for all iterates). *Let*

$$c(\boldsymbol{\alpha}) \triangleq \inf_{t \geq 0} \alpha_t.$$

If $c(\boldsymbol{\alpha}) > 0$ almost surely, then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, for all $t \geq 0$,

$$\Delta_t \leq \frac{\Delta_0}{\delta} \exp(-c_U t c(\boldsymbol{\alpha})).$$

Proof. Proof sketch. We start from the time-uniform bound in Theorem D.5 and lower bound $\sum_{s < t} \alpha_s$ by $t c(\boldsymbol{\alpha})$ under the uniform condition.

On the $(1 - \delta)$ -probability event from Theorem D.5, for all $t \geq 0$,

$$\Delta_t \leq \frac{\Delta_0}{\delta} \exp \left(-c_U \sum_{s=0}^{t-1} \alpha_s \right).$$

If $c(\boldsymbol{\alpha}) > 0$ almost surely, then almost surely for every $s \geq 0$, $\alpha_s \geq c(\boldsymbol{\alpha})$, hence for every $t \geq 0$,

$$\sum_{s=0}^{t-1} \alpha_s \geq \sum_{s=0}^{t-1} c(\boldsymbol{\alpha}) = t c(\boldsymbol{\alpha}).$$

Substituting this bound yields

$$\Delta_t \leq \frac{\Delta_0}{\delta} \exp(-c_U t c(\boldsymbol{\alpha})),$$

which matches the stated inequality and concludes the proof. $\square$

Relation between Corollaries D.8 and D.9. Corollary D.9 can be viewed as the special case of Corollary D.8 with $H = 1$. Indeed, if $c(\alpha) > 0$ almost surely, then $\alpha_t \geq c(\alpha)$ almost surely for all t , so Assumption D.7 holds with $H = 1$ and $\underline{\alpha} = c(\alpha)$:

$$\sum_{t=k}^k \alpha_t = \alpha_k \geq c(\alpha) = 1 \cdot \underline{\alpha}.$$

Substituting $H = 1$ and $\underline{\alpha} = c(\alpha)$ into Corollary D.8 gives, for all $k \geq 0$,

$$\Delta_k \leq \frac{\Delta_0}{\delta} \exp(-c_U k c(\alpha)),$$

which is exactly the bound in Corollary D.9.

D.7. Linear-rate convergence in expectation

This subsection complements the high-probability guarantees by establishing an exponential decay rate in expectation. The key point is to use a deterministic worst-case lower bound on the coverage coefficient once the sampling policies $\{\mathbb{P}_t\}_{t \geq 0}$ are fixed.

Theorem D.10 (Linear (exponential) convergence in expectation). *Let $\Delta_t := \mathcal{L}(\theta_t) - \mathcal{L}^* \geq 0$ and $c_U := \mu/(GL_{\max})$. Assume the conditions of Lemma D.4 hold. Fix the sampling policies $\{\mathbb{P}_t\}_{t \geq 0}$ and define the worst-case coverage constant*

$$c(\alpha) \triangleq \inf_{t \geq 0} \inf_{\theta} \alpha_t(\theta),$$

where $\alpha_t(\theta)$ denotes the coverage coefficient evaluated at parameter θ under $\mathbb{P}_t(\cdot | \theta)$. If $c(\alpha) > 0$, then for all $t \geq 0$,

$$\mathbb{E}[\Delta_t] \leq \mathbb{E}[\Delta_0] \exp(-c_U t c(\alpha)).$$

Moreover, if $c_U c(\alpha) \in (0, 1]$, then also

$$\mathbb{E}[\Delta_t] \leq \mathbb{E}[\Delta_0] (1 - c_U c(\alpha))^t.$$

Proof. Proof sketch. We take expectations in the one-step conditional contraction, use the deterministic lower bound $\alpha_t(\theta_t) \geq c(\alpha)$ to obtain a fixed multiplicative factor, and then iterate the resulting recursion.

By Lemma D.4, for all $t \geq 0$,

$$\mathbb{E}[\Delta_{t+1} | \mathcal{F}_t] \leq e^{-c_U \alpha_t} \Delta_t.$$

Taking total expectation and using the tower property,

$$\mathbb{E}[\Delta_{t+1}] = \mathbb{E}[\mathbb{E}[\Delta_{t+1} | \mathcal{F}_t]] \leq \mathbb{E}[e^{-c_U \alpha_t} \Delta_t].$$

By definition of $c(\alpha)$ and the fact that $\alpha_t = \alpha_t(\theta_t)$, we have $\alpha_t(\theta_t) \geq c(\alpha)$ for all t almost surely, hence

$$e^{-c_U \alpha_t} \leq e^{-c_U c(\alpha)} \quad \text{almost surely.}$$

Since $c(\alpha)$ depends only on the fixed policies $\{\mathbb{P}_t\}_{t \geq 0}$ (and the deterministic definition above), the factor $e^{-c_U c(\alpha)}$ is a constant and can be taken out of the expectation, giving

$$\mathbb{E}[\Delta_{t+1}] \leq \mathbb{E}[e^{-c_U c(\alpha)} \Delta_t] = e^{-c_U c(\alpha)} \mathbb{E}[\Delta_t].$$

Iterating the recursion yields

$$\mathbb{E}[\Delta_t] \leq (e^{-c_U c(\alpha)})^t \mathbb{E}[\Delta_0] = \mathbb{E}[\Delta_0] \exp(-c_U t c(\alpha)),$$

which proves the first claim.

For the second claim, assume $c_U c(\alpha) \in (0, 1]$ and use the contraction form in Lemma D.4:

$$\mathbb{E}[\Delta_{t+1} | \mathcal{F}_t] \leq (1 - c_U \alpha_t) \Delta_t \leq (1 - c_U c(\alpha)) \Delta_t,$$

where the last inequality uses $\alpha_t \geq c(\alpha)$ and $1 - c_U x$ being nonincreasing in x . Taking expectations and iterating gives

$$\mathbb{E}[\Delta_t] \leq (1 - c_U c(\alpha))^t \mathbb{E}[\Delta_0],$$

which is exactly the stated geometric bound and concludes the proof. $\square$

E. High-probability Dynamic-Regret Analysis for Our Online Selector

This section develops a high-probability dynamic-regret bound for our online group selector. We first introduce the reward model and the corresponding notions of dynamic regret and non-stationarity. Then, subsequent subsections derive concentration and regret guarantees under the same unified notation.

E.1. Setup and Notation

This subsection specifies the reward model for the online group selector, defines the associated dynamic regret against the ideal post-hoc time-varying Oracle strategy, and introduces a variation budget that quantifies non-stationarity of the normalized rewards.

Let $G \in \mathbb{N}$ be the number of parameter groups (arms). For each time $t \in \{1, \dots, T\}$ and arm $g \in [G] := \{1, \dots, G\}$, define the (ideal) group energy

$$w_g(\theta_t) := \frac{\|\nabla_g \mathcal{L}(\theta_t)\|_2^2}{L_g} \geq 0, \quad (18)$$

where $L_g > 0$ is the group Lipschitz constant and θ_t denotes the model parameters before taking the t -th update (e.g., t -th training step). In the main text, $w_g(\theta)$ is presented in a simplified form for readability by suppressing the (group-wise) Lipschitz constants. The two definitions are equivalent, and we use the unified notation $w_g(\theta)$ throughout this section.

Define the total energy and the *regularized normalized reward*

$$W_t := \sum_{j=1}^G w_j(\theta_t), \quad r_t^{(\varepsilon)}(g) := \frac{w_g(\theta_t)}{W_t + \varepsilon}, \quad \varepsilon > 0. \quad (19)$$

Then $0 \leq r_t^{(\varepsilon)}(g) \leq 1$ for all t, g since $0 \leq w_g(\theta_t) \leq W_t$ implies

$$0 \leq \frac{w_g(\theta_t)}{W_t + \varepsilon} \leq \frac{w_g(\theta_t)}{w_g(\theta_t) + \varepsilon} \leq 1.$$

Define the (dynamic) best arm $g_t^* \in \arg \max_{g \in [G]} r_t^{(\varepsilon)}(g)$ and the dynamic regret

$$R_T^{(\varepsilon)} := \sum_{t=1}^T \left(r_t^{(\varepsilon)}(g_t^*) - r_t^{(\varepsilon)}(g_t) \right), \quad (20)$$

where g_t is the arm chosen by the bandit rule at time t .

We measure non-stationarity using the variation budget

$$V_T^{(\varepsilon)} := \sum_{t=2}^T \max_{g \in [G]} |r_t^{(\varepsilon)}(g) - r_{t-1}^{(\varepsilon)}(g)|. \quad (21)$$

E.2. Assumptions

This subsection states the statistical and exploration assumptions used to obtain high-probability dynamic-regret bounds for the online group selector. We adopt the same mini-batch notation as in the main text, in particular the split $\mathcal{B}_t = \mathcal{B}_{t,1} \cup \mathcal{B}_{t,2}$ and the stochastic gradient notation $\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,i})$.

Assumption E.1 (Two-mini-batch unbiased energy estimator). Conditioned on the past sigma-field $\mathcal{F}_{t-1}$ and on selecting group g (i.e., $g_t = g$), we can draw two independent mini-batches $\mathcal{B}_{t,1}$ and $\mathcal{B}_{t,2}$ at the training step t , leading to two conditionally independent stochastic group-gradients $\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,1})$ and $\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,2})$ satisfying

$$\mathbb{E}[\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,i}) \mid \mathcal{F}_{t-1}, g_t = g] = \nabla_g \mathcal{L}(\theta_t), \quad i \in \{1, 2\}, \quad (22)$$

where $\nabla_g \mathcal{L}(\theta_t)$ is the full-batch gradient.

Define the (possibly signed) scalar estimator

$$\tilde{w}_g(\theta_t) := \frac{\langle \nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,1}), \nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,2}) \rangle}{L_g}. \quad (23)$$

Assumption E.2 (Boundedness). There exists $w_{\max} > 0$ such that for all $t \leq T$ and all $g \in [G]$,

$$|\tilde{w}_g(\theta_t)| \leq w_{\max} \quad \text{a.s.} \quad (24)$$

Consequently, $0 \leq w_g(\theta_t) \leq w_{\max}$ and $0 \leq W_t \leq G w_{\max}$, where $w_g(\theta_t) := \|\nabla_g \mathcal{L}(\theta_t)\|_2^2 / L_g$ and $W_t := \sum_{j=1}^G w_j(\theta_t)$.

Assumption E.3 (Forced exploration (time-varying)). The sampling probabilities satisfy a time-varying uniform lower bound: there exists a predictable sequence $\{\gamma_t\}_{t \leq T} \subset (0, 1]$ such that

$$\mathbb{P}_t(g) \geq P_{\min,t}, \quad P_{\min,t} := \frac{\gamma_t}{G}, \quad \forall t \leq T, \forall g \in [G]. \quad (25)$$

A sufficient construction is the time-varying argmax-plus-uniform mixture

$$\mathbb{P}_t = (1 - \gamma_t) e_{\hat{g}_t} + \gamma_t \cdot \frac{1}{G} \mathbf{1},$$

where $e_{\hat{g}_t} \in \mathbb{R}^G$ denotes the one-hot vector that puts unit mass on $\hat{g}_t$ and zero on all other groups.

Remark E.4 (Why time-varying exploration). A detailed anytime choice of (λ_t, γ_t) and the corresponding exploration lower bound are given in Sec. E.7, especially (82)–(83). Time-varying exploration is needed to match the effective window induced by λ_t : keeping γ fixed does not affect the unbiasedness of the reward estimator, but it can excessive uniform exploration (inflating regret) under discounting. Designing a joint schedule for (λ_t, γ_t) that simultaneously ensures adequate exploration counts and controlled non-stationarity drift as well as a controlled total exploration cost is one of the main technical difficulties in our analysis.

E.3. Bandit Algorithm (Discounted UCB on Regularized Normalized Rewards)

This subsection specifies our discounted-UCB selector for choosing a parameter group online. It maintains exponentially discounted sufficient statistics of the unbiased energy estimates, constructs high-probability confidence radii for both the (discounted) energy and the induced regularized normalized reward, and then selects groups by an upper-confidence rule combined with time-varying forced exploration.

Discounted statistics. We use (possibly) time-varying exponential discounting as in Algorithm 2. For each group $g \in [G]$, maintain the discounted count $N_t(g)$ and discounted sum $S_t(g)$ recursively by

$$\begin{aligned} N_{t+1}(g) &:= \lambda_t N_t(g) + \mathbf{1}\{g_t = g\}, \\ S_{t+1}(g) &:= \lambda_t S_t(g) + \mathbf{1}\{g_t = g\} \tilde{w}_g(\theta_t), \quad t \geq 1. \end{aligned} \quad (26)$$

With initialization $N_1(g) = S_1(g) = 0$, for every $t \geq 2$,

$$N_t(g) = \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}, \quad S_t(g) = \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\} \tilde{w}_g(\theta_s).$$

Plug-in energy and regularized normalized reward. When $N_t(g) = 0$, set $\hat{w}_t(g) := w_{\max}$ (optimism for unseen groups); otherwise set

$$\hat{w}_t(g) := \frac{S_t(g)}{N_t(g)}. \quad (27)$$

Project to nonnegative energies:

$$\hat{w}_t^+(g) := \max\{\hat{w}_t(g), 0\} \in [0, w_{\max}], \quad \widehat{W}_t := \sum_{j=1}^G \hat{w}_t^+(j), \quad \hat{r}_t^{(\varepsilon)}(g) := \frac{\hat{w}_t^+(g)}{\widehat{W}_t + \varepsilon}. \quad (28)$$

Confidence radii. Our goal is to build a high-probability band for each discounted energy estimate $\hat{w}_t(g)$ around its discounted conditional mean, and then propagate this band through the nonlinear map $w \mapsto w^+ / (W^+ + \varepsilon)$ to obtain a

valid confidence radius for the plug-in reward $\hat{r}_t^{(\varepsilon)}(g)$. For readability, the detailed derivations are moved to Sec. H; we summarize the resulting radii below.

Fix a target confidence level $\delta \in (0, 1)$ and set $\delta' := \delta/(2GT)$. For each (t, g) with $N_t(g) > 0$, define the discounted conditional mean

$$m_s(g) := \mathbb{E}[\tilde{w}_g(\theta_s) \mid \mathcal{F}_{s-1}, g_s = g], \quad \bar{w}_t(g) := \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\} m_s(g),$$

so that $\bar{w}_t(g)$ is $\mathcal{F}_{t-1}$ -measurable.

Energy radius. We define the per-group energy radius

$$\beta_t(g) := \begin{cases} 2\sqrt{2} w_{\max} \sqrt{\frac{\log\left(\frac{4GT}{\delta}\right)}{N_t(g)}}, & N_t(g) > 0, \\ +\infty, & N_t(g) = 0. \end{cases} \quad (29)$$

With this choice, for each fixed (t, g) with $N_t(g) > 0$, $\Pr(|\hat{w}_t(g) - \bar{w}_t(g)| \leq \beta_t(g)) \geq 1 - \delta'$. A union bound over all $t \in [T]$ and $g \in [G]$ implies that with probability at least $1 - \delta/2$,

$$|\hat{w}_t(g) - \bar{w}_t(g)| \leq \beta_t(g) \quad \text{for all } (t, g) \text{ with } N_t(g) > 0.$$

Reward radius. Define $\bar{W}_t := \sum_{j=1}^G \bar{w}_t(j)$ and $\bar{r}_t^{(\varepsilon)}(g) := \bar{w}_t(g)/(\bar{W}_t + \varepsilon)$. On the same high-probability event, the reward radius

$$\eta_t(g) := \frac{\beta_t(g)}{\varepsilon} + \frac{w_{\max}}{\varepsilon^2} \sum_{j=1}^G \beta_t(j) \quad (30)$$

satisfies $|\hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g)| \leq \eta_t(g)$ for all (t, g) . Here $\beta_t(g)$ quantifies uncertainty in the discounted *energy* estimate $\hat{w}_t(g)$, whereas $\eta_t(g)$ is the induced uncertainty after projection and normalization, which is the quantity needed to form a valid UCB index.

UCB index. Let

$$\text{UCB}_t(g) := \hat{r}_t^{(\varepsilon)}(g) + \eta_t(g). \quad (31)$$

This index is an upper-confidence estimate of the (discounted) regularized normalized reward: $\hat{r}_t^{(\varepsilon)}(g)$ is the plug-in estimate and $\eta_t(g)$ provides the exploration bonus that accounts for uncertainty.

Action selection (with time-varying exploration). Let $\hat{g}_t \in \arg \max_{g \in [G]} \text{UCB}_t(g)$. Choose an exploration schedule $\{\gamma_t\}_{t=1}^T$ with $\gamma_t \in (0, 1/2]$ and define

$$\mathbb{P}_t = (1 - \gamma_t) e_{\hat{g}_t} + \gamma_t \cdot \frac{1}{G} \mathbf{1}, \quad (32)$$

where $e_{\hat{g}_t} \in \mathbb{R}^G$ is the one-hot vector that puts unit mass on $\hat{g}_t$. Therefore, $\mathbb{P}_t(g) \geq \gamma_t/G$ for all $g \in [G]$.

Lemma E.5 (Geometric-sum upper bounds for nondecreasing $\{\lambda_t\}$). *Assume $\{\lambda_t\}_{t \geq 1} \subset (0, 1)$ is nondecreasing. Then for any integers $1 \leq s < t$,*

$$\prod_{i=s+1}^t \lambda_i \leq \lambda_t^{t-s}.$$

Consequently, for any $t \geq 1$,

$$\sum_{k=0}^{t-1} \prod_{i=t-k+1}^t \lambda_i \leq \sum_{k=0}^{\infty} \lambda_t^k = \frac{1}{1 - \lambda_t}.$$

Proof. Since $\lambda_i \leq \lambda_t$ for all $i \leq t$ (nondecreasing), we have

$$\prod_{i=s+1}^t \lambda_i \leq \prod_{i=s+1}^t \lambda_t = \lambda_t^{t-s}.$$

Summing over k gives the second inequality and the geometric-series identity, which concludes the proof. $\square$

Algorithm 2 Bandit-based online group selector

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1: Input: regularizer  $\varepsilon > 0$ , discount schedule  $\{\lambda_t\}_{t \geq 1}$ , exploration schedule  $\{\gamma_t\}_{t \geq 1}$ , confidence  $\delta \in (0, 1)$ , horizon  $T$ .
2: Initialize:  $N_1(g) \leftarrow 0$ ,  $S_1(g) \leftarrow 0$  for all  $g \in [G]$ .
3: for  $t = 1, 2, \dots, T$  do
4:   for  $g = 1, 2, \dots, G$  do
5:     if  $N_t(g) = 0$  then  $\hat{w}_t(g) \leftarrow w_{\max}$  else  $\hat{w}_t(g) \leftarrow S_t(g)/N_t(g)$ .
6:      $\hat{w}_t^+(g) \leftarrow \max\{\hat{w}_t(g), 0\}$ .
7:   end for
8:    $\bar{W}_t \leftarrow \sum_{j=1}^G \hat{w}_t^+(j)$ .
9:   for  $g = 1, 2, \dots, G$  do
10:     $\hat{r}_t^{(\varepsilon)}(g) \leftarrow \hat{w}_t^+(g)/(\bar{W}_t + \varepsilon)$ .
11:    if  $N_t(g) = 0$  then  $\beta_t(g) \leftarrow +\infty$  else  $\beta_t(g) \leftarrow 2\sqrt{2} w_{\max} \sqrt{\log(\frac{4GT}{\delta}) / N_t(g)}$ .
12:  end for
13:   $B_t \leftarrow \sum_{j=1}^G \beta_t(j)$ .
14:  for  $g = 1, 2, \dots, G$  do
15:     $\eta_t(g) \leftarrow \beta_t(g)/\varepsilon + (w_{\max}/\varepsilon^2) B_t$ .
16:     $\text{UCB}_t(g) \leftarrow \hat{r}_t^{(\varepsilon)}(g) + \eta_t(g)$ .
17:  end for
18:   $\hat{g}_t \in \arg \max_{g \in [G]} \text{UCB}_t(g)$ .
19:  Sample  $g_t \sim \mathbb{P}_t = (1 - \gamma_t)e_{\hat{g}_t} + \gamma_t \text{Unif}([G])$ .
20:  Observe and estimate mini-batch energy  $\tilde{w}_{g_t}(\theta_t)$  (Assumption E.1).
21:  for  $g = 1, 2, \dots, G$  do
22:     $N_{t+1}(g) \leftarrow \lambda_t N_t(g) + \mathbf{1}\{g_t = g\}$ .
23:     $S_{t+1}(g) \leftarrow \lambda_t S_t(g) + \mathbf{1}\{g_t = g\} \tilde{w}_g(\theta_t)$ .
24:  end for
25: end for
    
```

E.4. Part 1: Unbiasedness and Concentration for $\tilde{w}_g(\theta_t)$ and $\hat{w}_t(g)$

This subsection establishes two basic statistical facts used throughout the regret analysis: (i) the two-mini-batch energy estimator is conditionally unbiased for the full-batch group energy; and (ii) its discounted empirical averages concentrate around their conditional means.

Lemma E.6 (Unbiasedness of the two-mini-batch estimator). *Under Assumption E.1, for every $t \leq T$ and every $g \in [G]$,*

$$\mathbb{E}[\tilde{w}_g(\theta_t) \mid \mathcal{F}_{t-1}, g_t = g] = \frac{\|\nabla_g \mathcal{L}(\theta_t)\|_2^2}{L_g} = w_g(\theta_t). \quad (33)$$

Proof. Fix any $t \leq T$ and any $g \in [G]$, and condition on $(\mathcal{F}_{t-1}, g_t = g)$. By Assumption E.1, the two mini-batches $\mathcal{B}_{t,1}$ and $\mathcal{B}_{t,2}$ are conditionally independent, and

$$\mathbb{E}[\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,i}) \mid \mathcal{F}_{t-1}, g_t = g] = \nabla_g \mathcal{L}(\theta_t), \quad i \in \{1, 2\}.$$

Write the inner product component-wise and apply conditional independence:

$$\begin{aligned}
 & \mathbb{E}[\langle \nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,1}), \nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,2}) \rangle \mid \mathcal{F}_{t-1}, g_t = g] \\
 &= \sum_k \mathbb{E}[(\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,1}))_k (\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,2}))_k \mid \mathcal{F}_{t-1}, g_t = g] \\
 &= \sum_k \mathbb{E}[(\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,1}))_k \mid \mathcal{F}_{t-1}, g_t = g] \mathbb{E}[(\nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,2}))_k \mid \mathcal{F}_{t-1}, g_t = g] \\
 &= \sum_k (\nabla_g \mathcal{L}(\theta_t))_k^2 = \langle \nabla_g \mathcal{L}(\theta_t), \nabla_g \mathcal{L}(\theta_t) \rangle = \|\nabla_g \mathcal{L}(\theta_t)\|_2^2.
 \end{aligned}$$

Dividing both sides by L_g and using the definition $\tilde{w}_g(\theta_t) := \langle \nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,1}), \nabla_g \mathcal{L}(\theta_t; \mathcal{B}_{t,2}) \rangle / L_g$ yields (33), which concludes the proof. $\square$

Lemma E.7 (Azuma–Hoeffding for discounted averages). *Assume Assumptions E.1–E.2. Fix any group $g \in [G]$ and any time $t \leq T$ such that $N_t(g) > 0$. Define the discounted true average*

$$\bar{w}_t(g) := \frac{\sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\} w_g(\theta_s)}{N_t(g)}. \quad (34)$$

Then for any $\delta' \in (0, 1)$,

$$\mathbb{P} \left(|\hat{w}_t(g) - \bar{w}_t(g)| \geq 2w_{\max} \sqrt{\frac{2 \log(2/\delta')}{N_t(g)}} \right) \leq \delta'. \quad (35)$$

Proof. Fix the group g and time t with $N_t(g) > 0$. For each $s \in \{1, \dots, t-1\}$ define

$$a_s := \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}.$$

Each a_s is $\mathcal{F}_{s-1}$ -measurable and $a_s \in [0, 1]$. Moreover,

$$N_t(g) = \sum_{s=1}^{t-1} a_s, \quad S_t(g) = \sum_{s=1}^{t-1} a_s \tilde{w}_g(\theta_s), \quad \hat{w}_t(g) = \frac{S_t(g)}{N_t(g)}.$$

Define

$$X_s := a_s (\tilde{w}_g(\theta_s) - w_g(\theta_s)), \quad s = 1, \dots, t-1.$$

By Assumption E.1 and the fact that $a_s = 0$ on $\{g_s \neq g\}$, we have $\mathbb{E}[X_s \mid \mathcal{F}_{s-1}] = 0$.

By Assumption E.2, $|\tilde{w}_g(\theta_s)| \leq w_{\max}$ a.s., and by definition $0 \leq w_g(\theta_s) \leq w_{\max}$, hence

$$|\tilde{w}_g(\theta_s) - w_g(\theta_s)| \leq |\tilde{w}_g(\theta_s)| + |w_g(\theta_s)| \leq w_{\max} + w_{\max} = 2w_{\max}.$$

Multiplying by $a_s \in [0, 1]$ gives

$$|X_s| = a_s |\tilde{w}_g(\theta_s) - w_g(\theta_s)| \leq 2a_s w_{\max}.$$

Also,

$$\sum_{s=1}^{t-1} X_s = \sum_{s=1}^{t-1} a_s (\tilde{w}_g(\theta_s) - w_g(\theta_s)) = \sum_{s=1}^{t-1} a_s \tilde{w}_g(\theta_s) - \sum_{s=1}^{t-1} a_s w_g(\theta_s).$$

Recalling $S_t(g) = \sum_{s=1}^{t-1} a_s \tilde{w}_g(\theta_s)$ and $N_t(g) = \sum_{s=1}^{t-1} a_s$, we have

$$\begin{aligned} \hat{w}_t(g) - \bar{w}_t(g) &= \frac{S_t(g)}{N_t(g)} - \frac{\sum_{s=1}^{t-1} a_s w_g(\theta_s)}{N_t(g)} \\ &= \frac{\sum_{s=1}^{t-1} a_s \tilde{w}_g(\theta_s)}{N_t(g)} - \frac{\sum_{s=1}^{t-1} a_s w_g(\theta_s)}{N_t(g)} \\ &= \frac{\sum_{s=1}^{t-1} (a_s \tilde{w}_g(\theta_s) - a_s w_g(\theta_s))}{N_t(g)} \\ &= \frac{\sum_{s=1}^{t-1} a_s (\tilde{w}_g(\theta_s) - w_g(\theta_s))}{N_t(g)} = \frac{\sum_{s=1}^{t-1} X_s}{N_t(g)}. \end{aligned}$$

We expect to utilize Azuma–Hoeffding inequality, we now first verify the bounded-increment condition needed for Azuma–Hoeffding. We already have:

$$|X_s| \leq a_s (w_{\max} + w_{\max}) = 2a_s w_{\max}.$$

Therefore, the partial sums $M_u := \sum_{s=1}^u X_s$ form a martingale with respect to $\{\mathcal{F}_u\}$ and satisfy $|M_u - M_{u-1}| = |X_u| \leq 2a_u w_{\max}$ a.s. Applying Azuma–Hoeffding to $\{M_u\}$ yields that for any $u > 0$,

$$\mathbb{P}\left(\left|\sum_{s=1}^{t-1} X_s\right| \geq u\right) \leq 2 \exp\left(-\frac{u^2}{2 \sum_{s=1}^{t-1} (2a_s w_{\max})^2}\right) = 2 \exp\left(-\frac{u^2}{8w_{\max}^2 \sum_{s=1}^{t-1} a_s^2}\right).$$

Since $a_s \in [0, 1]$ implies $a_s^2 \leq a_s$, we have

$$\sum_{s=1}^{t-1} a_s^2 \leq \sum_{s=1}^{t-1} a_s = N_t(g),$$

and therefore

$$\mathbb{P}\left(\left|\sum_{s=1}^{t-1} X_s\right| \geq u\right) \leq 2 \exp\left(-\frac{u^2}{8w_{\max}^2 N_t(g)}\right).$$

Set

$$u := 2w_{\max} \sqrt{2N_t(g) \log(2/\delta')}.$$

Then

$$u^2 = 8w_{\max}^2 N_t(g) \log(2/\delta'), \quad \frac{u^2}{8w_{\max}^2 N_t(g)} = \log(2/\delta').$$

Substituting into the bound yields

$$2 \exp\left(-\frac{u^2}{8w_{\max}^2 N_t(g)}\right) = 2 \exp(-\log(2/\delta')) = 2\left(\frac{2}{\delta'}\right)^{-1} = \delta'.$$

Hence

$$\mathbb{P}\left(\left|\sum_{s=1}^{t-1} X_s\right| \geq 2w_{\max} \sqrt{2N_t(g) \log(2/\delta')}\right) \leq \delta'.$$

Dividing the inside event by $N_t(g) > 0$ and using $\hat{w}_t(g) - \bar{w}_t(g) = (\sum_{s=1}^{t-1} X_s)/N_t(g)$ gives

$$\mathbb{P}\left(|\hat{w}_t(g) - \bar{w}_t(g)| \geq \frac{2w_{\max} \sqrt{2N_t(g) \log(2/\delta')}}{N_t(g)}\right) \leq \delta'.$$

Finally,

$$\frac{2w_{\max} \sqrt{2N_t(g) \log(2/\delta')}}{N_t(g)} = 2w_{\max} \sqrt{\frac{2 \log(2/\delta')}{N_t(g)}},$$

which matches the threshold in (35). This concludes the proof. $\square$

Lemma E.8 (Uniform good event). *Let $\delta \in (0, 1)$ and define $\delta' := \delta/(2GT)$. Define the event*

$$\mathcal{E} := \bigcap_{t=1}^T \bigcap_{g=1}^G \{N_t(g) = 0 \text{ or } |\hat{w}_t(g) - \bar{w}_t(g)| \leq \beta_t(g)\},$$

where $\bar{w}_t(g)$ is defined in (34) and $\beta_t(g)$ is defined in (29). Then $\mathbb{P}(\mathcal{E}) \geq 1 - \frac{\delta}{2} \geq 1 - \delta$.

Proof. For any fixed (t, g) with $N_t(g) > 0$, Lemma E.7 with parameter δ' implies

$$\mathbb{P}\left(|\hat{w}_t(g) - \bar{w}_t(g)| > \beta_t(g)\right) \leq \delta'.$$

If $N_t(g) = 0$, then the corresponding clause in the intersection defining $\mathcal{E}$ holds by construction.

Applying the union bound over all $t \in [T]$ and $g \in [G]$ yields

$$\mathbb{P}(\mathcal{E}^c) \leq \sum_{t=1}^T \sum_{g=1}^G \mathbb{P}\left(|\hat{w}_t(g) - \bar{w}_t(g)| > \beta_t(g)\right) \leq \sum_{t=1}^T \sum_{g=1}^G \delta' = GT \cdot \frac{\delta}{2GT} = \frac{\delta}{2},$$

hence $\mathbb{P}(\mathcal{E}) \geq 1 - \frac{\delta}{2} \geq 1 - \delta$, which concludes the proof. $\square$

E.5. Part 2: From Energy Estimation to Reward Estimation (Computing η_t)

This subsection transfers concentration from the (discounted) group-energy estimates $\widehat{w}_t(g)$ to the regularized normalized rewards $\widehat{r}_t^{(\varepsilon)}(g)$, and makes explicit the additional drift term induced by non-stationarity.

Lemma E.9 (Projection to nonnegative energies is non-expansive). *For any $x \in \mathbb{R}$ and any $u \geq 0$,*

$$|\max\{x, 0\} - u| \leq |x - u|.$$

Proof. Let $\Pi(x) := \max\{x, 0\}$. We show $|\Pi(x) - u| \leq |x - u|$ for any $u \geq 0$ by cases.

Case 1: $x \geq 0$. Then $\Pi(x) = x$ and $|\Pi(x) - u| = |x - u|$.

Case 2: $x < 0$. Then $\Pi(x) = 0$ and $|\Pi(x) - u| = |0 - u| = u$. Since $x < 0 \leq u$, we have $u - x \geq u$, hence

$$|x - u| = u - x \geq u = |\Pi(x) - u|.$$

This concludes the proof. $\square$

Lemma E.10 (Algebraic bound for regularized ratios). *Let $\varepsilon > 0$, and let $A, X \geq 0$ and $a, x \geq 0$. Then*

$$\left| \frac{a}{A + \varepsilon} - \frac{x}{X + \varepsilon} \right| \leq \frac{|a - x|}{\varepsilon} + \frac{x}{\varepsilon^2} |A - X|. \quad (36)$$

Proof. Start from a common denominator:

$$\begin{aligned} \frac{a}{A + \varepsilon} - \frac{x}{X + \varepsilon} &= \frac{a(X + \varepsilon) - x(A + \varepsilon)}{(A + \varepsilon)(X + \varepsilon)} \\ &= \frac{aX + a\varepsilon - xA - x\varepsilon}{(A + \varepsilon)(X + \varepsilon)} \\ &= \frac{(a - x)(X + \varepsilon) + x(X - A)}{(A + \varepsilon)(X + \varepsilon)}. \end{aligned}$$

Taking absolute values and using the triangle inequality gives

$$\left| \frac{a}{A + \varepsilon} - \frac{x}{X + \varepsilon} \right| \leq \frac{|a - x|(X + \varepsilon)}{(A + \varepsilon)(X + \varepsilon)} + \frac{x|X - A|}{(A + \varepsilon)(X + \varepsilon)}.$$

Next, use the lower bounds $(A + \varepsilon) \geq \varepsilon$ and $(X + \varepsilon) \geq \varepsilon$, which imply

$$\frac{X + \varepsilon}{(A + \varepsilon)(X + \varepsilon)} = \frac{1}{A + \varepsilon} \leq \frac{1}{\varepsilon}, \quad \frac{1}{(A + \varepsilon)(X + \varepsilon)} \leq \frac{1}{\varepsilon^2}.$$

Substituting these bounds yields

$$\left| \frac{a}{A + \varepsilon} - \frac{x}{X + \varepsilon} \right| \leq \frac{|a - x|}{\varepsilon} + \frac{x}{\varepsilon^2} |X - A|,$$

which is exactly (36) and concludes the proof. $\square$

Lemma E.11 (From $\widehat{w}_t$ concentration to $\widehat{r}_t^{(\varepsilon)}$ concentration). *Assume Assumptions E.1–E.2 and work on the good event $\mathcal{E}$ of Lemma E.8. Define the discounted-average energies $\overline{w}_t(g)$ as in (34), and define the corresponding discounted-average regularized normalized rewards*

$$\overline{W}_t := \sum_{j=1}^G \overline{w}_t(j), \quad \overline{r}_t^{(\varepsilon)}(g) := \frac{\overline{w}_t(g)}{\overline{W}_t + \varepsilon}, \quad (37)$$

where $\overline{w}_t(g) \geq 0$ since $w_g(\theta_s) \geq 0$ and $\overline{w}_t(g)$ is a convex combination. Then for all $t \leq T$ and all $g \in [G]$,

$$\left| \widehat{r}_t^{(\varepsilon)}(g) - \overline{r}_t^{(\varepsilon)}(g) \right| \leq \eta_t(g), \quad (38)$$

where $\eta_t(g)$ is defined in (30). Consequently,

$$\left| \hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g) \right| \leq \eta_t(g) + \text{Drift}_t, \quad (39)$$

where the drift term is

$$\text{Drift}_t := \max_{g \in [G]} \left| \bar{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g) \right|. \quad (40)$$

Proof. Proof sketch. We proceed in four steps. First, we control the numerator error by projecting the estimated energy onto $\mathbb{R}_+$. Second, we bound the denominator error by summing the per-group energy deviations. Third, we translate these two bounds into a bound on the regularized ratio defining $\hat{r}_t^{(\varepsilon)}$. Finally, we relate the discounted-average reward to the instantaneous reward by adding a drift term that captures non-stationarity.

Fix any $t \leq T$ and any $g \in [G]$.

Step 1: Bound the numerator error $|\hat{w}_t^+(g) - \bar{w}_t(g)|$. Because $\bar{w}_t(g) \geq 0$, apply Lemma E.9 with $x = \hat{w}_t(g)$ and $u = \bar{w}_t(g)$ to obtain

$$|\hat{w}_t^+(g) - \bar{w}_t(g)| \leq |\hat{w}_t(g) - \bar{w}_t(g)|. \quad (41)$$

On the good event $\mathcal{E}$, we have $|\hat{w}_t(g) - \bar{w}_t(g)| \leq \beta_t(g)$ when $N_t(g) > 0$ by definition of $\mathcal{E}$. When $N_t(g) = 0$, $\beta_t(g) = +\infty$ and the bound holds trivially. Hence, on $\mathcal{E}$,

$$|\hat{w}_t^+(g) - \bar{w}_t(g)| \leq \beta_t(g) \quad \text{for all } t, g. \quad (42)$$

Step 2: Bound the denominator error $|\widehat{W}_t - \overline{W}_t|$. By definition,

$$\widehat{W}_t - \overline{W}_t = \sum_{j=1}^G \hat{w}_t^+(j) - \sum_{j=1}^G \bar{w}_t(j) = \sum_{j=1}^G (\hat{w}_t^+(j) - \bar{w}_t(j)).$$

Therefore, by the triangle inequality and (42),

$$|\widehat{W}_t - \overline{W}_t| \leq \sum_{j=1}^G |\hat{w}_t^+(j) - \bar{w}_t(j)| \leq \sum_{j=1}^G \beta_t(j). \quad (43)$$

Step 3: Convert to reward error using the ratio lemma. Apply Lemma E.10 with

$$a = \hat{w}_t^+(g), \quad A = \widehat{W}_t, \quad x = \bar{w}_t(g), \quad X = \overline{W}_t.$$

Because $\bar{w}_t(g) \geq 0$ and $\overline{W}_t \geq 0$, Lemma E.10 applies and yields

$$\left| \frac{\hat{w}_t^+(g)}{\widehat{W}_t + \varepsilon} - \frac{\bar{w}_t(g)}{\overline{W}_t + \varepsilon} \right| \leq \frac{|\hat{w}_t^+(g) - \bar{w}_t(g)|}{\varepsilon} + \frac{\bar{w}_t(g)}{\varepsilon^2} |\widehat{W}_t - \overline{W}_t|. \quad (44)$$

Recognize the left-hand side as $|\hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g)|$ by (28) and (37). Moreover, by Assumption E.2 we have $0 \leq w_g(\theta_s) \leq w_{\max}$, and thus $\bar{w}_t(g)$, being a convex combination of $\{w_g(\theta_s)\}_{s \leq t-1}$, satisfies

$$0 \leq \bar{w}_t(g) \leq w_{\max}. \quad (45)$$

Substituting (42), (43), and (45) into (44) gives

$$\left| \hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g) \right| \leq \frac{\beta_t(g)}{\varepsilon} + \frac{w_{\max}}{\varepsilon^2} \sum_{j=1}^G \beta_t(j) = \eta_t(g),$$

which proves (38).

Step 4: Add the drift term to compare with the instantaneous reward. By the triangle inequality,

$$\left| \hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g) \right| \leq \left| \hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g) \right| + \left| \bar{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g) \right|.$$

Apply (38) to the first term and (40) to the second term to obtain

$$\left| \hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g) \right| \leq \eta_t(g) + \text{Drift}_t,$$

which is exactly (39) and concludes the proof. $\square$

E.6. Part 3: Controlling the drift term via a variation budget

This subsection upper-bounds the cumulative drift $\sum_{t=1}^T \text{Drift}_t$ by a variation budget that quantifies the non-stationarity of the regularized normalized rewards $\{r_t^{(\varepsilon)}\}_{t \leq T}$.

From Part 2, for all $t \leq T$ and all $g \in [G]$,

$$\left| \hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g) \right| \leq \eta_t(g) + \text{Drift}_t, \quad \text{Drift}_t := \max_{g \in [G]} \left| \bar{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g) \right|,$$

where $\bar{r}_t^{(\varepsilon)}(g)$ is the exponentially discounted average reward defined in (37).

To control $\sum_{t=1}^T \text{Drift}_t$ rigorously, we introduce a non-stationarity measure for $\{r_t^{(\varepsilon)}\}$.

Definition E.12 (Variation budget for the regularized normalized rewards). Define the one-step variation

$$\Gamma_t := \|r_t^{(\varepsilon)} - r_{t-1}^{(\varepsilon)}\|_\infty = \max_{g \in [G]} |r_t^{(\varepsilon)}(g) - r_{t-1}^{(\varepsilon)}(g)|, \quad t = 2, 3, \dots, T, \quad (46)$$

and the total variation budget

$$V_T^{(\varepsilon)} := \sum_{t=2}^T \Gamma_t. \quad (47)$$

To facilitate a better understanding and readability, a simplified (time-homogeneous) drift bound is stated first because it exposes the core mechanism under a constant discount factor λ . The next next result will generalize this argument to the time-varying schedule $\{\lambda_t\}$ used by our anytime design.

Lemma E.13 (Deterministic drift bound for the action-dependent self-normalized average). *Let $W_{\max} := G w_{\max}$ and define the minimum discounted count:*

$$\underline{N}_T := \min_{2 \leq t \leq T} \min_{g \in [G]} N_t(g), \quad N_t(g) := \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\}.$$

Then for every $t \geq 2$,

$$\text{Drift}_t = \|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_\infty \leq \frac{C_{\text{drift}}}{(1-\lambda) \underline{N}_T} \sum_{u=2}^t \lambda^{t-u} \Gamma_u, \quad (48)$$

and consequently,

$$\sum_{t=1}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1-\lambda)^2 \underline{N}_T} V_T^{(\varepsilon)}. \quad (49)$$

Here $\Gamma_t := \|r_t^{(\varepsilon)} - r_{t-1}^{(\varepsilon)}\|_\infty$ and $V_T^{(\varepsilon)} := \sum_{t=2}^T \Gamma_t$ as in (47), and

$$C_{\text{drift}} := \left(\frac{1}{\varepsilon} + \frac{G w_{\max}}{\varepsilon^2} \right) \cdot (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right). \quad (50)$$

Moreover, defining $N_t^{\min} := \min_{g \in [G]} N_t(g)$, we also have the refined (tighter) bound

$$\sum_{t=1}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{1-\lambda} \sum_{t=2}^T \frac{1}{N_t^{\min}} \sum_{u=2}^t \lambda^{t-u} \Gamma_u. \quad (51)$$

Proof. Proof sketch. We proceed in four steps. First, we show that the normalization map $x \mapsto x/(\sum_j x(j) + \varepsilon)$ is Lipschitz in $\|\cdot\|_\infty$ over $[0, w_{\max}]^G$, with an explicit constant depending on $(\varepsilon, G, w_{\max})$. Second, we relate changes in the energy vector w_t to changes in the reward vector $r_t^{(\varepsilon)}$, obtaining a bound of the form $\|w_t - w_{t-1}\|_\infty \leq C(W_{\max} + \varepsilon)\Gamma_t$ for an explicit constant C . Third, we express the discounted average $\bar{r}_t^{(\varepsilon)}$ as a self-normalized, action-dependent exponential moving average, and bound $\|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_\infty$ by a geometrically weighted sum of one-step variations $\{\Gamma_u\}_{u \leq t}$ scaled by the inverse minimum discounted count $\underline{N}_T$. Finally, we sum the resulting pointwise bound over $t = 1, \dots, T$ and evaluate the geometric series to obtain (49).

Step 1 (A Lipschitz bound for the normalization map). Fix $x, y \in [0, w_{\max}]^G$ and define

$$S(x) := \sum_{j=1}^G x(j), \quad S(y) := \sum_{j=1}^G y(j).$$

Fix any $g \in [G]$ and consider

$$\frac{x(g)}{S(x) + \varepsilon} - \frac{y(g)}{S(y) + \varepsilon}.$$

Bring to a common denominator and reorganize the numerator:

$$\begin{aligned} \frac{x(g)}{S(x) + \varepsilon} - \frac{y(g)}{S(y) + \varepsilon} &= \frac{x(g)(S(y) + \varepsilon) - y(g)(S(x) + \varepsilon)}{(S(x) + \varepsilon)(S(y) + \varepsilon)} \\ &= \frac{(x(g) - y(g))(S(y) + \varepsilon) + y(g)(S(y) - S(x))}{(S(x) + \varepsilon)(S(y) + \varepsilon)}. \end{aligned}$$

Taking absolute values and applying the triangle inequality yields

$$\begin{aligned} \left| \frac{x(g)}{S(x) + \varepsilon} - \frac{y(g)}{S(y) + \varepsilon} \right| &\leq \frac{|x(g) - y(g)|(S(y) + \varepsilon)}{(S(x) + \varepsilon)(S(y) + \varepsilon)} + \frac{y(g)|S(x) - S(y)|}{(S(x) + \varepsilon)(S(y) + \varepsilon)} \\ &= \frac{|x(g) - y(g)|}{S(x) + \varepsilon} + \frac{y(g)}{(S(x) + \varepsilon)(S(y) + \varepsilon)} |S(x) - S(y)| \\ &\leq \frac{|x(g) - y(g)|}{\varepsilon} + \frac{y(g)}{\varepsilon^2} |S(x) - S(y)|. \end{aligned}$$

Next, since $y(g) \leq w_{\max}$ and

$$|S(x) - S(y)| = \left| \sum_{j=1}^G (x(j) - y(j)) \right| \leq \sum_{j=1}^G |x(j) - y(j)| \leq \sum_{j=1}^G \|x - y\|_\infty = G\|x - y\|_\infty,$$

and also $|x(g) - y(g)| \leq \|x - y\|_\infty$, we obtain

$$\begin{aligned} \left| \frac{x(g)}{S(x) + \varepsilon} - \frac{y(g)}{S(y) + \varepsilon} \right| &\leq \frac{1}{\varepsilon} \|x - y\|_\infty + \frac{w_{\max}}{\varepsilon^2} \cdot G\|x - y\|_\infty \\ &= \left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) \|x - y\|_\infty. \end{aligned}$$

Taking the maximum over $g \in [G]$ gives

$$\left\| \frac{x}{S(x) + \varepsilon} - \frac{y}{S(y) + \varepsilon} \right\|_\infty \leq \left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) \|x - y\|_\infty. \quad (52)$$

Now apply (52) with $x = \bar{w}_t$ and $y = w_t$ (so that $\bar{r}_t^{(\varepsilon)} = \bar{w}_t/(\sum_j \bar{w}_t(j) + \varepsilon)$ and $r_t^{(\varepsilon)} = w_t/(\sum_j w_t(j) + \varepsilon)$):

$$\text{Drift}_t = \|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_\infty \leq \left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) \|\bar{w}_t - w_t\|_\infty. \quad (53)$$

Step 2 (Relate $\|w(\theta_t) - w(\theta_{t-1})\|_\infty$ to Γ_t). Recall $r_t^{(\varepsilon)}(g) = \frac{w_g(\theta_t)}{W_t + \varepsilon}$ where

$$W_t := \sum_{j=1}^G w_j(\theta_t).$$

Thus for each g ,

$$w_g(\theta_t) = (W_t + \varepsilon) r_t^{(\varepsilon)}(g). \quad (54)$$

Define $s_t := \sum_{j=1}^G r_t^{(\varepsilon)}(j)$. Then

$$s_t = \sum_{j=1}^G \frac{w_j(\theta_t)}{W_t + \varepsilon} = \frac{W_t}{W_t + \varepsilon}, \quad \Rightarrow \quad 1 - s_t = \frac{\varepsilon}{W_t + \varepsilon}.$$

Hence

$$W_t + \varepsilon = \frac{\varepsilon}{1 - s_t}. \quad (55)$$

Because $0 \leq w_j(\theta_t) \leq w_{\max}$, we have $0 \leq W_t \leq W_{\max} := Gw_{\max}$, thus

$$W_t + \varepsilon \leq W_{\max} + \varepsilon, \quad 1 - s_t = \frac{\varepsilon}{W_t + \varepsilon} \geq \frac{\varepsilon}{W_{\max} + \varepsilon},$$

and similarly $1 - s_{t-1} \geq \frac{\varepsilon}{W_{\max} + \varepsilon}$.

Fix any $g \in [G]$. Using (54) at times t and $t-1$,

$$\begin{aligned} w_g(\theta_t) - w_g(\theta_{t-1}) &= (W_t + \varepsilon) r_t^{(\varepsilon)}(g) - (W_{t-1} + \varepsilon) r_{t-1}^{(\varepsilon)}(g) \\ &= (W_t + \varepsilon) (r_t^{(\varepsilon)}(g) - r_{t-1}^{(\varepsilon)}(g)) + ((W_t + \varepsilon) - (W_{t-1} + \varepsilon)) r_{t-1}^{(\varepsilon)}(g) \\ &= (W_t + \varepsilon) (r_t^{(\varepsilon)}(g) - r_{t-1}^{(\varepsilon)}(g)) + (W_t - W_{t-1}) r_{t-1}^{(\varepsilon)}(g). \end{aligned}$$

Take absolute values and use $|r_{t-1}^{(\varepsilon)}(g)| \leq 1$:

$$|w_g(\theta_t) - w_g(\theta_{t-1})| \leq (W_t + \varepsilon) |r_t^{(\varepsilon)}(g) - r_{t-1}^{(\varepsilon)}(g)| + |W_t - W_{t-1}|. \quad (56)$$

For the first term, use $W_t + \varepsilon \leq W_{\max} + \varepsilon$ and $|r_t^{(\varepsilon)}(g) - r_{t-1}^{(\varepsilon)}(g)| \leq \Gamma_t$:

$$(W_t + \varepsilon) |r_t^{(\varepsilon)}(g) - r_{t-1}^{(\varepsilon)}(g)| \leq (W_{\max} + \varepsilon) \Gamma_t. \quad (57)$$

For the second term, express W_t in terms of s_t : from $s_t = \frac{W_t}{W_t + \varepsilon}$ we get

$$W_t = s_t(W_t + \varepsilon) = s_t \frac{\varepsilon}{1 - s_t} = \frac{\varepsilon s_t}{1 - s_t}.$$

Therefore

$$\begin{aligned} |W_t - W_{t-1}| &= \left| \frac{\varepsilon s_t}{1 - s_t} - \frac{\varepsilon s_{t-1}}{1 - s_{t-1}} \right| = \varepsilon \left| \frac{s_t}{1 - s_t} - \frac{s_{t-1}}{1 - s_{t-1}} \right| \\ &= \varepsilon \left| \frac{s_t(1 - s_{t-1}) - s_{t-1}(1 - s_t)}{(1 - s_t)(1 - s_{t-1})} \right| = \varepsilon \frac{|s_t - s_{t-1}|}{(1 - s_t)(1 - s_{t-1})}. \end{aligned}$$

Bound the numerator:

$$|s_t - s_{t-1}| = \left| \sum_{j=1}^G (r_t^{(\varepsilon)}(j) - r_{t-1}^{(\varepsilon)}(j)) \right| \leq \sum_{j=1}^G \Gamma_t = G\Gamma_t.$$

Bound the denominator from below using $1 - s_t \geq \frac{\varepsilon}{W_{\max} + \varepsilon}$ and $1 - s_{t-1} \geq \frac{\varepsilon}{W_{\max} + \varepsilon}$:

$$(1 - s_t)(1 - s_{t-1}) \geq \left(\frac{\varepsilon}{W_{\max} + \varepsilon} \right)^2.$$

Hence

$$|W_t - W_{t-1}| \leq \varepsilon \cdot \frac{G\Gamma_t}{\left(\frac{\varepsilon}{W_{\max} + \varepsilon} \right)^2} = G \frac{(W_{\max} + \varepsilon)^2}{\varepsilon} \Gamma_t. \quad (58)$$

Combine (56), (57), and (58):

$$\begin{aligned} |w_g(\theta_t) - w_g(\theta_{t-1})| &\leq (W_{\max} + \varepsilon)\Gamma_t + G \frac{(W_{\max} + \varepsilon)^2}{\varepsilon} \Gamma_t \\ &= (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right) \Gamma_t. \end{aligned}$$

Taking the maximum over $g \in [G]$ yields

$$\|w(\theta_t) - w(\theta_{t-1})\|_{\infty} \leq (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right) \Gamma_t. \quad (59)$$

Step 3 (Action-dependent discounted averaging yields a $1/N_T$ factor). For each group $g \in [G]$, the self-normalized discounted average is

$$\bar{w}_t(g) = \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} w_g(\theta_s), \quad N_t(g) := \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\}.$$

Therefore, we write $\bar{w}_t(g)$ using its definition and then subtract $w_g(\theta_t)$:

$$\begin{aligned} \bar{w}_t(g) - w_g(\theta_t) &= \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} w_g(\theta_s) - w_g(\theta_t) \\ &= \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} w_g(\theta_s) - \frac{1}{N_t(g)} \cdot N_t(g) \cdot w_g(\theta_t) \\ &= \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} w_g(\theta_s) - \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} w_g(\theta_t) \\ &= \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} (w_g(\theta_s) - w_g(\theta_t)). \end{aligned}$$

Taking absolute values and using the triangle inequality,

$$|\bar{w}_t(g) - w_g(\theta_t)| \leq \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} |w_g(\theta_s) - w_g(\theta_t)|. \quad (60)$$

For any $1 \leq s \leq t-1$, expand the difference as a telescoping sum:

$$w_g(\theta_s) - w_g(\theta_t) = \sum_{u=s+1}^t (w_g(\theta_{u-1}) - w_g(\theta_u)).$$

Taking absolute values and applying the triangle inequality again,

$$|w_g(\theta_s) - w_g(\theta_t)| \leq \sum_{u=s+1}^t |w_g(\theta_{u-1}) - w_g(\theta_u)| \leq \sum_{u=s+1}^t \|w(\theta_u) - w(\theta_{u-1})\|_{\infty}.$$

Substituting into (60) gives

$$|\bar{w}_t(g) - w_g(\theta_t)| \leq \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} \sum_{u=s+1}^t \|w(\theta_u) - w(\theta_{u-1})\|_\infty.$$

Swap the order of summation (collect the coefficient of each $\|w(\theta_u) - w(\theta_{u-1})\|_\infty$):

$$|\bar{w}_t(g) - w_g(\theta_t)| \leq \frac{1}{N_t(g)} \sum_{u=2}^t \|w(\theta_u) - w(\theta_{u-1})\|_\infty \sum_{s=1}^{u-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\}. \quad (61)$$

Upper bound the inner sum using $\mathbf{1}\{g_s = g\} \leq 1$:

$$\sum_{s=1}^{u-1} \lambda^{t-1-s} \mathbf{1}\{g_s = g\} \leq \sum_{s=1}^{u-1} \lambda^{t-1-s}.$$

Compute the geometric sum:

$$\sum_{s=1}^{u-1} \lambda^{t-1-s} = \lambda^{t-u} \sum_{k=0}^{u-2} \lambda^k \leq \lambda^{t-u} \sum_{k=0}^{\infty} \lambda^k = \frac{\lambda^{t-u}}{1-\lambda}.$$

Plugging this into (61) yields

$$|\bar{w}_t(g) - w_g(\theta_t)| \leq \frac{1}{(1-\lambda)N_t(g)} \sum_{u=2}^t \lambda^{t-u} \|w(\theta_u) - w(\theta_{u-1})\|_\infty. \quad (62)$$

Now introduce the clipped minimum discounted count

$$\underline{N}_T := \min_{2 \leq t \leq T} \min_{g \in [G]} N_t(g).$$

Since $1/N_t(g) \leq 1/\underline{N}_T$, for all $T \geq t \geq 2$,

$$|\bar{w}_t(g) - w_g(\theta_t)| \leq \frac{1}{(1-\lambda)\underline{N}_T} \sum_{u=2}^t \lambda^{t-u} \|w(\theta_u) - w(\theta_{u-1})\|_\infty. \quad (63)$$

Taking the maximum over g gives

$$\|\bar{w}_t - w(\theta_t)\|_\infty \leq \frac{1}{(1-\lambda)\underline{N}_T} \sum_{u=2}^t \lambda^{t-u} \|w(\theta_u) - w(\theta_{u-1})\|_\infty. \quad (64)$$

Step 4 (Combine all steps and sum over t). Combine (53) and (64):

$$\text{Drift}_t \leq \left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) \cdot \frac{1}{(1-\lambda)\underline{N}_T} \sum_{u=2}^t \lambda^{t-u} \|w(\theta_u) - w(\theta_{u-1})\|_\infty. \quad (65)$$

Now apply (59) to each $\|w(\theta_u) - w(\theta_{u-1})\|_\infty$:

$$\|w(\theta_u) - w(\theta_{u-1})\|_\infty \leq (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right) \Gamma_u.$$

Substitute this into (65):

$$\begin{aligned} \text{Drift}_t &\leq \left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) \cdot \frac{1}{(1-\lambda)\underline{N}_T} \sum_{u=2}^t \lambda^{t-u} (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right) \Gamma_u \\ &= \underbrace{\left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right)}_{C_{\text{drift}}} \cdot \frac{1}{(1-\lambda)\underline{N}_T} \sum_{u=2}^t \lambda^{t-u} \Gamma_u. \end{aligned} \quad (66)$$

This yields the explicit constant definition (matching (50)):

$$C_{\text{drift}} = \left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right). \quad (67)$$

Finally, sum (66) over $t = 2, \dots, T$:

$$\sum_{t=2}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1-\lambda)\underline{N}_T} \sum_{t=2}^T \sum_{u=2}^t \lambda^{t-u} \Gamma_u.$$

Swap the order of summation: for each fixed u , t ranges from u to T :

$$\sum_{t=2}^T \sum_{u=2}^t \lambda^{t-u} \Gamma_u = \sum_{u=2}^T \Gamma_u \sum_{t=u}^T \lambda^{t-u}.$$

Compute the inner geometric sum:

$$\sum_{t=u}^T \lambda^{t-u} = \sum_{k=0}^{T-u} \lambda^k \leq \sum_{k=0}^{\infty} \lambda^k = \frac{1}{1-\lambda}.$$

Therefore

$$\sum_{t=2}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1-\lambda)\underline{N}_T} \sum_{u=2}^T \Gamma_u \cdot \frac{1}{1-\lambda} = \frac{C_{\text{drift}}}{(1-\lambda)^2 \underline{N}_T} \sum_{u=2}^T \Gamma_u.$$

Recall $V_T^{(\varepsilon)} := \sum_{u=2}^T \Gamma_u$, so we obtain

$$\sum_{t=1}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1-\lambda)^2 \underline{N}_T} V_T^{(\varepsilon)}. \quad (68)$$

This proves the lemma up to (49) and (50).

Step 5 (A refined summed drift bound without $\underline{N}_T$). Starting from the pointwise bound (62), take the maximum over $g \in [G]$ and use $\max_{g \in [G]} \frac{1}{N_t(g)} = \frac{1}{\min_{g \in [G]} N_t(g)} =: \frac{1}{N_t^{\min}}$ to obtain

$$\|\bar{w}_t - w(\theta_t)\|_{\infty} \leq \frac{1}{(1-\lambda)N_t^{\min}} \sum_{u=2}^t \lambda^{t-u} \|w(\theta_u) - w(\theta_{u-1})\|_{\infty}.$$

Combining this display with (53) and (59) yields, for all $t \geq 2$,

$$\text{Drift}_t \leq \frac{C_{\text{drift}}}{(1-\lambda)N_t^{\min}} \sum_{u=2}^t \lambda^{t-u} \Gamma_u.$$

Summing over $t = 2, \dots, T$ gives

$$\sum_{t=2}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{1-\lambda} \sum_{t=2}^T \frac{1}{N_t^{\min}} \sum_{u=2}^t \lambda^{t-u} \Gamma_u.$$

Since $\text{Drift}_1 = 0$, the same bound holds for $\sum_{t=1}^T \text{Drift}_t$, proving (51). $\square$

Next lemma is the time-varying counterpart of Lemma E.13. It is the version used by our anytime schedules, where the discount factor evolves with time. The proof largely reuses the constant- λ proof, with geometric weights replaced by products and controlled via $\bar{\lambda}_T$.

Lemma E.14 (Time-varying discount extension of Lemma E.13). *Let $W_{\max} := G w_{\max}$ and let $\{\lambda_t\}_{t \geq 1} \subset (0, 1)$ be an arbitrary (possibly time-varying) discount schedule. For each $g \in [G]$ and $t \geq 2$, define the discounted count and discounted average*

$$N_t(g) := \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}, \quad \bar{w}_t(g) := \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\} w_g(\theta_s),$$

and let $\underline{N}_T := \min_{2 \leq t \leq T} \min_{g \in [G]} N_t(g)$. Let

$$r_t^{(\varepsilon)}(g) := \frac{w_g(\theta_t)}{W_t + \varepsilon}, \quad W_t := \sum_{j=1}^G w_j(\theta_t), \quad \bar{r}_t^{(\varepsilon)}(g) := \frac{\bar{w}_t(g)}{\bar{W}_t + \varepsilon}, \quad \bar{W}_t := \sum_{j=1}^G \bar{w}_t(j),$$

denote the instantaneous and discounted normalized rewards, and define

$$\text{Drift}_t := \|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_{\infty}, \quad \Gamma_t := \|r_t^{(\varepsilon)} - r_{t-1}^{(\varepsilon)}\|_{\infty}, \quad V_T^{(\varepsilon)} := \sum_{t=2}^T \Gamma_t.$$

Set $\bar{\lambda}_T := \max_{1 \leq t \leq T} \lambda_t$ (so $1 - \bar{\lambda}_T = \min_{1 \leq t \leq T} (1 - \lambda_t)$). Then for every $t \geq 2$,

$$\text{Drift}_t \leq \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T) \underline{N}_T} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \Gamma_u, \quad (69)$$

and consequently,

$$\sum_{t=1}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T)^2 \underline{N}_T} V_T^{(\varepsilon)}. \quad (70)$$

Here, one may take the same explicit constant as in Lemma E.13:

$$C_{\text{drift}} := \left(\frac{1}{\varepsilon} + \frac{G w_{\max}}{\varepsilon^2} \right) (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right).$$

In particular, if $\lambda_t \equiv \lambda$ is constant, then $\bar{\lambda}_T = \lambda$ and $\prod_{j=u}^{t-1} \lambda_j = \lambda^{t-u}$, so (69)–(70) reduce to Lemma E.13.

Moreover, defining $N_t^{\min} := \min_{g \in [G]} N_t(g)$, we also have the refined bound

$$\sum_{t=1}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{1 - \bar{\lambda}_T} \sum_{t=2}^T \frac{1}{N_t^{\min}} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \Gamma_u \leq \frac{C_{\text{drift}}}{1 - \bar{\lambda}_T} \sum_{t=2}^T \frac{1}{N_t^{\min}} \sum_{u=2}^t (\bar{\lambda}_T)^{t-u} \Gamma_u. \quad (71)$$

Proof. Proof sketch. We first use a Lipschitz property of the regularized normalization map $x \mapsto x/(\mathbf{1}^\top x + \varepsilon)$ on $[0, w_{\max}]^G$ to reduce $\text{Drift}_t = \|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_{\infty}$ to a bound on $\|\bar{w}_t - w(\theta_t)\|_{\infty}$. We then rewrite each coordinate difference $\bar{w}_t(g) - w_g(\theta_t)$ as a self-normalized discounted sum of past increments, where the weights involve the products $\prod_{j=u}^{t-1} \lambda_j$. Next, we upper bound the accumulated weights by factoring out $\prod_{j=u}^{t-1} \lambda_j$ and controlling the remaining sum by a geometric series with ratio $\bar{\lambda}_T$. Finally, we relate the energy increments $\|w(\theta_u) - w(\theta_{u-1})\|_{\infty}$ to the reward variation Γ_u , which yields the pointwise bound (69); summing over t and using another geometric-series bound gives (70).

Fix $t \geq 2$.

Step 1: Reduce Drift_t to $\|\bar{w}_t - w(\theta_t)\|_{\infty}$ (same as Lemma E.13). Define the regularized normalization map $\phi(x) := x/(\mathbf{1}^\top x + \varepsilon)$ on $[0, w_{\max}]^G$. The Lipschitz calculation used in Lemma E.13 (see (52)–(53)) gives

$$\text{Drift}_t = \|\phi(\bar{w}_t) - \phi(w(\theta_t))\|_{\infty} \leq \left(\frac{1}{\varepsilon} + \frac{G w_{\max}}{\varepsilon^2} \right) \|\bar{w}_t - w(\theta_t)\|_{\infty}. \quad (72)$$

Step 2: Express $|\bar{w}_t(g) - w_g(\theta_t)|$ as a weighted sum of increments. Fix a group $g \in [G]$. Introduce the (nonnegative) weights

$$a_{s,t}(g) := \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}, \quad s = 1, \dots, t-1,$$

where $a_{s,t}(g)$ denotes how much the group g at time s contributes to the time t ($t > s$) when we compute the discounted count of group g (i.e., the denominator). So that

$$N_t(g) = \sum_{s=1}^{t-1} a_{s,t}(g),$$

which denotes the the cumulative contribution of group g at time $[1, t)$ to the time t , i.e., the definition of the discounted count of the group g before the time t , and

$$\bar{w}_t(g) = \frac{1}{N_t(g)} \sum_{s=1}^{t-1} a_{s,t}(g) w_g(\theta_s).$$

Then, by expanding the difference and regrouping terms,

$$\begin{aligned} w_g(\theta_t) - \bar{w}_t(g) &= w_g(\theta_t) - \frac{1}{N_t(g)} \sum_{s=1}^{t-1} a_{s,t}(g) w_g(\theta_s) \\ &= \frac{N_t(g)}{N_t(g)} w_g(\theta_t) - \frac{1}{N_t(g)} \sum_{s=1}^{t-1} a_{s,t}(g) w_g(\theta_s) \\ &= \frac{1}{N_t(g)} \left(N_t(g) w_g(\theta_t) - \sum_{s=1}^{t-1} a_{s,t}(g) w_g(\theta_s) \right) \\ &= \frac{1}{N_t(g)} \left(\sum_{s=1}^{t-1} a_{s,t}(g) w_g(\theta_t) - \sum_{s=1}^{t-1} a_{s,t}(g) w_g(\theta_s) \right) \quad \left(\text{since } N_t(g) = \sum_{s=1}^{t-1} a_{s,t}(g) \right) \\ &= \frac{1}{N_t(g)} \sum_{s=1}^{t-1} a_{s,t}(g) (w_g(\theta_t) - w_g(\theta_s)). \end{aligned} \tag{73}$$

For each $s \leq t-1$, telescope

$$w_g(\theta_t) - w_g(\theta_s) = \sum_{u=s+1}^t (w_g(\theta_u) - w_g(\theta_{u-1})),$$

plug into (73), take absolute values, and use $|w_g(\theta_u) - w_g(\theta_{u-1})| \leq \|w(\theta_u) - w(\theta_{u-1})\|_\infty$:

$$\begin{aligned} |w_g(\theta_t) - \bar{w}_t(g)| &\leq \frac{1}{N_t(g)} \sum_{s=1}^{t-1} a_{s,t}(g) |w_g(\theta_t) - w_g(\theta_s)| \\ &\leq \frac{1}{N_t(g)} \sum_{s=1}^{t-1} a_{s,t}(g) \sum_{u=s+1}^t |w_g(\theta_u) - w_g(\theta_{u-1})| \\ &\leq \frac{1}{N_t(g)} \sum_{s=1}^{t-1} a_{s,t}(g) \sum_{u=s+1}^t \|w(\theta_u) - w(\theta_{u-1})\|_\infty \\ &= \frac{1}{N_t(g)} \sum_{s=1}^{t-1} \sum_{u=s+1}^t a_{s,t}(g) \|w(\theta_u) - w(\theta_{u-1})\|_\infty \\ &= \frac{1}{N_t(g)} \sum_{u=2}^t \sum_{s=1}^{u-1} a_{s,t}(g) \|w(\theta_u) - w(\theta_{u-1})\|_\infty \\ &= \frac{1}{N_t(g)} \sum_{u=2}^t \|w(\theta_u) - w(\theta_{u-1})\|_\infty \sum_{s=1}^{u-1} a_{s,t}(g). \end{aligned} \tag{74}$$

where the penultimate step swaps the order of summation (for fixed u , the indices that contribute satisfy $s \leq u - 1$).

Step 3: Bound the inner weight sum $\sum_{s=1}^{u-1} a_{s,t}(g)$. Fix $u \in \{2, \dots, t\}$. For every $s \leq u - 1$, factor the product in $a_{s,t}(g)$ as

$$\prod_{j=s+1}^{t-1} \lambda_j = \left(\prod_{j=u}^{t-1} \lambda_j \right) \left(\prod_{j=s+1}^{u-1} \lambda_j \right),$$

so

$$\begin{aligned} \sum_{s=1}^{u-1} a_{s,t}(g) &= \left(\prod_{j=u}^{t-1} \lambda_j \right) \sum_{s=1}^{u-1} \left(\prod_{j=s+1}^{u-1} \lambda_j \right) \mathbf{1}\{g_s = g\} \\ &\leq \left(\prod_{j=u}^{t-1} \lambda_j \right) \sum_{s=1}^{u-1} \left(\prod_{j=s+1}^{u-1} \lambda_j \right). \end{aligned} \quad (75)$$

Rewrite the last sum by grouping terms according to the length of the product:

$$\sum_{s=1}^{u-1} \left(\prod_{j=s+1}^{u-1} \lambda_j \right) = \sum_{k=0}^{u-2} \left(\prod_{j=u-k}^{u-1} \lambda_j \right),$$

where $\prod_{j=u}^{u-1} \lambda_j := 1$.

Since each $\lambda_j \leq \bar{\lambda}_T$, we have $\prod_{j=u-k}^{u-1} \lambda_j \leq (\bar{\lambda}_T)^k$, and therefore

$$\sum_{s=1}^{u-1} \left(\prod_{j=s+1}^{u-1} \lambda_j \right) \leq \sum_{k=0}^{u-2} (\bar{\lambda}_T)^k \leq \sum_{k=0}^{\infty} (\bar{\lambda}_T)^k = \frac{1}{1 - \bar{\lambda}_T}. \quad (76)$$

Combining (75) and (76) yields

$$\sum_{s=1}^{u-1} a_{s,t}(g) \leq \frac{1}{1 - \bar{\lambda}_T} \left(\prod_{j=u}^{t-1} \lambda_j \right). \quad (77)$$

Step 4: Obtain a bound on $\|\bar{w}_t - w(\theta_t)\|_{\infty}$. Substitute (77) into (74) to obtain, for every $g \in [G]$,

$$|w_g(\theta_t) - \bar{w}_t(g)| \leq \frac{1}{(1 - \bar{\lambda}_T) N_t(g)} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \|w(\theta_u) - w(\theta_{u-1})\|_{\infty}.$$

Using $N_t(g) \geq \underline{N}_T$ and taking the maximum over $g \in [G]$ gives

$$\|\bar{w}_t - w(\theta_t)\|_{\infty} \leq \frac{1}{(1 - \bar{\lambda}_T) \underline{N}_T} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \|w(\theta_u) - w(\theta_{u-1})\|_{\infty}. \quad (78)$$

Step 5: Convert $\|w(\theta_u) - w(\theta_{u-1})\|_{\infty}$ to Γ_u (same as Lemma E.13). The deterministic conversion used in Lemma E.13 (see (59)) yields, for every $u \geq 2$,

$$\|w(\theta_u) - w(\theta_{u-1})\|_{\infty} \leq (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right) \Gamma_u. \quad (79)$$

Step 6: Prove the pointwise drift bound (69). Plug (78) and (79) into (72):

$$\begin{aligned} \text{Drift}_t &\leq \left(\frac{1}{\varepsilon} + \frac{Gw_{\max}}{\varepsilon^2} \right) \cdot \frac{1}{(1 - \bar{\lambda}_T) \underline{N}_T} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) (W_{\max} + \varepsilon) \left(1 + \frac{G(W_{\max} + \varepsilon)}{\varepsilon} \right) \Gamma_u \\ &= \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T) \underline{N}_T} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \Gamma_u, \end{aligned}$$

which is exactly (69).

Step 7: Sum over t to obtain (70). Sum (69) over $t = 2, \dots, T$ and swap the order of summation:

$$\begin{aligned} \sum_{t=2}^T \text{Drift}_t &\leq \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T) \underline{N}_T} \sum_{t=2}^T \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \Gamma_u \\ &= \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T) \underline{N}_T} \sum_{u=2}^T \Gamma_u \sum_{t=u}^T \left(\prod_{j=u}^{t-1} \lambda_j \right). \end{aligned}$$

For each fixed u , bound the inner sum by a geometric series: since $\lambda_j \leq \bar{\lambda}_T$,

$$\prod_{j=u}^{t-1} \lambda_j \leq (\bar{\lambda}_T)^{t-u}, \quad \Rightarrow \quad \sum_{t=u}^T \left(\prod_{j=u}^{t-1} \lambda_j \right) \leq \sum_{k=0}^{\infty} (\bar{\lambda}_T)^k = \frac{1}{1 - \bar{\lambda}_T}.$$

Therefore,

$$\sum_{t=2}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T)^2 \underline{N}_T} \sum_{u=2}^T \Gamma_u = \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T)^2 \underline{N}_T} V_T^{(\varepsilon)}.$$

Since $\text{Drift}_1 = 0$ by definition (no past to average), the same bound holds for $\sum_{t=1}^T \text{Drift}_t$, which proves (70).

Step 8: A refined summed drift bound without $\underline{N}_T$. From the displayed inequality in Step 4 (before using $N_t(g) \geq \underline{N}_T$), we have for every $g \in [G]$,

$$|w_g(\theta_t) - \bar{w}_t(g)| \leq \frac{1}{(1 - \bar{\lambda}_T) N_t(g)} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \|w(\theta_u) - w(\theta_{u-1})\|_{\infty}.$$

Taking the maximum over $g \in [G]$ and using $\max_{g \in [G]} \frac{1}{N_t(g)} = \frac{1}{\min_{g \in [G]} N_t(g)} =: \frac{1}{N_t^{\min}}$ gives

$$\|\bar{w}_t - w(\theta_t)\|_{\infty} \leq \frac{1}{(1 - \bar{\lambda}_T) N_t^{\min}} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \|w(\theta_u) - w(\theta_{u-1})\|_{\infty}.$$

Combining this display with (72) and (79) yields, for all $t \geq 2$,

$$\text{Drift}_t \leq \frac{C_{\text{drift}}}{(1 - \bar{\lambda}_T) N_t^{\min}} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \Gamma_u.$$

Summing over $t = 2, \dots, T$ gives

$$\sum_{t=2}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{1 - \bar{\lambda}_T} \sum_{t=2}^T \frac{1}{N_t^{\min}} \sum_{u=2}^t \left(\prod_{j=u}^{t-1} \lambda_j \right) \Gamma_u.$$

Since $\prod_{j=u}^{t-1} \lambda_j \leq (\bar{\lambda}_T)^{t-u}$, we further obtain

$$\sum_{t=2}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{1 - \bar{\lambda}_T} \sum_{t=2}^T \frac{1}{N_t^{\min}} \sum_{u=2}^t (\bar{\lambda}_T)^{t-u} \Gamma_u.$$

As $\text{Drift}_1 = 0$, this proves (71). □

E.7. An anytime choice of (λ_t, γ_t) ensuring $\underline{N}_t = \Omega(\log t)$ and sublinear drift

This subsection specifies an anytime schedule for the discount factors $\{\lambda_t\}_{t \geq 1}$ and exploration levels $\{\gamma_t\}_{t \geq 1}$. The goal is to guarantee that the minimum discounted count $\underline{N}_t$ grows at least logarithmically in t , while the induced drift term remains sublinear.

Anytime schedules. Fix a confidence parameter $\delta \in (0, 1)$ and any $\kappa > 0$. For all $t \geq 1$, define

$$1 - \lambda_t := \frac{1}{(\log(e + t))^{1+\kappa}} \iff \lambda_t = 1 - \frac{1}{(\log(e + t))^{1+\kappa}}. \quad (80)$$

We also introduce an inflation factor $\phi : \mathbb{N} \rightarrow [1, \infty)$ that is nondecreasing in t ; in this paper we take

$$\phi(t) := (\log(e + t))^{\kappa/2}. \quad (81)$$

Define the time-varying exploration level

$$\gamma_t := \min \left\{ \frac{1}{2}, c(1 - \lambda_t) \phi(t) \log \left(\frac{G\pi^2 t^2}{\delta} \right) \right\}, \quad c > 0. \quad (82)$$

We assume the action selection distribution satisfies the (already used) exploration lower bound

$$\mathbb{P}_t(g) \geq \frac{\gamma_t}{G}, \quad \forall t \geq 1, \forall g \in [G]. \quad (83)$$

A sufficient construction is the argmax-plus-uniform mixture

$$\mathbb{P}_t = (1 - \gamma_t) e_{\hat{g}_t} + \gamma_t \cdot \frac{1}{G} \mathbf{1},$$

where $e_g \in \mathbb{R}^G$ denotes the g -th standard basis vector and $\mathbf{1} \in \mathbb{R}^G$ is the all-ones vector.

Discounted counts with time-varying λ_t . Define the discounted count process recursively by

$$N_{t+1}(g) := \lambda_t N_t(g) + \mathbf{1}\{g_t = g\}, \quad N_1(g) := 0, \quad (84)$$

which is equivalent to the explicit expansion

$$N_t(g) = \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}, \quad t \geq 2. \quad (85)$$

Remark E.15 (Time-varying vs. constant discount factors). Throughout Sec. E, F, and the subsequent mismatch analysis starting from Lemma G.1—we allow a predictable time-varying discount schedule $\{\lambda_t\}_{t \geq 1}$, leading to the recursion (84). The constant-discount algorithm is recovered as the special case $\lambda_t \equiv \lambda$ for all t . We use

$$\bar{\lambda}_T := \max_{1 \leq t \leq T} \lambda_t$$

to denote the worst-case (largest) discount factor up to time T . Accordingly, bounds are expressed in terms of $(1 - \bar{\lambda}_T)$, which captures the smallest discount/exploration scale over $t \leq T$.

Discussion on the construction of $1 - \lambda_t$. We provide additional intuition for our choice of the schedule $\{1 - \lambda_t\}_{t \geq 1}$ and state a simple sufficient condition that captures its role at a higher level. To keep the main regret analysis streamlined, the extended discussion is deferred to Sec. E.7.1. (Note that $\mathcal{W}_t$ below denotes a *window length*, and should not be confused with the total energy defined earlier in the setup.)

Lemma E.16 (A deterministic lower bound via a recent window). *Fix any $t \geq 3$ and define the window length*

$$\mathcal{W}_t := \left\lceil \frac{1}{1 - \lambda_{t-1}} \right\rceil. \quad (86)$$

Then for every $g \in [G]$ we have the deterministic bound

$$N_t(g) \geq \left(\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j \right) \cdot \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbf{1}\{g_s = g\}. \quad (87)$$

Proof. Start from the expansion (85):

$$N_t(g) = \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}.$$

All terms are nonnegative, so restricting the sum to the indices $s \in \{t - \mathcal{W}_t, \dots, t - 1\}$ gives

$$N_t(g) \geq \sum_{s=t-\mathcal{W}_t}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}.$$

For any $s \in \{t - \mathcal{W}_t, \dots, t - 1\}$, we have $s + 1 \geq t - \mathcal{W}_t + 1$, hence

$$\prod_{j=s+1}^{t-1} \lambda_j \geq \prod_{j=t-\mathcal{W}_t+1}^{t-1} \lambda_j = \frac{\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j}{\lambda_{t-\mathcal{W}_t}} \geq \prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j,$$

where the last inequality uses $\lambda_{t-\mathcal{W}_t} \leq 1$. Therefore, for each such s ,

$$\left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\} \geq \left(\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}.$$

Summing over $s = t - \mathcal{W}_t, \dots, t - 1$ yields (87), which concludes the proof. $\square$

We next show that, under our anytime schedule, the cumulative discount over the most recent $\mathcal{W}_t$ steps is bounded below by an absolute constant. This lower bound ensures that the window argument in Lemma E.16 does not vanish due to discounting, which is a key ingredient for obtaining uniform (in t) lower bounds on discounted counts and, consequently, for controlling $\underline{N}_T$ and the drift terms later.

Lemma E.17 (Lower bounding the window discount factor). *Assume $\kappa > 0$. Let (λ_t) be defined in (80), and let $\mathcal{W}_t$ be defined by (86). Define*

$$c_0 := \inf_{j \geq 1} \lambda_j, \quad C_\kappa := \sup_{t \geq 3} \frac{\mathcal{W}_t}{(\log(e + \ell_t))^{1+\kappa}}, \quad \ell_t := \max\{1, t - \mathcal{W}_t\},$$

and set $c_{\text{win}} := C_\kappa / c_0$. Then for all $t \geq 3$,

$$\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j \geq \exp(-c_{\text{win}}). \quad (88)$$

Proof. Proof sketch. We aim to lower bound the product $\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j$ by converting it into a sum via logarithms:

$$\log \left(\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j \right) = \sum_{j=t-\mathcal{W}_t}^{t-1} \log \lambda_j.$$

We first derive a uniform pointwise lower bound for each term, of the form $\log \lambda_j \geq -(1 - \lambda_j)/c_0$, where $c_0 = \inf_{j \geq 1} \lambda_j > 0$. Summing this bound over the window reduces the problem to controlling the deficit sum $\sum_{j=t-\mathcal{W}_t}^{t-1} (1 - \lambda_j)$. We then upper bound this deficit sum by (window length) $\times$ (largest deficit within the window), which introduces the ratio $\mathcal{W}_t / (\log(e + \ell_t))^{1+\kappa}$ (with $\ell_t = \max\{1, t - \mathcal{W}_t\}$ ensuring valid indices). Finally, we show this ratio is bounded uniformly over all t (large t by growth-rate comparison, and small t by finiteness), so the whole logarithmic sum is bounded below by a negative constant $-c_{\text{win}}$. Exponentiating completes the sketch and yields the product lower bound $\exp(-c_{\text{win}})$.

Step 0: Make the index range well-defined. By convention, define $\lambda_j := 1$ for all $j \leq 0$ (no discounting prior to $t = 1$). With $\ell_t := \max\{1, t - \mathcal{W}_t\}$, this implies

$$\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j = \prod_{j=\ell_t}^{t-1} \lambda_j,$$

since all factors with $j < 1$ equal 1.

Step 1: A pointwise lower bound on $\log \lambda_j$. For any $x \in (0, 1)$, define $f(x) := \log(1 - x) + \frac{x}{1-x}$. Then $f(0) = 0$ and

$$f'(x) = -\frac{1}{1-x} + \frac{1}{(1-x)^2} = \frac{x}{(1-x)^2} \geq 0,$$

so $f(x) \geq 0$ for all $x \in (0, 1)$, i.e.,

$$\log(1-x) \geq -\frac{x}{1-x}, \quad \forall x \in (0, 1). \quad (89)$$

For $j \geq 1$, we have $0 < \lambda_j < 1$ and can apply (89) with $x := 1 - \lambda_j$:

$$\log \lambda_j = \log(1 - (1 - \lambda_j)) \geq -\frac{1 - \lambda_j}{\lambda_j}.$$

Step 2: Use the global lower bound c_0 . Since $\kappa > 0$, the map $j \mapsto (\log(e + j))^{-(1+\kappa)}$ is strictly decreasing, hence $\lambda_j = 1 - (\log(e + j))^{-(1+\kappa)}$ is non-decreasing in j . Therefore,

$$c_0 = \inf_{j \geq 1} \lambda_j = \lambda_1 \in (0, 1),$$

and

$$\lambda_j \geq c_0, \quad \forall j \geq 1.$$

Consequently $\frac{1}{\lambda_j} \leq \frac{1}{c_0}$, and since $-(1 - \lambda_j) < 0$, multiplying by this negative quantity reverses the inequality:

$$-\frac{1 - \lambda_j}{\lambda_j} \geq -\frac{1 - \lambda_j}{c_0}.$$

Combining with Step 1 yields

$$\log \lambda_j \geq -\frac{1 - \lambda_j}{c_0}, \quad \forall j \geq 1. \quad (90)$$

Step 3: Sum over the window and bound the deficit sum. Summing (90) over $j = \ell_t, \dots, t - 1$ gives

$$\sum_{j=\ell_t}^{t-1} \log \lambda_j \geq -\frac{1}{c_0} \sum_{j=\ell_t}^{t-1} (1 - \lambda_j).$$

For $j \geq \ell_t$, since $\log(e + j) \geq \log(e + \ell_t)$ and $u \mapsto u^{-(1+\kappa)}$ is decreasing on $(0, \infty)$,

$$1 - \lambda_j = \frac{1}{(\log(e + j))^{1+\kappa}} \leq \frac{1}{(\log(e + \ell_t))^{1+\kappa}}.$$

Moreover, the number of terms in the sum satisfies $t - \ell_t \leq \mathcal{W}_t$ by the definition $\ell_t = \max\{1, t - \mathcal{W}_t\}$. Hence

$$\sum_{j=\ell_t}^{t-1} (1 - \lambda_j) \leq \frac{t - \ell_t}{(\log(e + \ell_t))^{1+\kappa}} \leq \frac{\mathcal{W}_t}{(\log(e + \ell_t))^{1+\kappa}} \leq C_\kappa,$$

where

$$C_\kappa := \sup_{t \geq 3} \frac{\mathcal{W}_t}{(\log(e + \ell_t))^{1+\kappa}}.$$

Thus, combining the above bounds yields

$$\sum_{j=\ell_t}^{t-1} \log \lambda_j \geq -\frac{1}{c_0} \sum_{j=\ell_t}^{t-1} (1 - \lambda_j) \geq -\frac{C_\kappa}{c_0}.$$

Step 4: Verify that $C_\kappa < \infty$. We show the supremum definition C_κ is finite. Since $\mathcal{W}_t = \lceil (\log(e + t - 1))^{1+\kappa} \rceil = O((\log t)^{1+\kappa})$ and $(\log t)^{1+\kappa} = o(t)$, there exists a threshold $t_0 = t_0(\kappa) \geq 3$ such that for all $t \geq t_0$,

$$\mathcal{W}_t \leq \frac{t}{2}. \quad (91)$$

For such t , we have $\ell_t = t - \mathcal{W}_t \geq t/2$, so

$$\frac{\mathcal{W}_t}{(\log(e + \ell_t))^{1+\kappa}} \leq \frac{(\log(e + t - 1))^{1+\kappa} + 1}{(\log(e + t/2))^{1+\kappa}} \leq \frac{2(\log(e + t - 1))^{1+\kappa}}{(\log(e + t/2))^{1+\kappa}}.$$

Also, for all $t \geq 3$,

$$\log(e + t - 1) \leq \log(2(e + t/2)) = \log 2 + \log(e + t/2) \leq 2 \log(e + t/2),$$

since $\log(e + t/2) \geq 1$. Therefore,

$$\frac{\mathcal{W}_t}{(\log(e + \ell_t))^{1+\kappa}} \leq 2 \cdot 2^{1+\kappa} = 2^{2+\kappa}, \quad \forall t \geq t_0.$$

For the finitely many $t \in \{3, 4, \dots, t_0 - 1\}$, the ratio $\frac{\mathcal{W}_t}{(\log(e + \ell_t))^{1+\kappa}}$ attains a finite maximum, denoted C_κ^{fin} . Thus $C_\kappa \leq \max\{2^{2+\kappa}, C_\kappa^{\text{fin}}\} < \infty$.

Step 5: Conclude the product bound. Putting Steps 0–3 together,

$$\sum_{j=t-\mathcal{W}_t}^{t-1} \log \lambda_j = \sum_{j=\ell_t}^{t-1} \log \lambda_j \geq -\frac{C_\kappa}{c_0} = -c_{\text{win}}.$$

Exponentiating both sides (and using monotonicity of $\exp$) yields

$$\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j \geq \exp(-c_{\text{win}}),$$

which is exactly (88). $\square$

Lemma E.18 (A high-probability lower bound on $N_t(g)$ via Freedman). *Fix any $\delta \in (0, 1)$ and let (λ_t, γ_t) be defined in (80)–(82). Assume (83). Let c_{win} be the (absolute-in- t) constant from Lemma E.17. Then there exists a numerical constant $c_0 > 0$ such that if $c \geq c_0$ in (82), then with probability at least $1 - \delta$, the following statements hold simultaneously.*

(i) **Pointwise lower bound (strict, abstract form).** For all $t \geq 3$ and all $g \in [G]$,

$$N_t(g) \geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \phi(t - \mathcal{W}_t) \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right), \quad (92)$$

where $\mathcal{W}_t := \left\lceil \frac{1}{1 - \lambda_{t-1}} \right\rceil$ is the deterministic window length used in Lemma E.16.

(i) **Concrete substitution (still strict).** Under the specific choice $\phi(t) = (\log(e + t))^{\kappa/2}$ adopted in (81), (92) specializes to

$$N_t(g) \geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} (\log(e + t - \mathcal{W}_t))^{\kappa/2} \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right).$$

(ii) **Minima over groups and over time (strict, abstract form).** Define the terminal minimum discounted count

$$N_T^{\min} := \min_{g \in [G]} N_T(g),$$

and define the tail-minimum discounted count over $[t_0(T), T]$ by

$$t_0(T) := \left\lceil \sqrt{T} \right\rceil, \quad \underline{N}_{t_0(T):T} := \min_{t_0(T) \leq t \leq T} \min_{g \in [G]} N_t(g). \quad (93)$$

Then

$$\begin{aligned} N_T^{\min} &\geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \phi(T - \mathcal{W}_T) \log\left(\frac{G\pi^2(T - \mathcal{W}_T)^2}{\delta}\right), \\ \underline{N}_{t_0(T):T} &\geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \phi(t_0(T) - \mathcal{W}_T) \log\left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta}\right). \end{aligned} \quad (94)$$

Explanation for the second line in (94). By the assumed monotonicity of $t \mapsto (1 - \lambda_t)$ under (80), we have $\mathcal{W}_t = \lceil 1/(1 - \lambda_{t-1}) \rceil \leq \lceil 1/(1 - \lambda_{T-1}) \rceil = \mathcal{W}_T$ for all $t \leq T$, hence for every $t \in [t_0(T), T]$, $t - \mathcal{W}_t \geq t - \mathcal{W}_T \geq t_0(T) - \mathcal{W}_T$. Since $\phi(\cdot)$ is nondecreasing and $u \mapsto \log(\frac{G\pi^2 u^2}{\delta})$ is nondecreasing for $u \geq 1$, the pointwise lower bound in part (i) implies the tail-minimum lower bound in the second line of (94).

(ii) **Concrete substitution and the desired $\log(\sqrt{T} - \text{polylog}(T))$ form (strict).** Under (80) with $1 - \lambda_t = (\log(e+t))^{-(1+\kappa)}$, we have

$$\mathcal{W}_T = \left\lceil \frac{1}{1 - \lambda_{T-1}} \right\rceil = \lceil (\log(e + T - 1))^{1+\kappa} \rceil.$$

Therefore, defining

$$\mathfrak{m}_T := t_0(T) - \mathcal{W}_T = \lceil \sqrt{T} \rceil - \lceil (\log(e + T - 1))^{1+\kappa} \rceil,$$

the second line of (94) can be written as the strict bound

$$\underline{N}_{t_0(T):T} \geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \phi(\mathfrak{m}_T) \log\left(\frac{G\pi^2 \mathfrak{m}_T^2}{\delta}\right),$$

which is exactly of the form $\log(\sqrt{T} - \text{polylog}(T))$ through $\mathfrak{m}_T$.

Heuristic scale (for intuition only). Since $\mathfrak{m}_T = \sqrt{T} - \text{polylog}(T)$, we have $\log(\mathfrak{m}_T^2) = \log T + O(1)$ in the usual asymptotic sense, so the logarithmic factor in the strict bound may be intuitively read as being of order $\log T$; this heuristic reading is only for interpretation, while subsequent derivations may keep the strict factor $\log(\frac{G\pi^2 \mathfrak{m}_T^2}{\delta})$ unchanged.

Proof. Proof sketch. We first restrict the discounted count $N_t(g)$ to its most recent $\mathcal{W}_t$ rounds, which yields a discount-product factor times a window count $M_{t,g}$. A deterministic window-product bound lower bounds the product factor by $\exp(-c_{\text{win}})$ uniformly in t . It therefore suffices to lower bound $M_{t,g}$. We write $M_{t,g}$ as its conditional mean $\mu_{t,g}$ plus a martingale fluctuation and control the left tail of the fluctuation via Freedman's inequality, choosing the deviation level $x = \mu_{t,g}/2$. A deterministic calculation lower bounds $\mu_{t,g}$ by a multiple of $c \log(G\pi^2 t^2/\delta)$ using the exploration lower bound $\mathbb{P}_s(g) \geq \gamma_s/G$ and the fact that the cap in γ_s is inactive once s is large enough. Finally, we allocate failure probability $\delta/(G\pi^2 t^2)$ to each pair (t, g) and take a union bound over all $t \geq 3$ and $g \in [G]$ to prove (i). Statement (ii) then follows immediately by taking minima.

Step 1 (reduce $N_t(g)$ to a window count). Fix any $t \geq 3$ and any $g \in [G]$. Define

$$M_{t,g} := \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbf{1}\{g_s = g\}.$$

By Lemma E.16,

$$N_t(g) \geq \left(\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j \right) M_{t,g}.$$

By Lemma E.17, $\prod_{j=t-\mathcal{W}_t}^{t-1} \lambda_j \geq \exp(-c_{\text{win}})$ for all $t \geq 3$. Therefore,

$$N_t(g) \geq \exp(-c_{\text{win}}) M_{t,g}. \quad (95)$$

Step 2 (martingale decomposition for $M_{t,g}$). Let $\{\mathcal{F}_s\}_{s \geq 0}$ be the natural filtration. Define martingale differences

$$Z_s := \mathbf{1}\{g_s = g\} - \mathbb{P}_s(g), \quad s \geq 1.$$

Then $\mathbb{E}[Z_s \mid \mathcal{F}_{s-1}] = 0$ and $|Z_s| \leq 1$ almost surely. Let

$$S_{t,g} := \sum_{s=t-\mathcal{W}_t}^{t-1} Z_s, \quad \mu_{t,g} := \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbb{P}_s(g).$$

By construction,

$$M_{t,g} = \mu_{t,g} + S_{t,g}. \quad (96)$$

Hence for any $x > 0$,

$$\Pr(M_{t,g} \leq \mu_{t,g} - x) = \Pr(S_{t,g} \leq -x). \quad (97)$$

Step 3 (Freedman inequality on $S_{t,g}$). We use the following standard form of Freedman's inequality.

Freedman (bounded-increment, one-sided form). Let $\{D_s\}_{s \geq 1}$ be a martingale difference sequence with respect to a filtration $\{\mathcal{F}_s\}_{s \geq 0}$, i.e., $\mathbb{E}[D_s \mid \mathcal{F}_{s-1}] = 0$ for all s , and assume $D_s \leq 1$ almost surely for all s . Define the predictable quadratic variation

$$V_n := \sum_{s=1}^n \mathbb{E}[D_s^2 \mid \mathcal{F}_{s-1}].$$

Then for any $x > 0$,

$$\Pr\left(\sum_{s=1}^n D_s \geq x\right) \leq \exp\left(-\frac{x^2}{2(V_n + x/3)}\right).$$

Applying this inequality to $-D_s$ yields the corresponding left-tail bound

$$\Pr\left(\sum_{s=1}^n D_s \leq -x\right) \leq \exp\left(-\frac{x^2}{2(V_n + x/3)}\right).$$

We now verify the conditions and specialize the inequality to our window sum. Recall $Z_s := \mathbf{1}\{g_s = g\} - \mathbb{P}_s(g)$. By construction, $\mathbb{E}[Z_s \mid \mathcal{F}_{s-1}] = 0$. Moreover, since $\mathbf{1}\{g_s = g\} \in \{0, 1\}$ and $\mathbb{P}_s(g) \in [0, 1]$, we have $Z_s \in [-1, 1]$ and hence $-Z_s \leq 1$ almost surely.

Define the window sum $S_{t,g} := \sum_{s=t-\mathcal{W}_t}^{t-1} Z_s$ and its predictable quadratic variation

$$V_{t,g} := \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbb{E}[Z_s^2 \mid \mathcal{F}_{s-1}].$$

To upper bound $V_{t,g}$, fix s and set $p := \mathbb{P}_s(g)$. Conditioned on $\mathcal{F}_{s-1}$, the variable $\mathbf{1}\{g_s = g\}$ is Bernoulli(p), so

$$\mathbb{E}[Z_s^2 \mid \mathcal{F}_{s-1}] = \mathbb{E}[(\mathbf{1}\{g_s = g\} - p)^2 \mid \mathcal{F}_{s-1}] = \text{Var}(\mathbf{1}\{g_s = g\} \mid \mathcal{F}_{s-1}) = p(1-p).$$

Since $p \in [0, 1]$, we have $1-p \leq 1$ and thus $p(1-p) \leq p$. Therefore

$$\mathbb{E}[Z_s^2 \mid \mathcal{F}_{s-1}] \leq p = \mathbb{P}_s(g).$$

Summing over $s = t - \mathcal{W}_t, \dots, t-1$ gives

$$V_{t,g} \leq \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbb{P}_s(g) = \mu_{t,g}. \quad (98)$$

Finally, apply Freedman's inequality to the martingale difference sequence $D_s := -Z_s$ over the window $s = t - \mathcal{W}_t, \dots, t-1$. Since $D_s \leq 1$ almost surely and $\sum D_s = -S_{t,g}$, for any $x > 0$,

$$\Pr(S_{t,g} \leq -x) = \Pr(-S_{t,g} \geq x) \leq \exp\left(-\frac{x^2}{2(V_{t,g} + x/3)}\right).$$

Combining this with (98) yields

$$\Pr(S_{t,g} \leq -x) \leq \exp\left(-\frac{x^2}{2(V_{t,g} + x/3)}\right) \leq \exp\left(-\frac{x^2}{2(\mu_{t,g} + x/3)}\right). \quad (99)$$

Step 4 (deterministic lower bound on $\mu_{t,g}$). Let

$$L_t := \log\left(\frac{G\pi^2 t^2}{\delta}\right), \quad A := \frac{G\pi^2}{\delta}.$$

Since $G \geq 1$ and $\delta \in (0, 1)$, we have

$$A = \frac{G\pi^2}{\delta} \geq \frac{\pi^2}{\delta} > \pi^2 > 1,$$

and therefore $\log A \geq 0$. In particular, $L_t = \log A + 2 \log t > 0$ for all $t \geq 1$. By (83), $\mathbb{P}_s(g) \geq \gamma_s/G$ for all s and g , hence

$$\mu_{t,g} = \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbb{P}_s(g) \geq \frac{1}{G} \sum_{s=t-\mathcal{W}_t}^{t-1} \gamma_s. \quad (100)$$

Claim 1 (the cap is inactive on a tail). Recall $A := G\pi^2/\delta$. Under the specific choice $\phi(t) = (\log(e+t))^{\kappa/2}$ in (81), define

$$u_{\text{cap}} := \max\left\{(8c)^{2/\kappa}, (4c \log A)^{1/(1+\kappa/2)}\right\}, \quad t_{\text{cap}} := \max\{3, \lceil \exp(u_{\text{cap}}) \rceil\}.$$

Then for all $s \geq t_{\text{cap}}$,

$$\gamma_s = c(1 - \lambda_s) \phi(s) \log\left(\frac{G\pi^2 s^2}{\delta}\right). \quad (101)$$

Proof. Fix any $s \geq t_{\text{cap}}$ and set $u := \log(e+s)$. Since $s \geq \lceil \exp(u_{\text{cap}}) \rceil$, we have $s \geq \exp(u_{\text{cap}})$ and hence $u = \log(e+s) \geq \log s \geq u_{\text{cap}}$. By (80) and (81), we have

$$1 - \lambda_s = \frac{1}{u^{1+\kappa}}, \quad \phi(s) = u^{\kappa/2}.$$

Moreover, since $\log(As^2) = \log A + 2 \log s$ and $\log s \leq \log(e+s) = u$, we obtain

$$\log(As^2) \leq \log A + 2u.$$

Therefore,

$$\begin{aligned} c(1 - \lambda_s) \phi(s) \log(As^2) &= c \cdot \frac{1}{u^{1+\kappa}} \cdot u^{\kappa/2} \cdot \log(As^2) = c \cdot \frac{\log(As^2)}{u^{1+\kappa/2}} \\ &\leq c \cdot \frac{\log A + 2u}{u^{1+\kappa/2}} = c \left(\frac{\log A}{u^{1+\kappa/2}} + \frac{2}{u^{\kappa/2}} \right). \end{aligned}$$

We bound the two terms separately. Since $u \geq u_{\text{cap}} \geq (4c \log A)^{1/(1+\kappa/2)}$, we have

$$c \cdot \frac{\log A}{u^{1+\kappa/2}} \leq c \cdot \frac{\log A}{4c \log A} = \frac{1}{4}.$$

Since $u \geq u_{\text{cap}} \geq (8c)^{2/\kappa}$, we have $u^{\kappa/2} \geq 8c$ and hence

$$c \cdot \frac{2}{u^{\kappa/2}} \leq c \cdot \frac{2}{8c} = \frac{1}{4}.$$

Combining these bounds gives

$$c(1 - \lambda_s) \phi(s) \log(As^2) \leq \frac{1}{2},$$

By the definition of γ_s in (82), this inequality implies that the minimum $\min\{\frac{1}{2}, c(1 - \lambda_s)\phi(s) \log(\frac{G\pi^2 s^2}{\delta})\}$ attains its second argument, i.e.,

$$\gamma_s = c(1 - \lambda_s)\phi(s) \log\left(\frac{G\pi^2 s^2}{\delta}\right),$$

which is exactly (101). $\square$

Claim 2 (a window-mean lower bound). Fix any $t \geq t_{\text{cap}} + \mathcal{W}_t$ and any $g \in [G]$. Then the window mean satisfies

$$\mu_{t,g} \geq \frac{c}{G} \phi(t - \mathcal{W}_t) \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right). \quad (102)$$

In particular, since $\phi(\cdot) \geq 1$, we also have the weaker consequence

$$\mu_{t,g} \geq \frac{c}{G} \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right).$$

Proof. Recall $\mu_{t,g} = \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbb{P}_s(g)$ and the exploration lower bound $\mathbb{P}_s(g) \geq \gamma_s/G$ from (83), hence

$$\mu_{t,g} = \sum_{s=t-\mathcal{W}_t}^{t-1} \mathbb{P}_s(g) \geq \frac{1}{G} \sum_{s=t-\mathcal{W}_t}^{t-1} \gamma_s. \quad (103)$$

By the assumption $t - \mathcal{W}_t \geq t_{\text{cap}}$ and Claim 1, for every $s \in \{t - \mathcal{W}_t, \dots, t - 1\}$ the cap is inactive, so

$$\gamma_s = c(1 - \lambda_s)\phi(s) \log\left(\frac{G\pi^2 s^2}{\delta}\right).$$

Moreover, since $s \mapsto \phi(s)$ and $s \mapsto \log\left(\frac{G\pi^2 s^2}{\delta}\right)$ are nondecreasing, for all such s we have

$$\phi(s) \geq \phi(t - \mathcal{W}_t), \quad \log\left(\frac{G\pi^2 s^2}{\delta}\right) \geq \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right).$$

Therefore,

$$\sum_{s=t-\mathcal{W}_t}^{t-1} \gamma_s = c \sum_{s=t-\mathcal{W}_t}^{t-1} (1 - \lambda_s)\phi(s) \log\left(\frac{G\pi^2 s^2}{\delta}\right) \geq c\phi(t - \mathcal{W}_t) \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right) \sum_{s=t-\mathcal{W}_t}^{t-1} (1 - \lambda_s).$$

Finally, by the definition $\mathcal{W}_t = \lceil 1/(1 - \lambda_{t-1}) \rceil$ and the monotonicity of $u \mapsto 1 - \lambda_u$ under (80), we have for every $s \leq t - 1$ that $(1 - \lambda_s) \geq (1 - \lambda_{t-1})$, hence

$$\sum_{s=t-\mathcal{W}_t}^{t-1} (1 - \lambda_s) \geq \mathcal{W}_t(1 - \lambda_{t-1}) \geq 1.$$

Combining the last three displays with (103) yields (102), which concludes the proof. $\square$

Step 5 (choose x , apply Freedman, and union bound). We now determine the failure probability budget. Let $\mathcal{T} := \{t \in \mathbb{Z}_{\geq 3} : t > \mathcal{W}_t\}$. Note that for any $t \notin \mathcal{T}$ (i.e., $t \leq \mathcal{W}_t$), the claim in (92) is vacuous (or the log term is undefined), so we restrict our attention to $t \in \mathcal{T}$. Since $\mathcal{W}_t = o(t)$ (specifically $\mathcal{W}_t \approx \text{polylog}(t)$), we have $\lim_{t \rightarrow \infty} \frac{t^2}{(t - \mathcal{W}_t)^2} = 1$. By the limit comparison test with the convergent series $\sum t^{-2}$, the series

$$S := \sum_{t \in \mathcal{T}} \frac{1}{(t - \mathcal{W}_t)^2}$$

converges to a finite constant. We assign a failure probability budget $\delta_{t,g}$ to each pair (t, g) with $t \in \mathcal{T}$ and $g \in [G]$:

$$\delta_{t,g} := \frac{\delta}{GS(t - \mathcal{W}_t)^2}.$$

Summing over all $g \in [G]$ and $t \in \mathcal{T}$, we have $\sum_{t,g} \delta_{t,g} = \sum_{t \in \mathcal{T}} G \cdot \frac{\delta}{GS(t - \mathcal{W}_t)^2} = \delta$.

Now, fix any $t \in \mathcal{T}$ and $g \in [G]$. Set $x := \frac{1}{2}\mu_{t,g}$. Substituting this into (99) yields

$$\Pr(S_{t,g} \leq -x) \leq \exp\left(-\frac{x^2}{2(\mu_{t,g} + x/3)}\right) = \exp\left(-\frac{(\mu_{t,g}/2)^2}{(7/3)\mu_{t,g}}\right) = \exp\left(-\frac{3}{28}\mu_{t,g}\right).$$

By (97), $\Pr(M_{t,g} \leq \frac{1}{2}\mu_{t,g}) = \Pr(S_{t,g} \leq -x)$. To ensure $\Pr(M_{t,g} \leq \frac{1}{2}\mu_{t,g}) \leq \delta_{t,g}$, it suffices to show

$$\frac{3}{28}\mu_{t,g} \geq \log\left(\frac{1}{\delta_{t,g}}\right) = \log\left(\frac{GS(t - \mathcal{W}_t)^2}{\delta}\right).$$

Using the lower bound for $\mu_{t,g}$ from Step 4 (Claim 2), and noting $\phi(\cdot) \geq 1$, we have

$$\mu_{t,g} \geq \frac{c}{G} \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right).$$

Let $u := (t - \mathcal{W}_t)^2$. The condition becomes

$$\frac{3c}{28G} \log\left(\frac{G\pi^2 u}{\delta}\right) \geq \log\left(\frac{GSu}{\delta}\right).$$

Since $\log(\frac{GSu}{\delta}) = \log(\frac{GS}{\delta}) + \log u$ and similarly for the LHS, by choosing the constant c sufficiently large (specifically, large enough to dominate the ratio of the coefficients and the constant offsets $\log S$ vs $\log \pi^2$), this inequality holds for all $u \geq 1$. Consequently, with probability at least $1 - \delta$, we have $M_{t,g} \geq \frac{1}{2}\mu_{t,g}$ for all $t \in \mathcal{T}, g \in [G]$.

Finally, combining this with (95) and Claim 2, we obtain

$$N_t(g) \geq e^{-c_{\text{win}}} M_{t,g} \geq e^{-c_{\text{win}}} \frac{1}{2}\mu_{t,g} \geq e^{-c_{\text{win}}} \frac{c}{2G} \phi(t - \mathcal{W}_t) \log\left(\frac{G\pi^2(t - \mathcal{W}_t)^2}{\delta}\right).$$

Adjusting the constant $c/2G$ to $c/4G$ (relaxing the bound to match the statement) proves part (i).

For part (ii), applying part (i) at $t = T$ and taking the minimum over $g \in [G]$ gives

$$N_T^{\min} = \min_{g \in [G]} N_T(g) \geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \phi(T - \mathcal{W}_T) \log\left(\frac{G\pi^2(T - \mathcal{W}_T)^2}{\delta}\right).$$

For the second statement, since part (i) holds for every $t \in [t_0(T), T]$, we take the minimum over this interval. By the monotonicity explained in the lemma statement (i.e., $t - \mathcal{W}_t \geq t_0(T) - \mathcal{W}_T$ and ϕ is nondecreasing), we have

$$\underline{N}_{t_0(T):T} = \min_{t_0(T) \leq t \leq T} \min_{g \in [G]} N_t(g) \geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \phi(t_0(T) - \mathcal{W}_T) \log\left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta}\right).$$

With $t_0(T) = \lceil \sqrt{T} \rceil$ and $\mathcal{W}_T = \text{polylog}(T)$, the argument $t_0(T) - \mathcal{W}_T \approx \sqrt{T}$. Thus, the log factor scales as $\log(T)$, and $\phi(\cdot)$ contributes the polylog factor, confirming the desired growth rate. This concludes the proof. $\square$

Corollary E.19 (Cumulative drift under the anytime schedules). *Fix any $\delta \in (0, 1)$ and adopt the anytime schedules (80)–(82). Let $t_0(T)$ be any integer-valued function satisfying $3 \leq t_0(T) \leq T$, and let*

$$\underline{N}_{t_0(T):T} := \min_{t_0(T) \leq t \leq T} \min_{g \in [G]} N_t(g)$$

be the corresponding tail-minimum discounted count. Define the constant

$$C'_{\text{drift}} := \frac{4G \exp(c_{\text{win}})}{c} C_{\text{drift}}.$$

On the event of Lemma E.18, for every integer $T \geq 3$ such that $t_0(T) > \mathcal{W}_T$,

$$\sum_{t=1}^T \text{Drift}_t \leq t_0(T) + \frac{C'_{\text{drift}}}{(1 - \lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta}\right)}. \quad (104)$$

Concrete instantiation. Under (80), we have $(1 - \lambda_T)^{-2} = (\log(e + T))^{2+2\kappa}$. Taking $t_0(T) = \lceil \sqrt{T} \rceil$ and defining

$$\mathfrak{m}_T := t_0(T) - \mathcal{W}_T = \lceil \sqrt{T} \rceil - \left\lceil \frac{1}{1 - \lambda_{T-1}} \right\rceil,$$

Eq. (104) specializes (for every $T \geq 3$ such that $\mathfrak{m}_T \geq 1$) to

$$\sum_{t=1}^T \text{Drift}_t \leq \lceil \sqrt{T} \rceil + C'_{\text{drift}} (\log(e + T))^{2+2\kappa} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2 \mathfrak{m}_T^2}{\delta}\right)}. \quad (105)$$

Moreover, for the anytime choice (80) we have

$$\mathcal{W}_T = \left\lceil \frac{1}{1 - \lambda_{T-1}} \right\rceil = \lceil (\log(e + T - 1))^{1+\kappa} \rceil,$$

so $\mathfrak{m}_T = \lceil \sqrt{T} \rceil - \lceil (\log(e + T - 1))^{1+\kappa} \rceil = \sqrt{T} - \text{polylog}(T)$. In particular, for all sufficiently large T we have $\mathfrak{m}_T \geq \frac{1}{2}\sqrt{T}$, and hence

$$\log\left(\frac{G\pi^2 \mathfrak{m}_T^2}{\delta}\right) = \log\left(\frac{G\pi^2}{\delta}\right) + \log(\mathfrak{m}_T^2) = \log\left(\frac{G\pi^2}{\delta}\right) + \log T + O(1) = \Theta(\log T).$$

Hence a sufficient condition for $\sum_{t=1}^T \text{Drift}_t = o(T)$ is

$$V_T^{(\varepsilon)} = o\left(\frac{T \log T}{(\log(e + T))^{2+2\kappa}}\right),$$

which is an $o(T/\text{polylog}(T))$ -type requirement; see Remark G.9 for further discussion on the growth regimes of $V_T^{(\varepsilon)}$.

Proof. Proof sketch. We split $\sum_{t=1}^T \text{Drift}_t$ into an early part $t < t_0(T)$ and a tail part $t \geq t_0(T)$. The early part is bounded by $t_0(T)$ using $\text{Drift}_t \leq 1$. For the tail part, we start from the refined drift sum inequality (71), pull out the tail-minimum discounted count $\underline{N}_{t_0(T):T}$, and then swap summations to evaluate the resulting geometric series, which contributes an extra factor $(1 - \lambda_T)^{-1}$. Finally, Lemma E.18 provides a lower bound on $\underline{N}_{t_0(T):T}$ in terms of $\phi(t_0(T) - \mathcal{W}_T) \log(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta})$; since $\phi(\cdot) \geq 1$, we may drop it to obtain the simplified bound used in (104). We then verify that the stated sufficient condition indeed implies $\sum_{t=1}^T \text{Drift}_t = o(T)$.

Step 1: Introduce the splitting time and decompose the time horizon. Fix an integer-valued splitting function $t_0(T)$ with $3 \leq t_0(T) \leq T$. We decompose the cumulative drift into an *early* part and a *tail* part: the early part $t \leq t_0(T) - 1$ will be controlled by a crude worst-case bound on each Drift_t , while the tail part $t \geq t_0(T)$ will be controlled using the tail-minimum discounted count $\underline{N}_{t_0(T):T}$ and a geometric-series summation. Concretely,

$$\sum_{t=1}^T \text{Drift}_t = \sum_{t=1}^{t_0(T)-1} \text{Drift}_t + \sum_{t=t_0(T)}^T \text{Drift}_t.$$

Step 2: Bound the early part by $t_0(T)$. For each t , both vectors $r_t^{(\varepsilon)}$ and $\bar{r}_t^{(\varepsilon)}$ are group-indexed quantities with entries in $[0, 1]$: for every $g \in [G]$, we have $r_t^{(\varepsilon)}(g) \in [0, 1]$ and $\bar{r}_t^{(\varepsilon)}(g) \in [0, 1]$. Therefore, for every $g \in [G]$,

$$|\bar{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq 1,$$

and taking the maximum over g yields

$$\text{Drift}_t = \|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_\infty = \max_{g \in [G]} |\bar{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq 1.$$

Summing this crude bound over the early range $t = 1, \dots, t_0(T) - 1$ gives

$$\sum_{t=1}^{t_0(T)-1} \text{Drift}_t \leq \sum_{t=1}^{t_0(T)-1} 1 = t_0(T) - 1 \leq t_0(T).$$

Step 3: Tail bound via $\underline{N}_{t_0(T):T}$. Let $N_t^{\min} := \min_{g \in [G]} N_t(g)$. Using the refined drift sum inequality (71) and the monotonicity of λ_t under (80) (so that $\bar{\lambda}_T = \lambda_T$), we have

$$\sum_{t=t_0(T)}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{1 - \lambda_T} \sum_{t=t_0(T)}^T \frac{1}{N_t^{\min}} \sum_{u=2}^t (\lambda_T)^{t-u} \Gamma_u.$$

On the tail $t \in [t_0(T), T]$, we have $N_t^{\min} \geq \underline{N}_{t_0(T):T}$ by definition, hence

$$\sum_{t=t_0(T)}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1 - \lambda_T) \underline{N}_{t_0(T):T}} \sum_{t=t_0(T)}^T \sum_{u=2}^t (\lambda_T)^{t-u} \Gamma_u.$$

Swap the order of summation: for each fixed u , the index t ranges over $t = \max\{t_0(T), u\}, \dots, T$. Thus

$$\sum_{t=t_0(T)}^T \sum_{u=2}^t (\lambda_T)^{t-u} \Gamma_u = \sum_{u=2}^T \Gamma_u \sum_{t=\max\{t_0(T), u\}}^T (\lambda_T)^{t-u} \leq \sum_{u=2}^T \Gamma_u \sum_{k=0}^{\infty} (\lambda_T)^k = \frac{1}{1 - \lambda_T} \sum_{u=2}^T \Gamma_u = \frac{V_T^{(\varepsilon)}}{1 - \lambda_T}.$$

Therefore,

$$\sum_{t=t_0(T)}^T \text{Drift}_t \leq \frac{C_{\text{drift}}}{(1 - \lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\underline{N}_{t_0(T):T}}. \quad (106)$$

Step 4: Plug the lower bound on $\underline{N}_{t_0(T):T}$. On the event of Lemma E.18, the second line of (94) gives

$$\underline{N}_{t_0(T):T} \geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \phi(t_0(T) - \mathcal{W}_T) \log \left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta} \right).$$

Since $\phi(\cdot) \geq 1$, we can relax the bound by removing the ϕ term:

$$\underline{N}_{t_0(T):T} \geq \exp(-c_{\text{win}}) \cdot \frac{c}{4G} \log \left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta} \right).$$

Substituting this lower bound into (106) (note that $\underline{N}_{t_0(T):T}$ appears in the denominator, so using a valid lower bound preserves the $\leq$ direction) yields

$$\sum_{t=t_0(T)}^T \text{Drift}_t \leq \frac{4G \exp(c_{\text{win}}) C_{\text{drift}}}{c(1 - \lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log \left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta} \right)}.$$

Combining with Step 2 gives (104).

Step 5: Verify sublinearity for $t_0(T) = \lceil \sqrt{T} \rceil$. With $t_0(T) = \lceil \sqrt{T} \rceil$, the first term in (105) is $\lceil \sqrt{T} \rceil = O(\sqrt{T}) = o(T)$. For the second term, recall that under (80) we have $(1 - \lambda_T)^{-2} = (\log(e + T))^{2+2\kappa}$. Moreover, for this choice of $t_0(T)$ we have $m_T = t_0(T) - \mathcal{W}_T = \sqrt{T} - \text{polylog}(T)$, hence for all sufficiently large T , $m_T \geq \frac{1}{2}\sqrt{T}$ and therefore

$$\log \left(\frac{G\pi^2 m_T^2}{\delta} \right) = \log \left(\frac{G\pi^2}{\delta} \right) + \log(m_T^2) \geq \log \left(\frac{G\pi^2}{\delta} \right) + \log \left(\frac{T}{4} \right) = \log T + O(1).$$

Under the stated condition $V_T^{(\varepsilon)} = o \left(\frac{T \log T}{(\log(e + T))^{2+2\kappa}} \right)$, it follows that

$$C'_{\text{drift}} (\log(e + T))^{2+2\kappa} \cdot \frac{V_T^{(\varepsilon)}}{\log \left(\frac{G\pi^2 m_T^2}{\delta} \right)} = o(T).$$

Therefore $\sum_{t=1}^T \text{Drift}_t = o(T)$, which concludes the proof of Corollary E.19. $\square$

Remark E.20 (Choosing the splitting time $t_0(T)$). The choice $t_0(T) = \lceil \sqrt{T} \rceil$ is mainly for convenience: it yields an early-time contribution at most $O(\sqrt{T})$ and ensures

$$\log\left(\frac{G\pi^2 m_T^2}{\delta}\right) = \Theta(\log T), \quad \text{with } m_T := t_0(T) - \mathcal{W}_T, \quad (107)$$

whenever $m_T \geq 1$ (as required by (104)). More generally, (104) highlights a trade-off between the early-time term $t_0(T)$ and the tail term that scales like

$$\frac{V_T^{(\varepsilon)}}{(1 - \lambda_T)^2 \log\left(\frac{G\pi^2 m_T^2}{\delta}\right)}.$$

In principle, one can choose $t_0(T)$ to balance these contributions, which may tighten constants or slightly relax the required growth of $V_T^{(\varepsilon)}$. Any choice from a constant function up to $t_0(T) = o(T)$ is admissible, but it leads to different sufficient conditions on the environment; moreover, the feasibility condition $m_T = t_0(T) - \mathcal{W}_T \geq 1$ must be satisfied.

To minimize the total drift, one may balance the early-time and tail terms by solving the implicit relation

$$t_0(T) \asymp \frac{V_T^{(\varepsilon)}}{(1 - \lambda_T)^2 \log\left(\frac{G\pi^2 m_T^2}{\delta}\right)}, \quad m_T = t_0(T) - \mathcal{W}_T \geq 1.$$

Since $t_0(T)$ affects the tail contribution only through a logarithm, the square-root choice is typically near-optimal up to polylogarithmic factors. Moreover, for specific growth models of $V_T^{(\varepsilon)}$, the solution to the balancing equation can be written in closed form using the Lambert W function, but the resulting expression is cumbersome and thus less transparent for presentation.

Choosing a faster-growing split such as $t_0(T) = \lceil T^{0.99} \rceil$ increases the early-time term substantially, while the logarithmic factor $\log(\frac{G\pi^2 m_T^2}{\delta})$ changes only by a constant factor (once m_T is polynomial in T); therefore, the overall impact on the sufficient condition for $V_T^{(\varepsilon)}$ is typically marginal. For this reason, we keep the simple square-root choice in the main statement, despite the fact that more precise choices of $t_0(T)$ could further optimize the drift.

Discussion: An integrated view via the effective window length $\psi(t)$. Define the *effective window length*

$$\psi(t) := \frac{1}{1 - \lambda_t}.$$

Recall that $N_t(g)$ is a discounted count of how frequently group g has been selected in the recent past, with an effective horizon of order $\psi(t)$. Under time-varying uniform exploration (i.e., $\mathbb{P}_s(g) \geq \gamma_s/G$ for all s and g), the *typical* discounted exploration mass accumulated by group g over this effective window is heuristically of order

$$\mathbb{E}[N_t(g)] \approx \sum_{k \geq 0} \lambda_t^k \cdot \frac{\gamma_{t-k}}{G} \approx \frac{1}{G} \cdot \frac{\gamma_t}{1 - \lambda_t} = \frac{\gamma_t}{G} \psi(t),$$

where the last approximation uses that γ_{t-k} varies slowly over the window scale $\psi(t)$. This suggests that a sufficient way to enforce a high-probability lower bound $N_t(g) \gtrsim \log t$ (uniformly over g) is to choose γ_t of order

$$\gamma_t \asymp \frac{G \log t}{\psi(t)} \phi(t),$$

which makes the heuristic mean exploration mass scale as $\mathbb{E}[N_t(g)] \approx \log t \cdot \phi(t)$. With such a choice, the *exploration cost term* scales like

$$\sum_{t \leq T} \gamma_t \asymp \sum_{t \leq T} \frac{\log t}{\psi(t)} \phi(t).$$

For the drift contribution, the quantity that enters (104) is the tail-minimum discounted count

$$\underline{N}_{t_0(T):T} := \min_{t_0(T) \leq t \leq T} \min_{g \in [G]} N_t(g).$$

On the high-probability event of Lemma E.18, we have the logarithmic growth

$$\underline{N}_{t_0(T):T} = \Omega \left(\log \left(\frac{G\pi^2 t_0(T)^2}{\delta} \right) \right) = \Omega(\log t_0(T)),$$

and in particular, taking $t_0(T) = \lceil \sqrt{T} \rceil$ yields $\underline{N}_{t_0(T):T} = \Omega(\log T)$. Plugging this tail-minimum lower bound into (104) gives the heuristic scaling

$$\sum_{t \leq T} \text{Drift}_t \lesssim \frac{V_T^{(\varepsilon)}}{\underline{N}_{t_0(T):T}} \cdot \psi(T)^2 \asymp \frac{V_T^{(\varepsilon)} \psi(T)^2}{\log t_0(T)},$$

and for $t_0(T) = \lceil \sqrt{T} \rceil$ this becomes $\asymp V_T^{(\varepsilon)} \psi(T)^2 / \log T$. Therefore, to ensure $R^{(\varepsilon)}(T, \delta) = o(T)$ in Definition E.25, it suffices to enforce simultaneously

$$\sum_{t \leq T} \frac{\log t}{\psi(t)} \phi(t) = o(T), \quad V_T^{(\varepsilon)} \psi(T)^2 = o(T \log t_0(T)),$$

together with the (independent) condition needed by the bias term (III) in (112).

Remark E.21 (Why we consider the effective window length). The above effective-window reasoning is a standard design principle in non-stationary bandits. Discounted and sliding-window methods and their analyses routinely balance (i) exploration accumulated over an effective window and (ii) the amount of non-stationarity or variation that can occur within that window; see, e.g., discounted and sliding-window UCB style analyses (Garivier & Moulines, 2008; Besbes et al., 2015; Garivier & Moulines, 2011; Besbes et al., 2014; Bubeck & Cesa-Bianchi, 2012a). Our schedules (80)–(82) implement exactly this balance in an anytime manner (without requiring knowledge of the horizon T), and the high-probability lower bound $\underline{N}_{t_0(T):T} = \Omega(\log t_0(T))$ follows from a martingale concentration argument for discounted sums, instantiated in Lemma E.18. In short, λ_t sets the window length $\psi(t)$, γ_t ensures each group receives $\Omega(\log t)$ exploration mass inside that window, and the resulting drift bound becomes explicitly sublinear under a mild condition on $V_T^{(\varepsilon)}$.

E.7.1. MORE DISCUSSIONS ON THE CONSTRUCTION OF λ_t

This section records simple, verifiable conditions on the discount schedule $\{\lambda_t\}_{t \geq 1}$ that guarantee a uniform lower bound on window products. Such bounds are used to keep confidence-accumulation terms controlled by $(T, V_T^{(\varepsilon)})$ and polylogarithmic factors only.

Why the window-product condition (108) is not an extra assumption. Proposition E.28 involves

$$\Lambda_{T,H} := \min_{2 \leq u \leq T} \prod_{j=(u-H) \vee 1}^{u-1} \lambda_j,$$

and requires a uniform lower bound on $\Lambda_{T,H}$ to keep the confidence-accumulation term dependent only on $(T, V_T^{(\varepsilon)})$ and polylog factors. Concretely, we will use the constant bound

$$\Lambda_{T,H} \geq e^{-2}, \tag{108}$$

so that $\Lambda_{T,H}^{-1/2}$ is an absolute constant. The next lemma shows that (108) follows from a mild “admissibility” condition on the discount schedule. In words, (109) prevents the discounting from forgetting too much over the last H rounds, so the length- H window product cannot be too small.

Lemma E.22 (A simple sufficient condition for (108)). *Fix a window length $H \geq 1$. Suppose that for all $t \geq 2$,*

$$0 < 1 - \lambda_t \leq \frac{1}{2} \quad \text{and} \quad \sum_{j=(t-H) \vee 1}^{t-1} (1 - \lambda_j) \leq 1. \tag{109}$$

Then

$$\prod_{j=(t-H) \vee 1}^{t-1} \lambda_j \geq e^{-2} \quad \text{for all } t \geq 2, \tag{110}$$

and hence $\Lambda_{T,H} \geq e^{-2}$, which is exactly (108).

Proof. Fix $t \geq 2$ and write $x_j := 1 - \lambda_j \in [0, 1/2]$ for $j \in \{(t-H) \vee 1, \dots, t-1\}$. All sums below are taken over $j = (t-H) \vee 1, \dots, t-1$. Then

$$\log \left(\prod_{j=(t-H) \vee 1}^{t-1} \lambda_j \right) = \sum_j \log(1 - x_j).$$

On $[0, 1/2]$ we have the elementary bound $\log(1 - x) \geq -x - x^2$.³ Therefore,

$$\sum_j \log(1 - x_j) \geq - \sum_j x_j - \sum_j x_j^2 \geq -2 \sum_j x_j \geq -2,$$

where we used $x_j^2 \leq x_j$ (since $x_j \in [0, 1/2]$) and (109). Exponentiating yields (110), completing the proof. $\square$

Remark E.23 (Interpretation of (108)). Condition (108) is a compact way of requiring that, over any window of length H , the *total amount of forgetting* is $O(1)$. Indeed, with $x_j := 1 - \lambda_j$, (109) enforces $\sum_{j=(t-H) \vee 1}^{t-1} x_j \leq 1$, which guarantees that the discounted estimator still places a non-negligible aggregate weight on the most recent H samples. This is precisely what is needed in Proposition E.28 when comparing discounted and sliding-window type quantities.

Relation to Lemma E.17. Lemma E.17 lower bounds a *different* window product, namely the product over the time-varying window length $\mathcal{W}_t$, by $\exp(-c_{\text{win}})$ in (88). In contrast, (108) concerns the product over a fixed length H used in Proposition E.28, and (109) provides a convenient sufficient condition ensuring the constant bound e^{-2} .

Examples. A few standard choices satisfy (109) (hence (108)): (i) constant discount $\lambda_t \equiv \lambda$ with $1 - \lambda \leq 1/H$; (ii) piecewise-constant schedules (e.g., doubling epochs) with $H(1 - \lambda) \leq 1$ within each epoch; (iii) predictable anytime schedules with $\lambda_t = 1 - 1/\psi(t)$ and increasing $\psi(\cdot)$, whenever $\sum_{j=t-H}^{t-1} 1/\psi(j) \leq 1$ for the relevant t .

E.8. Part 4: High-probability dynamic regret bound (Discounted-UCB)

We now prove a high-probability dynamic regret bound for Algorithm 2 with respect to the *instantaneous regularized normalized rewards* $r_t^{(\varepsilon)}(\cdot)$ in Theorem E.26. The proof proceeds by first introducing a convenient upper bound functional $R^{(\varepsilon)}(T, \delta)$ that matches the term-by-term decomposition used in the analysis (martingale fluctuation, explicit exploration cost, confidence radii, and drift), and then showing that $\widehat{\text{Reg}}_T^{(\varepsilon)} \leq R^{(\varepsilon)}(T, \delta)$ holds with probability at least $1 - \delta$. Finally, we upper bound the exploration cost term $\sum_{t \leq T} \frac{\gamma_t}{1 - \gamma_t}$ under the anytime schedule (Corollary E.27) and control the confidence-accumulation term $\sum_{t \leq T} \eta_t(\hat{g}_t)$ via a discounted-UCB summation bound (Proposition E.28). In the last subsection, we have already handled the remaining task, *i.e.*, the drift term.

Definition E.24 (Dynamic regret w.r.t. $r_t^{(\varepsilon)}$). Define the (pathwise) dynamic regret up to time T as

$$\widehat{\text{Reg}}_T^{(\varepsilon)} := \sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t) \right), \quad r_{t,\star}^{(\varepsilon)} := \max_{g \in [G]} r_t^{(\varepsilon)}(g). \quad (111)$$

Definition E.25 (Bandit regret upper bound function). For any horizon $T \geq 1$ and confidence level $\delta \in (0, 1)$, define

$$R^{(\varepsilon)}(T, \delta) := \frac{1}{1 - \gamma_{\max}} \sqrt{2T \log \frac{2}{\delta}} + \sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t} + 2 \sum_{t=1}^T \eta_t(\hat{g}_t) + 2 \sum_{t=1}^T \text{Drift}_t, \quad (112)$$

$$\hat{g}_t \in \arg\max_{g \in [G]} \text{UCB}_t(g).$$

Here, $\gamma_{\max} := \max_{1 \leq t \leq T} \gamma_t$, and $\eta_t(\cdot)$ and Drift_t are as in Lemma E.11. The definition of $\text{UCB}_t(g)$ can be found in Algorithm 2 and in the proof of Theorem E.26.

Theorem E.26 (High-probability dynamic regret bound). Fix $\delta \in (0, 1)$ and run Algorithm 2 with the time-varying mixture policy (32), where $\gamma_t \in (0, 1/2]$ for all t . Assume the concentration conditions required to construct $\eta_t(g)$ hold (e.g., Assumptions E.1–E.2), and define Drift_t and $\eta_t(g)$ as in Lemma E.11 and (30). Then, with probability at least $1 - \delta$,

$$\widehat{\text{Reg}}_T^{(\varepsilon)} := \sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t) \right) \leq R^{(\varepsilon)}(T, \delta), \quad (113)$$

³Define $f(x) := \log(1 - x) + x + x^2$ on $[0, 1/2]$, note $f(0) = 0$, and compute $f'(x) = -\frac{1}{1-x} + 1 + 2x = \frac{x(1-2x)}{1-x} \geq 0$ on $[0, 1/2]$.

where $R^{(\varepsilon)}(T, \delta)$ is defined in Definition E.25.

Proof. Proof sketch. We work on a uniform concentration event under which the UCB estimates are accurate up to $\eta_t(\cdot)$ plus a drift term Drift_t . On this event, we first relate the instantaneous optimum $r_{t,\star}^{(\varepsilon)}$ to the UCB-greedy group $\hat{g}_t$ via an optimistic comparison. We then use the time-varying mixture policy to compare $r_t^{(\varepsilon)}(\hat{g}_t)$ to the conditional expectation of $r_t^{(\varepsilon)}(g_t)$, introducing the factor $1/(1 - \gamma_t)$. Finally, we sum over time and apply Azuma–Hoeffding to control the martingale fluctuation $r_t^{(\varepsilon)}(g_t) - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}]$, and we collect the remaining terms into $R^{(\varepsilon)}(T, \delta)$.

Step 1: Define a high-probability “good event”. Let $\mathcal{E}$ be the uniform event from Lemma E.8 ensuring $|\hat{w}_t(g) - \bar{w}_t(g)| \leq \beta_t(g)$ for all (t, g) with $N_t(g) > 0$. By Lemma E.8, $\mathbb{P}(\mathcal{E}) \geq 1 - \delta/2$.

On $\mathcal{E}$, Lemma E.11 guarantees that for all $t \leq T$ and all $g \in [G]$,

$$|\hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq \eta_t(g) + \text{Drift}_t. \quad (114)$$

Step 2: One-step optimistic comparison. Fix $t \in \{1, \dots, T\}$ and work on $\mathcal{E}$. Let $g_t^* \in \arg \max_{g \in [G]} r_t^{(\varepsilon)}(g)$ be any oracle group.

By (31) and (114), for any group $g \in [G]$ we have

$$r_t^{(\varepsilon)}(g) \leq \hat{r}_t^{(\varepsilon)}(g) + \eta_t(g) + \text{Drift}_t = \text{UCB}_t(g) + \text{Drift}_t. \quad (115)$$

Similarly, again by (114),

$$\hat{r}_t^{(\varepsilon)}(g) \leq r_t^{(\varepsilon)}(g) + \eta_t(g) + \text{Drift}_t. \quad (116)$$

In particular, for the oracle group g_t^* ,

$$r_{t,\star}^{(\varepsilon)} = r_t^{(\varepsilon)}(g_t^*) \leq \text{UCB}_t(g_t^*) + \text{Drift}_t. \quad (117)$$

By definition of $\hat{g}_t$ as an argmax of $\text{UCB}_t(\cdot)$,

$$\text{UCB}_t(g_t^*) \leq \text{UCB}_t(\hat{g}_t). \quad (118)$$

Combining (117) and (118) gives

$$r_{t,\star}^{(\varepsilon)} \leq \text{UCB}_t(\hat{g}_t) + \text{Drift}_t = \hat{r}_t^{(\varepsilon)}(\hat{g}_t) + \eta_t(\hat{g}_t) + \text{Drift}_t. \quad (119)$$

Step 3: Relate the chosen group to the greedy group under time-varying exploration. Algorithm 2 samples g_t from the mixture distribution $\mathbb{P}_t$ in (32), namely

$$\mathbb{P}_t = (1 - \gamma_t) e_{\hat{g}_t} + \gamma_t \cdot \frac{1}{G} \mathbf{1}, \quad t \geq 1,$$

where $e_{\hat{g}_t}$ is the point mass on $\hat{g}_t$ and $\mathbf{1}$ is the all-ones vector in $\mathbb{R}^G$. Therefore, conditioning on $\mathcal{F}_{t-1}$,

$$\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] = (1 - \gamma_t) r_t^{(\varepsilon)}(\hat{g}_t) + \gamma_t \cdot \frac{1}{G} \sum_{g=1}^G r_t^{(\varepsilon)}(g) \geq (1 - \gamma_t) r_t^{(\varepsilon)}(\hat{g}_t). \quad (120)$$

Rearranging (120) yields

$$r_t^{(\varepsilon)}(\hat{g}_t) \leq \frac{1}{1 - \gamma_t} \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right]. \quad (121)$$

Now apply (116) with $g = \hat{g}_t$:

$$\hat{r}_t^{(\varepsilon)}(\hat{g}_t) \leq r_t^{(\varepsilon)}(\hat{g}_t) + \eta_t(\hat{g}_t) + \text{Drift}_t.$$

Substitute this into (119) to obtain

$$\begin{aligned} r_{t,\star}^{(\varepsilon)} &\leq (r_t^{(\varepsilon)}(\hat{g}_t) + \eta_t(\hat{g}_t) + \text{Drift}_t) + \eta_t(\hat{g}_t) + \text{Drift}_t \\ &= r_t^{(\varepsilon)}(\hat{g}_t) + 2\eta_t(\hat{g}_t) + 2\text{Drift}_t. \end{aligned} \quad (122)$$

Finally, apply (121) to the first term in (122):

$$r_{t,\star}^{(\varepsilon)} \leq \frac{1}{1-\gamma_t} \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] + 2\eta_t(\hat{g}_t) + 2\text{Drift}_t. \quad (123)$$

Rearrange (123) by subtracting $r_t^{(\varepsilon)}(g_t)$ on both sides:

$$r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t) \leq \frac{1}{1-\gamma_t} \left(\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] - r_t^{(\varepsilon)}(g_t) \right) + \left(\frac{1}{1-\gamma_t} - 1 \right) r_t^{(\varepsilon)}(g_t) + 2\eta_t(\hat{g}_t) + 2\text{Drift}_t. \quad (124)$$

Since $r_t^{(\varepsilon)}(g_t) \in [0, 1]$ (Assumption E.2),

$$\left(\frac{1}{1-\gamma_t} - 1 \right) r_t^{(\varepsilon)}(g_t) = \frac{\gamma_t}{1-\gamma_t} r_t^{(\varepsilon)}(g_t) \leq \frac{\gamma_t}{1-\gamma_t}.$$

Hence

$$r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t) \leq \frac{1}{1-\gamma_t} \left(\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] - r_t^{(\varepsilon)}(g_t) \right) + \frac{\gamma_t}{1-\gamma_t} + 2\eta_t(\hat{g}_t) + 2\text{Drift}_t. \quad (125)$$

Step 4: Sum over t and remove conditional expectations via Azuma. Sum (125) over $t = 1, \dots, T$ to obtain

$$\widehat{\text{Reg}}_T^{(\varepsilon)} \leq \sum_{t=1}^T \frac{1}{1-\gamma_t} \left(\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] - r_t^{(\varepsilon)}(g_t) \right) + \sum_{t=1}^T \frac{\gamma_t}{1-\gamma_t} + 2 \sum_{t=1}^T \eta_t(\hat{g}_t) + 2 \sum_{t=1}^T \text{Drift}_t. \quad (126)$$

Define the (weighted) martingale differences

$$Z_t := \frac{1}{1-\gamma_t} \left(\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] - r_t^{(\varepsilon)}(g_t) \right), \quad t = 1, \dots, T.$$

Since $\hat{g}_t$ (hence γ_t and the mixture distribution in (32)) is $\mathcal{F}_{t-1}$ -measurable, we have $\mathbb{E}[Z_t \mid \mathcal{F}_{t-1}] = 0$, i.e., $\{Z_t\}$ is a martingale difference sequence. Moreover, by Assumption E.2, $r_t^{(\varepsilon)}(g) \in [0, 1]$ for all g , so

$$\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] \in [0, 1] \quad \text{and} \quad \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] - r_t^{(\varepsilon)}(g_t) \in [-1, 1].$$

Hence, almost surely,

$$|Z_t| \leq \frac{1}{1-\gamma_t} \leq \frac{1}{1-\gamma_{\max}}, \quad \gamma_{\max} := \max_{1 \leq t \leq T} \gamma_t.$$

Applying Azuma–Hoeffding to $\sum_{t=1}^T Z_t$ with increment bound $1/(1-\gamma_{\max})$, we obtain

$$\mathbb{P} \left(\sum_{t=1}^T Z_t > \frac{1}{1-\gamma_{\max}} \sqrt{2T \log \frac{2}{\delta}} \right) \leq \frac{\delta}{2}. \quad (127)$$

Equivalently, with probability at least $1 - \delta/2$,

$$\sum_{t=1}^T \frac{1}{1-\gamma_t} \left(\mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] - r_t^{(\varepsilon)}(g_t) \right) = \sum_{t=1}^T Z_t \leq \frac{1}{1-\gamma_{\max}} \sqrt{2T \log \frac{2}{\delta}}.$$

Step 5: Union bound and conclude (113). On the event in Step 1 we have (114) simultaneously for all (t, g) , and by Lemma E.8 this event has probability at least $1 - \delta/2$. On the Azuma event from (127), which has probability at least $1 - \delta/2$, plugging the above bound into (126) yields

$$\widehat{\text{Reg}}_T^{(\varepsilon)} \leq \frac{1}{1 - \gamma_{\max}} \sqrt{2T \log \frac{2}{\delta}} + \sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t} + 2 \sum_{t=1}^T \eta_t(\hat{g}_t) + 2 \sum_{t=1}^T \text{Drift}_t.$$

The two events need not be independent; the union bound applies regardless, hence their intersection has probability at least $1 - \delta/2 - \delta/2 = 1 - \delta$. On this intersection, the right-hand side is exactly $R^{(\varepsilon)}(T, \delta)$ as defined in Definition E.25 (see (112)), so (113) holds with probability at least $1 - \delta$. This concludes the proof. $\square$

Corollary E.27 (Sublinear exploration cost under the anytime schedule). *Fix any $\kappa > 0$ and $c > 0$, and adopt the anytime schedules (80)–(82). In particular, define the deterministic inflation factor*

$$\phi(t) := (\log(e + t))^{\kappa/2}, \quad t \geq 1,$$

and set

$$\gamma_t := \min \left\{ \frac{1}{2}, c \phi(t) (1 - \lambda_t) \log \left(\frac{G\pi^2 t^2}{\delta} \right) \right\}, \quad t \geq 1. \quad (128)$$

Then for every horizon $T \geq 3$,

$$\sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t} \leq \underbrace{\frac{C_{\gamma,1} T}{(\log(e + T))^{\kappa/2}}}_{\text{main term}} + \underbrace{\frac{C_{\gamma,2} T}{(\log(e + T))^{1+\kappa/2}}}_{\text{lower order}} + \underbrace{C_{\gamma,3} \sqrt{T}}_{\text{finite-}T \text{ remainder}}, \quad (129)$$

where

$$C_{\gamma,1} := 4c 2^{\kappa/2}, \quad C_{\gamma,2} := c 2^{2+\kappa/2} \log \left(\frac{G\pi^2}{\delta} \right), \quad C_{\gamma,3} := 2c \left(2 + \log \left(\frac{G\pi^2}{\delta} \right) \right).$$

Consequently, $\sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t} = o(T)$ as $T \rightarrow \infty$.

Proof. Proof sketch. The goal is to show that the exploration overhead $\sum_{t \leq T} \frac{\gamma_t}{1 - \gamma_t}$ is sublinear under the anytime choice of γ_t . We proceed in four transparent steps. First, since $\gamma_t \leq 1/2$ by construction, we remove the denominator and reduce the problem to bounding $\sum_{t \leq T} \gamma_t$ up to a constant factor. Second, we plug in the explicit anytime form of $(1 - \lambda_t)$ and the inflation factor $\phi(t)$ to obtain a clean upper bound on γ_t as the sum of two terms: one of order $(\log(e + t))^{-\kappa/2}$ and a lower-order term of order $(\log(e + t))^{-(1+\kappa/2)}$. Third, we sum these two sequences over $t = 1, \dots, T$ using a simple splitting argument at $t \approx \sqrt{T}$, which yields a finite- T remainder of order $\sqrt{T}$ plus the desired polylogarithmic rates in T . Finally, we collect constants carefully to match (129), and then divide by T and let $T \rightarrow \infty$ to conclude $\sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t} = o(T)$.

Step 1 (remove the denominator). By construction, $\gamma_t \leq \frac{1}{2}$ for all t , hence $1 - \gamma_t \geq \frac{1}{2}$ and therefore

$$\frac{\gamma_t}{1 - \gamma_t} = \gamma_t \cdot \frac{1}{1 - \gamma_t} \leq 2\gamma_t.$$

Summing from $t = 1$ to T gives

$$\sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t} \leq 2 \sum_{t=1}^T \gamma_t. \quad (130)$$

Step 2 (upper bound γ_t by a clean $(\log)^{-\kappa/2}$ rate). From (80), for all $t \geq 1$,

$$1 - \lambda_t = \frac{1}{(\log(e + t))^{1+\kappa}}.$$

Also, by the definition of $\phi(t)$ in this corollary, for all $t \geq 1$,

$$\phi(t) (1 - \lambda_t) = (\log(e + t))^{\kappa/2} \cdot \frac{1}{(\log(e + t))^{1+\kappa}} = \frac{1}{(\log(e + t))^{1+\kappa/2}}.$$

Moreover, for all $t \geq 1$,

$$\log\left(\frac{G\pi^2 t^2}{\delta}\right) = \log\left(\frac{G\pi^2}{\delta}\right) + 2\log t \leq \log\left(\frac{G\pi^2}{\delta}\right) + 2\log(e+t).$$

Therefore, for all $t \geq 1$,

$$\begin{aligned} \gamma_t &\leq c\phi(t)(1-\lambda_t)\log\left(\frac{G\pi^2 t^2}{\delta}\right) \\ &= c \cdot \frac{1}{(\log(e+t))^{1+\kappa/2}} \cdot \log\left(\frac{G\pi^2 t^2}{\delta}\right) \\ &\leq c \cdot \frac{1}{(\log(e+t))^{1+\kappa/2}} \cdot \left(\log\left(\frac{G\pi^2}{\delta}\right) + 2\log(e+t)\right) \\ &= \frac{c\log(\frac{G\pi^2}{\delta})}{(\log(e+t))^{1+\kappa/2}} + \frac{2c}{(\log(e+t))^{\kappa/2}}. \end{aligned} \quad (131)$$

Step 3 (sum the two pieces carefully). Summing (131) over $t = 1, \dots, T$ yields

$$\sum_{t=1}^T \gamma_t \leq c\log\left(\frac{G\pi^2}{\delta}\right) \sum_{t=1}^T \frac{1}{(\log(e+t))^{1+\kappa/2}} + 2c \sum_{t=1}^T \frac{1}{(\log(e+t))^{\kappa/2}}. \quad (132)$$

We now bound the two sums in (132).

(i) *The $(1+\kappa/2)$ -sum.* Since $(\log(e+t))^{-(1+\kappa/2)} \leq 1$ for all t and the sequence is decreasing,

$$\sum_{t=1}^T \frac{1}{(\log(e+t))^{1+\kappa/2}} \leq \lfloor \sqrt{T} \rfloor + \sum_{t=\lfloor \sqrt{T} \rfloor+1}^T \frac{1}{(\log(e+\lfloor \sqrt{T} \rfloor+1))^{1+\kappa/2}} \leq \sqrt{T} + \frac{T}{(\log(e+\sqrt{T}))^{1+\kappa/2}}.$$

For $T \geq 3$, we have $e+T \leq (e+\sqrt{T})^2$, hence $\log(e+\sqrt{T}) \geq \frac{1}{2}\log(e+T)$ and thus

$$\frac{1}{(\log(e+\sqrt{T}))^{1+\kappa/2}} \leq \frac{2^{1+\kappa/2}}{(\log(e+T))^{1+\kappa/2}}.$$

Therefore,

$$\sum_{t=1}^T \frac{1}{(\log(e+t))^{1+\kappa/2}} \leq \sqrt{T} + \frac{2^{1+\kappa/2}}{(\log(e+T))^{1+\kappa/2}} T. \quad (133)$$

(ii) *The $(\kappa/2)$ -sum.* Similarly,

$$\sum_{t=1}^T \frac{1}{(\log(e+t))^{\kappa/2}} \leq \sqrt{T} + \frac{T}{(\log(e+\sqrt{T}))^{\kappa/2}}.$$

Again using $\log(e+\sqrt{T}) \geq \frac{1}{2}\log(e+T)$ for $T \geq 3$,

$$\frac{1}{(\log(e+\sqrt{T}))^{\kappa/2}} \leq \frac{2^{\kappa/2}}{(\log(e+T))^{\kappa/2}},$$

so

$$\sum_{t=1}^T \frac{1}{(\log(e+t))^{\kappa/2}} \leq \sqrt{T} + \frac{2^{\kappa/2}}{(\log(e+T))^{\kappa/2}} T. \quad (134)$$

Step 4 (combine everything, without hiding terms). Using (132) together with (133) and (134), we obtain

$$\begin{aligned} \sum_{t=1}^T \gamma_t &\leq 2c\left(\sqrt{T} + \frac{2^{\kappa/2}T}{(\log(e+T))^{\kappa/2}}\right) + c\log\left(\frac{G\pi^2}{\delta}\right)\left(\sqrt{T} + \frac{2^{1+\kappa/2}T}{(\log(e+T))^{1+\kappa/2}}\right) \\ &= \frac{2c \cdot 2^{\kappa/2}T}{(\log(e+T))^{\kappa/2}} + \frac{c \cdot 2^{1+\kappa/2}\log(\frac{G\pi^2}{\delta})T}{(\log(e+T))^{1+\kappa/2}} + c\left(2 + \log\left(\frac{G\pi^2}{\delta}\right)\right)\sqrt{T}. \end{aligned} \quad (135)$$

Multiplying (135) by 2 and using (130) yields

$$\sum_{t=1}^T \frac{\gamma_t}{1-\gamma_t} \leq \frac{4c \cdot 2^{\kappa/2} T}{(\log(e+T))^{\kappa/2}} + \frac{c \cdot 2^{2+\kappa/2} \log\left(\frac{G\pi^2}{\delta}\right) T}{(\log(e+T))^{1+\kappa/2}} + 2c \left(2 + \log\left(\frac{G\pi^2}{\delta}\right)\right) \sqrt{T},$$

which is exactly (129) with the stated constants.

Finally, since $\kappa > 0$, dividing both sides of (129) by T and letting $T \rightarrow \infty$ gives $\sum_{t=1}^T \frac{\gamma_t}{1-\gamma_t} = o(T)$, which concludes the proof. $\square$

Proposition E.28 (Discounted-UCB confidence accumulation bound for the greedy group). *Fix $\delta \in (0, 1)$ and consider Algorithm 2 in Section H under the anytime schedules (80)–(82) with $\kappa > 0$ and $c > 0$. Let $\{\lambda_t\}_{t \geq 1} \subset (0, 1)$ be predictable, and let discounted counts $N_t(g)$ be defined by the recursion in (26) (equivalently (85)).*

Let $\eta_t(g)$ be the confidence radius defined in (30) (with confidence parameter $\delta/2$ used in Section H), and let $\hat{g}_t$ be the greedy/UCB-selected group defined in Algorithm 2. Recall the non-stationarity budget $V_T^{(\varepsilon)}$ defined in (47). Define the logarithmic factor

$$L_T := \sqrt{\log\left(\frac{8GT}{\delta}\right)}. \quad (136)$$

In this proposition, we use the strengthened exploration schedule

$$\gamma_t := \min \left\{ \frac{1}{2}, c(1-\lambda_t) \log\left(\frac{G\pi^2 t^2}{\delta}\right) (\log(e+t))^{\kappa/2} \right\}, \quad t \geq 1. \quad (137)$$

Then, on the concentration event $\mathcal{E}_{\text{conc}}$ constructed in the proof of Theorem E.26, there exists a constant $C_\eta > 0$ (independent of T) such that (138) holds:

$$\sum_{t=1}^T \eta_t(\hat{g}_t) \leq C_\eta L_T \left[(V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right], \quad (138)$$

where, one may take $C_\eta = C_\eta(G, w_{\max}, \varepsilon, \kappa, c)$. In particular, for the choice of H specified in the proof ensuring $\Pr(\mathcal{E}_{\text{win}}(H)) \geq 1 - \delta/2$, a union bound implies that (138) holds with probability at least $1 - \delta$ on $\mathcal{E}_{\text{conc}} \cap \mathcal{E}_{\text{win}}(H)$.

Proof. We work on the event $\mathcal{E}_{\text{conc}}$ throughout.

Proof sketch. Step 1 relates the discounted counts $N_t(g)$ to non-discounted window counts $\tilde{N}_t^{(H)}(g)$ via a window product $\Lambda_{t,H}$. Step 2 proves that all window counts are at least a constant fraction of their predictable means and lower bounds those means using uniform exploration. Step 3 plugs the resulting deterministic lower bounds into (29)–(30) to upper bound $\eta_t(\hat{g}_t)$ pointwise. Step 4 sums the pointwise bounds over time and controls the initial segment using the capped-radius convention for $N_t(g) = 0$. Step 5 chooses H and upper bounds $\sum_t \Lambda_{t,H}^{-1/2}$ without any exponentially large factors, yielding (138).

Step 1 (Relating discounted counts to a length- H window). Fix an integer $H \in \{1, \dots, T\}$ (to be tuned later). Define the window (non-discounted) count

$$\tilde{N}_t^{(H)}(g) := \sum_{s=(t-H) \vee 1}^{t-1} \mathbf{1}\{g_s = g\}, \quad t \geq 2, g \in [G]. \quad (139)$$

For each $t \geq 2$, define the window product

$$\Lambda_{t,H} := \prod_{j=(t-H) \vee 1}^{t-1} \lambda_j. \quad (140)$$

Recall the discounted-count expansion (85): for any $t \geq 2$ and any $g \in [G]$,

$$N_t(g) = \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}. \quad (141)$$

Since $\lambda_j \in (0, 1)$, each summand in (141) is nonnegative, hence restricting the sum yields

$$N_t(g) \geq \sum_{s=(t-H) \vee 1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}. \quad (142)$$

Fix any $s \in \{(t-H) \vee 1, \dots, t-1\}$. Because $\{s+1, \dots, t-1\} \subseteq \{(t-H) \vee 1, \dots, t-1\}$, we have

$$\prod_{j=s+1}^{t-1} \lambda_j = \frac{\prod_{j=(t-H) \vee 1}^{t-1} \lambda_j}{\prod_{j=(t-H) \vee 1}^s \lambda_j} \geq \prod_{j=(t-H) \vee 1}^{t-1} \lambda_j = \Lambda_{t,H}, \quad (143)$$

where the inequality uses $\prod_{j=(t-H) \vee 1}^s \lambda_j \leq 1$ since each $\lambda_j \in (0, 1)$. Plugging (143) into (142) and factoring out $\Lambda_{t,H}$ gives

$$N_t(g) \geq \Lambda_{t,H} \tilde{N}_t^{(H)}(g). \quad (144)$$

Step 2 (A high-probability lower bound on window counts via uniform exploration). For $t \geq 2$ and $g \in [G]$, define the window mean

$$\mu_{t,g}^{(H)} := \sum_{s=(t-H) \vee 1}^{t-1} \mathbb{P}_s(g), \quad (145)$$

and the centered window sum

$$S_{t,g}^{(H)} := \sum_{s=(t-H) \vee 1}^{t-1} \left(\mathbf{1}\{g_s = g\} - \mathbb{P}_s(g) \right). \quad (146)$$

Then

$$\tilde{N}_t^{(H)}(g) = \mu_{t,g}^{(H)} + S_{t,g}^{(H)}. \quad (147)$$

Fix (t, g) with $t \geq 2$. Let $Z_s := \mathbf{1}\{g_s = g\} - \mathbb{P}_s(g)$. Then $\mathbb{E}[Z_s \mid \mathcal{F}_{s-1}] = 0$, and $|Z_s| \leq 1$ almost surely. Moreover,

$$\mathbb{E}[Z_s^2 \mid \mathcal{F}_{s-1}] = \mathbb{P}_s(g)(1 - \mathbb{P}_s(g)) \leq \mathbb{P}_s(g).$$

Hence the predictable quadratic variation satisfies

$$V_{t,g}^{(H)} := \sum_{s=(t-H) \vee 1}^{t-1} \mathbb{E}[Z_s^2 \mid \mathcal{F}_{s-1}] \leq \sum_{s=(t-H) \vee 1}^{t-1} \mathbb{P}_s(g) = \mu_{t,g}^{(H)}. \quad (148)$$

Applying Freedman's inequality gives, for any $x > 0$,

$$\Pr \left(S_{t,g}^{(H)} \leq -x \right) \leq \exp \left(- \frac{x^2}{2(V_{t,g}^{(H)} + x/3)} \right) \leq \exp \left(- \frac{x^2}{2(\mu_{t,g}^{(H)} + x/3)} \right), \quad (149)$$

where the second inequality uses (148). Choose $x := \mu_{t,g}^{(H)}/2$. Then $\mu_{t,g}^{(H)} + x/3 = (7/6)\mu_{t,g}^{(H)}$, hence

$$\Pr \left(S_{t,g}^{(H)} \leq -\frac{1}{2}\mu_{t,g}^{(H)} \right) \leq \exp \left(- \frac{3}{28}\mu_{t,g}^{(H)} \right). \quad (150)$$

By (147), the event $\{S_{t,g}^{(H)} \geq -\mu_{t,g}^{(H)}/2\}$ is equivalent to $\{\tilde{N}_t^{(H)}(g) \geq \mu_{t,g}^{(H)}/2\}$, hence

$$\Pr \left(\tilde{N}_t^{(H)}(g) \leq \frac{1}{2}\mu_{t,g}^{(H)} \right) \leq \exp \left(- \frac{3}{28}\mu_{t,g}^{(H)} \right). \quad (151)$$

We now lower bound $\mu_{t,g}^{(H)}$ using uniform exploration, without invoking any monotonicity of $\{\gamma_t\}$. By the mixture rule in Algorithm 2, for every $s \geq 1$ and $g \in [G]$,

$$\mathbb{P}_s(g) \geq \frac{\gamma_s}{G}. \quad (152)$$

Therefore, for every $t \geq 2$,

$$\mu_{t,g}^{(H)} \geq \frac{1}{G} \sum_{s=(t-H) \vee 1}^{t-1} \gamma_s. \quad (153)$$

We next derive a deterministic lower bound on γ_s valid for all $s \leq T$ under (137). For $s \leq T$ we have $\log(\frac{G\pi^2 s^2}{\delta}) \geq \log(\frac{G\pi^2}{\delta})$ and $\log(e+s) \leq \log(e+T)$, so

$$(1 - \lambda_s) (\log(e+s))^{\kappa/2} = \frac{(\log(e+s))^{\kappa/2}}{(\log(e+s))^{1+\kappa}} = \frac{1}{(\log(e+s))^{1+\kappa/2}} \geq \frac{1}{(\log(e+T))^{1+\kappa/2}}.$$

Hence

$$\gamma_s \geq \min \left\{ \frac{1}{2}, c \log \left(\frac{G\pi^2}{\delta} \right) \cdot \frac{1}{(\log(e+T))^{1+\kappa/2}} \right\} =: \underline{\gamma}_T, \quad \forall s \in [T]. \quad (154)$$

Plugging (154) into (153) gives, for all $t \geq H+1$,

$$\mu_{t,g}^{(H)} \geq \frac{H}{G} \underline{\gamma}_T, \quad \forall g \in [G]. \quad (155)$$

Define the good event

$$\mathcal{E}_{\text{win}}(H) := \left\{ \tilde{N}_t^{(H)}(g) \geq \frac{1}{2} \mu_{t,g}^{(H)} \text{ for all } t \in \{H+1, \dots, T\}, g \in [G] \right\}. \quad (156)$$

Taking a union bound over $(t, g) \in \{H+1, \dots, T\} \times [G]$ and using (151) yields

$$\Pr(\mathcal{E}_{\text{win}}(H)^c) \leq GT \exp \left(-\frac{3}{28} \cdot \frac{H}{G} \underline{\gamma}_T \right). \quad (157)$$

Remark for consistency with the global analysis. The non-stationarity budget $V_T^{(\varepsilon)}$ does not enter the concentration step above because $\mu_{t,g}^{(H)}$ depends only on the algorithmic sampling probabilities $\{\mathbb{P}_s(g)\}$, and the lower bound (152) holds deterministically for every environment; $V_T^{(\varepsilon)}$ will enter only through the choice of the proof parameter H in Step 5 when balancing estimation and non-stationarity.

Step 3 (Bounding $\eta_t(\hat{g}_t)$ via window counts). Fix any $t \in \{H+1, \dots, T\}$ and work on $\mathcal{E}_{\text{win}}(H)$. By definition (156), for every $g \in [G]$ we have $\tilde{N}_t^{(H)}(g) \geq \frac{1}{2} \mu_{t,g}^{(H)}$. Combining this with (155) yields

$$\tilde{N}_t^{(H)}(g) \geq \frac{H}{2G} \underline{\gamma}_T, \quad \forall g \in [G], \forall t \in \{H+1, \dots, T\}. \quad (158)$$

Using (144) and (158) gives

$$N_t(g) \geq \Lambda_{t,H} \cdot \frac{H}{2G} \underline{\gamma}_T, \quad \forall g \in [G], \forall t \in \{H+1, \dots, T\}. \quad (159)$$

Since $\lambda_j \in (0, 1)$, we have $\Lambda_{t,H} > 0$, hence $N_t(g) > 0$ on this range and $\beta_t(g)$ in (29) is finite.

From (29) and $\log(\frac{4GT}{\delta}) \leq L_T^2$, for any $t \in \{H+1, \dots, T\}$ and $g \in [G]$,

$$\beta_t(g) = 2\sqrt{2} w_{\max} \sqrt{\frac{\log(\frac{4GT}{\delta})}{N_t(g)}} \leq 2\sqrt{2} w_{\max} L_T \cdot \frac{1}{\sqrt{N_t(g)}}.$$

Using (159) yields

$$\beta_t(g) \leq 2\sqrt{2} w_{\max} L_T \cdot \sqrt{\frac{2G}{\Lambda_{t,H} H \underline{\gamma}_T}} = 4 w_{\max} L_T \cdot \sqrt{\frac{G}{\Lambda_{t,H} H \underline{\gamma}_T}}. \quad (160)$$

Summing (160) over $g \in [G]$ gives

$$\sum_{g=1}^G \beta_t(g) \leq 4w_{\max} L_T \cdot G \sqrt{\frac{G}{\Lambda_{t,H} H \underline{\gamma}_T}}. \quad (161)$$

Applying (30) at $g = \hat{g}_t$ and using (160)–(161) yields

$$\eta_t(\hat{g}_t) \leq C_1 \frac{L_T}{\sqrt{\Lambda_{t,H}}} \cdot \sqrt{\frac{G}{H \underline{\gamma}_T}}, \quad (162)$$

where one may take

$$C_1 := 4w_{\max} \left(\frac{1}{\varepsilon} + \frac{w_{\max} G}{\varepsilon^2} \right),$$

so $C_1 = C_1(G, w_{\max}, \varepsilon)$.

Step 4 (Summing over t and including the early times). We split the sum at time $2H$. This is purely a proof device: it allows a uniform and non-explosive control of $\Lambda_{t,H}$ for all $t \geq 2H$ in Step 5, and the initial segment $t \leq 2H$ will be handled by a capped-radius bound.

Early segment $t \leq 2H$. Define the capped radius

$$\beta_t^{\text{cap}}(g) := 2\sqrt{2} w_{\max} \sqrt{\frac{\log(\frac{4GT}{\delta})}{N_t(g) \vee 1}}, \quad t \in [T], \quad g \in [G]. \quad (163)$$

which equals (29) when $N_t(g) \geq 1$ and replaces the symbolic $+\infty$ (when $N_t(g) = 0$) by the finite value $2\sqrt{2} w_{\max} \sqrt{\log(\frac{4GT}{\delta})}$. Since $\log(\frac{4GT}{\delta}) \leq L_T^2$, we have $\beta_t^{\text{cap}}(g) \leq 2\sqrt{2} w_{\max} L_T$ for all (t, g) and thus

$$\beta_t^{\text{cap}}(\hat{g}_t) \leq 2\sqrt{2} w_{\max} L_T, \quad \sum_{g=1}^G \beta_t^{\text{cap}}(g) \leq 2\sqrt{2} w_{\max} G L_T.$$

Plugging these into (30) yields, for every $t \leq 2H$,

$$\eta_t(\hat{g}_t) \leq \frac{2\sqrt{2} w_{\max} L_T}{\varepsilon} + \frac{w_{\max}}{\varepsilon^2} \cdot 2\sqrt{2} w_{\max} G L_T = C_2 L_T,$$

where one may take $C_2 := 2\sqrt{2} w_{\max} \left(\frac{1}{\varepsilon} + \frac{w_{\max} G}{\varepsilon^2} \right)$. Therefore

$$\sum_{t=1}^{2H} \eta_t(\hat{g}_t) \leq 2H C_2 L_T. \quad (164)$$

Tail segment $t \geq 2H + 1$. On $\mathcal{E}_{\text{win}}(H)$, summing (162) over $t = 2H + 1, \dots, T$ gives

$$\sum_{t=2H+1}^T \eta_t(\hat{g}_t) \leq C_1 L_T \sqrt{\frac{G}{H \underline{\gamma}_T}} \sum_{t=2H+1}^T \frac{1}{\sqrt{\Lambda_{t,H}}}. \quad (165)$$

Combining (164) and (165), on $\mathcal{E}_{\text{win}}(H)$ we have

$$\sum_{t=1}^T \eta_t(\hat{g}_t) \leq 2H C_2 L_T + C_1 L_T \sqrt{\frac{G}{H \underline{\gamma}_T}} \sum_{t=2H+1}^T \frac{1}{\sqrt{\Lambda_{t,H}}}. \quad (166)$$

Step 5 (Choosing H , controlling $\mathcal{E}_{\text{win}}(H)$, and controlling $\Lambda_{t,H}$). *Step 5(a): ensuring $\Pr(\mathcal{E}_{\text{win}}(H)) \geq 1 - \delta/2$.* From (157), it suffices to have

$$GT \exp\left(-\frac{3}{28} \cdot \frac{H}{G} \underline{\gamma}_T\right) \leq \frac{\delta}{2}.$$

Equivalently,

$$\frac{H}{G} \underline{\gamma}_T \geq \frac{28}{3} \log\left(\frac{2GT}{\delta}\right).$$

Define

$$H_{\min} := \left\lceil \frac{28}{3} \cdot \frac{G}{\underline{\gamma}_T} \log\left(\frac{2GT}{\delta}\right) \right\rceil. \quad (167)$$

Then for any $H \geq H_{\min}$ we have

$$\Pr(\mathcal{E}_{\text{win}}(H)) \geq 1 - \frac{\delta}{2}. \quad (168)$$

Step 5(b): a non-explosive bound on $\Lambda_{t,H}^{-1/2}$ for $t \geq 2H + 1$. We first record the elementary inequality

$$\log u \geq -2(1 - u) \quad \text{for all } u \in [1/2, 1]. \quad (169)$$

Indeed, letting $h(u) := \log u + 2(1 - u)$ on $[1/2, 1]$, we have $h(1) = 0$ and $h'(u) = 1/u - 2 \geq 0$, hence $h(u) \geq 0$.

Under (80) we have $1 - \lambda_t = (\log(e + t))^{-(1+\kappa)} \leq 1/2$ for all $t \geq 2$, so $\lambda_t \in [1/2, 1]$ for all $t \geq 2$ and (169) applies. For $t \geq 2H + 1$ we have $(t - H) \vee 1 = t - H \geq H + 1 \geq 2$, hence

$$\log \Lambda_{t,H} = \sum_{j=t-H}^{t-1} \log \lambda_j \geq -2 \sum_{j=t-H}^{t-1} (1 - \lambda_j). \quad (170)$$

Since $j \mapsto (1 - \lambda_j)$ is non-increasing under (80), for all $j \in \{t - H, \dots, t - 1\}$ we have $(1 - \lambda_j) \leq (1 - \lambda_{t-H})$, hence

$$\sum_{j=t-H}^{t-1} (1 - \lambda_j) \leq H (1 - \lambda_{t-H}) = \frac{H}{(\log(e + t - H))^{1+\kappa}}. \quad (171)$$

Combining (170) and (171) yields

$$\Lambda_{t,H} \geq \exp\left(-\frac{2H}{(\log(e + t - H))^{1+\kappa}}\right), \quad \text{hence} \quad \frac{1}{\sqrt{\Lambda_{t,H}}} \leq \exp\left(\frac{H}{(\log(e + t - H))^{1+\kappa}}\right). \quad (172)$$

We now use the fact that on the tail $t \geq 2H + 1$ we have $t - H \geq t/2$. Therefore $\log(e + t - H) \geq \log(e + t/2)$. For every $t \geq 3$ we have $\log(e + t/2) \geq \frac{1}{2} \log(e + t)$, hence for all $t \geq 2H + 1 \geq 3$,

$$(\log(e + t - H))^{1+\kappa} \geq (\log(e + t/2))^{1+\kappa} \geq 2^{-(1+\kappa)} (\log(e + t))^{1+\kappa} \geq 2^{-(1+\kappa)} (\log(e + T))^{1+\kappa}.$$

Assume additionally that

$$H \leq (\log(e + T))^{1+\kappa}. \quad (173)$$

Then the exponent in (172) satisfies, for all $t \geq 2H + 1$,

$$\frac{H}{(\log(e + t - H))^{1+\kappa}} \leq \frac{H}{2^{-(1+\kappa)} (\log(e + T))^{1+\kappa}} \leq 2^{1+\kappa}.$$

Hence, defining the numerical constant

$$C_\Lambda(\kappa) := \exp(2^{1+\kappa}), \quad (174)$$

we obtain the uniform bound

$$\frac{1}{\sqrt{\Lambda_{t,H}}} \leq C_\Lambda(\kappa), \quad \forall t \in \{2H + 1, \dots, T\}. \quad (175)$$

Consequently,

$$\sum_{t=2H+1}^T \frac{1}{\sqrt{\Lambda_{t,H}}} \leq C_\Lambda(\kappa) (T - 2H) \leq C_\Lambda(\kappa) T. \quad (176)$$

Step 5(c): bounding $\underline{\gamma}_T$ and choosing H . Recall from (154) that for all $s \in [T]$,

$$\gamma_s \geq \underline{\gamma}_T \quad \text{where} \quad \underline{\gamma}_T := \min \left\{ \frac{1}{2}, c \log \left(\frac{G\pi^2}{\delta} \right) \cdot \frac{1}{(\log(e+T))^{1+\kappa/2}} \right\}.$$

Define the constant

$$c_0 := \min \left\{ \frac{1}{2}, c \log \left(\frac{G\pi^2}{\delta} \right) \right\},$$

so that (since $\min\{a, bx\} \geq \min\{a, b\}x$ for any $x \in (0, 1]$)

$$\underline{\gamma}_T = \min \left\{ \frac{1}{2}, \left(c \log \left(\frac{G\pi^2}{\delta} \right) \right) \cdot (\log(e+T))^{-(1+\kappa/2)} \right\} \geq c_0 (\log(e+T))^{-(1+\kappa/2)}. \quad (177)$$

Taking square-roots and reciprocals in (177) gives

$$\frac{1}{\sqrt{\underline{\gamma}_T}} \leq \frac{1}{\sqrt{c_0}} (\log(e+T))^{\frac{1}{2} + \frac{\kappa}{4}}. \quad (178)$$

We now choose H . Let

$$H_{\text{ns}} := \left\lceil \left(\frac{T}{V_T^{(\varepsilon)} \vee 1} \right)^{2/3} \right\rceil, \quad H_{\text{max}} := \lfloor (\log(e+T))^{1+\kappa} \rfloor.$$

Recall also H_{min} from (167). We set

$$H := (H_{\text{ns}} \wedge H_{\text{max}}) \vee H_{\text{min}}. \quad (179)$$

Then $H \geq H_{\text{min}}$ holds by construction, hence (168) applies. Moreover, $H \leq H_{\text{max}}$ whenever $H_{\text{min}} \leq H_{\text{max}}$; if $H_{\text{min}} > H_{\text{max}}$, we still keep the choice (179) (so $H = H_{\text{min}}$), and the bound remains valid (only looser) because the tail estimate below uses only inequalities that hold for all $H \in \{1, \dots, T\}$; in particular, we will explicitly indicate where $H \leq H_{\text{max}}$ is used.

Step 5(d): concluding the bound on $\mathcal{E}_{\text{win}}(H)$. We work on $\mathcal{E}_{\text{win}}(H)$. Starting from (166) and using (176), we obtain

$$\begin{aligned} \sum_{t=1}^T \eta_t(\hat{g}_t) &\leq 2HC_2L_T + C_1L_T \sqrt{\frac{G}{H\underline{\gamma}_T}} \sum_{t=2H+1}^T \frac{1}{\sqrt{\Lambda_{t,H}}} \\ &\leq 2HC_2L_T + C_1L_T \sqrt{\frac{G}{H\underline{\gamma}_T}} \cdot (C_\Lambda(\kappa) T) \quad (\text{by (176)}) \\ &= 2HC_2L_T + \left(C_1C_\Lambda(\kappa)\sqrt{G} \right) L_T \cdot \frac{T}{\sqrt{H}} \cdot \frac{1}{\sqrt{\underline{\gamma}_T}}. \end{aligned} \quad (180)$$

For convenience, define

$$\tilde{C} := C_1C_\Lambda(\kappa)\sqrt{G},$$

so (180) becomes

$$\sum_{t=1}^T \eta_t(\hat{g}_t) \leq 2HC_2L_T + \tilde{C} L_T \cdot \frac{T}{\sqrt{H}} \cdot \frac{1}{\sqrt{\underline{\gamma}_T}}. \quad (181)$$

Next, plug the explicit bound (178) into (181):

$$\begin{aligned} \sum_{t=1}^T \eta_t(\hat{g}_t) &\leq 2HC_2L_T + \tilde{C} L_T \cdot \frac{T}{\sqrt{H}} \cdot \frac{1}{\sqrt{c_0}} (\log(e+T))^{\frac{1}{2} + \frac{\kappa}{4}} \\ &= 2HC_2L_T + \frac{\tilde{C}}{\sqrt{c_0}} L_T \cdot \frac{T}{\sqrt{H}} \cdot (\log(e+T))^{\frac{1}{2} + \frac{\kappa}{4}}. \end{aligned} \quad (182)$$

We now upper bound the two H -dependent terms in (182) under the choice (179). There are two possible regimes:

Regime 1 (non-truncated): $H_{\text{ns}} \leq H_{\text{max}}$ and $H_{\text{ns}} \geq H_{\text{min}}$. In this case (179) gives $H = H_{\text{ns}}$. Since $H = \lceil (T/(V_T^{(\varepsilon)} \vee 1))^{2/3} \rceil$, we have the two deterministic bounds

$$H \geq \left(\frac{T}{V_T^{(\varepsilon)} \vee 1} \right)^{2/3}, \quad \text{and} \quad H \leq 1 + \left(\frac{T}{V_T^{(\varepsilon)} \vee 1} \right)^{2/3}. \quad (183)$$

From the first inequality in (183), taking square-roots and reciprocals yields

$$\frac{T}{\sqrt{H}} \leq T \left(\frac{V_T^{(\varepsilon)} \vee 1}{T} \right)^{1/3} = (V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3}. \quad (184)$$

Plugging (184) into the second term of (182) gives

$$\frac{\tilde{C}}{\sqrt{c_0}} L_T \cdot \frac{T}{\sqrt{H}} \cdot (\log(e+T))^{\frac{1}{2} + \frac{\kappa}{4}} \leq \frac{\tilde{C}}{\sqrt{c_0}} L_T (V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\frac{1}{2} + \frac{\kappa}{4}}. \quad (185)$$

For the first term of (182), the second inequality in (183) yields

$$2HC_2 L_T \leq 2C_2 L_T + 2C_2 L_T \left(\frac{T}{V_T^{(\varepsilon)} \vee 1} \right)^{2/3}. \quad (186)$$

Since $(V_T^{(\varepsilon)} \vee 1) \geq 1$, we have $(T/(V_T^{(\varepsilon)} \vee 1))^{2/3} \leq T^{2/3}$, hence the early-time contribution (186) is at most of order $L_T T^{2/3}$ and can be absorbed into the leading $L_T (V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4}$ term after adjusting constants (because $(V_T^{(\varepsilon)} \vee 1)^{1/3} (\log(e+T))^{\kappa/4} \geq 1$ for all $T \geq 3$).

Regime 2 (truncated): $H_{\text{ns}} > H_{\text{max}}$ and $H_{\text{max}} \geq H_{\text{min}}$. In this case (179) gives $H = H_{\text{max}} = \lfloor (\log(e+T))^{1+\kappa} \rfloor$. Then $H \geq \frac{1}{2} (\log(e+T))^{1+\kappa}$ for all T large enough, and thus

$$\frac{T}{\sqrt{H}} \leq \sqrt{2} \frac{T}{(\log(e+T))^{\frac{1}{2} + \frac{\kappa}{2}}}. \quad (187)$$

Plugging (187) into the second term of (182) yields

$$\begin{aligned} \frac{\tilde{C}}{\sqrt{c_0}} L_T \cdot \frac{T}{\sqrt{H}} \cdot (\log(e+T))^{\frac{1}{2} + \frac{\kappa}{4}} &\leq \frac{\sqrt{2} \tilde{C}}{\sqrt{c_0}} L_T \cdot \frac{T}{(\log(e+T))^{\frac{1}{2} + \frac{\kappa}{2}}} \cdot (\log(e+T))^{\frac{1}{2} + \frac{\kappa}{4}} \\ &= \frac{\sqrt{2} \tilde{C}}{\sqrt{c_0}} L_T \cdot \frac{T}{(\log(e+T))^{\kappa/4}}. \end{aligned} \quad (188)$$

For the first term in (182), $H = H_{\text{max}} \leq (\log(e+T))^{1+\kappa}$ implies

$$2HC_2 L_T \leq 2C_2 L_T (\log(e+T))^{1+\kappa}. \quad (189)$$

This is polylogarithmic in T times L_T , hence is dominated by the second term (188) once T is sufficiently large; for small T it can be absorbed into the constant by enlarging C_η .

Putting the two regimes into a single bound. Combining (185) (Regime 1) and (188) (Regime 2), and absorbing the early-time terms (186)–(189) into the same expression by adjusting constants, we obtain that on $\mathcal{E}_{\text{win}}(H)$,

$$\sum_{t=1}^T \eta_t(\hat{g}_t) \leq C_\eta L_T \left[(V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right],$$

which is exactly (138).

Finally, (168) implies $\Pr(\mathcal{E}_{\text{win}}(H)) \geq 1 - \delta/2$, and by construction $\Pr(\mathcal{E}_{\text{conc}}) \geq 1 - \delta/2$. A union bound gives $\Pr(\mathcal{E}_{\text{conc}} \cap \mathcal{E}_{\text{win}}(H)) \geq 1 - \delta$. Therefore, (138) holds with probability at least $1 - \delta$, which is the claimed statement. $\square$

Remark E.29 (How Proposition E.28 is used). Proposition E.28 provides a fully explicit control of term (III) in the regret interface $R^{(\varepsilon)}(T, \delta)$ (see Definition E.25 and Corollary E.27 below). Together with Corollary E.19 for the drift term (IV), this makes Sec. E.8 self-contained: the full asymptotic order of $R^{(\varepsilon)}(T, \delta)$ is explicit within Sec. E.8.

Corollary E.30 (Asymptotic form of Theorem E.26). *Let $R^{(\varepsilon)}(T, \delta)$ be the regret upper bound defined in (112). Then for every $T \geq 3$,*

$$R^{(\varepsilon)}(T, \delta) \leq \underbrace{\frac{1}{1 - \gamma_{\max}} \sqrt{2T \log \frac{2}{\delta}}}_{(I)} + \underbrace{\sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t}}_{(II)} + \underbrace{2 \sum_{t=1}^T \eta_t(\hat{g}_t)}_{(III)} + \underbrace{2 \sum_{t=1}^T \text{Drift}_t}_{(IV)}. \quad (190)$$

Assume the anytime schedules (80)–(82). Let L_T be as in (136), and recall that $V_T^{(\varepsilon)}$ is defined in (47). Then there exist absolute constants $C_1, C_2 > 0$ (depending only on universal numeric constants in Sec. E.8) such that, for every $T \geq 3$ satisfying $\lceil \sqrt{T} \rceil > \mathcal{W}_T$,

$$\begin{aligned} R^{(\varepsilon)}(T, \delta) &\leq C_1 \sqrt{T \log \frac{2}{\delta}} + C_2 \left(\frac{T}{(\log(e+T))^{\kappa/2}} + \frac{T}{(\log(e+T))^{1+\kappa/2}} + \sqrt{T} \right) \\ &\quad + 2C_\eta L_T \left[(V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right] \\ &\quad + 2 \lceil \sqrt{T} \rceil + \frac{2C'_{\text{drift}}}{(1 - \lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2(\lceil \sqrt{T} \rceil - \mathcal{W}_T)^2}{\delta}\right)}. \end{aligned} \quad (191)$$

In particular, under the anytime schedule,

$$1 - \lambda_t = \frac{1}{(\log(e+t))^{1+\kappa}} \implies (1 - \lambda_T)^{-2} = (\log(e+T))^{2+2\kappa}. \quad (192)$$

Moreover, if $V_T^{(\varepsilon)}$ satisfies

$$V_T^{(\varepsilon)} = o \left(\min \left\{ \frac{T}{L_T^3 (\log(e+T))^{3\kappa/4}}, \frac{T \log\left(\frac{G\pi^2(\lceil \sqrt{T} \rceil - \mathcal{W}_T)^2}{\delta}\right)}{(\log(e+T))^{2+2\kappa}} \right\} \right), \quad (193)$$

then

$$R^{(\varepsilon)}(T, \delta) = o(T), \quad \text{as } T \rightarrow \infty. \quad (194)$$

Proof. We bound each term in (190).

Term (I). Under (82), we always have $\gamma_t \leq \frac{1}{2}$, hence $\gamma_{\max} \leq \frac{1}{2}$ and $(1 - \gamma_{\max})^{-1} \leq 2$. Therefore

$$(I) = \frac{1}{1 - \gamma_{\max}} \sqrt{2T \log \frac{2}{\delta}} \leq 2 \sqrt{2T \log \frac{2}{\delta}} \leq C_1 \sqrt{T \log \frac{2}{\delta}}$$

for an absolute C_1 .

Term (II). By Corollary E.27, for every $T \geq 3$,

$$(II) = \sum_{t=1}^T \frac{\gamma_t}{1 - \gamma_t} \leq C_2 \left(\frac{T}{(\log(e+T))^{\kappa/2}} + \frac{T}{(\log(e+T))^{1+\kappa/2}} + \sqrt{T} \right),$$

where C_2 depends only on (c, κ) and the fixed logarithmic constants in the schedule.

Term (III). Proposition E.28 states that, on the same high-probability event as Theorem E.26,

$$\sum_{t=1}^T \eta_t(\hat{g}_t) \leq C_\eta L_T \left[(V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right].$$

Multiplying both sides by 2 yields

$$(III) = 2 \sum_{t=1}^T \eta_t(\hat{g}_t) \leq 2C_\eta L_T \left[(V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right].$$

Term (IV). We invoke Corollary E.19 with the concrete choice $t_0(T) = \lceil \sqrt{T} \rceil$. On the event of Lemma E.18, for every $T \geq 3$ such that $\lceil \sqrt{T} \rceil > \mathcal{W}_T$,

$$\sum_{t=1}^T \text{Drift}_t \leq \lceil \sqrt{T} \rceil + \frac{C'_{\text{drift}}}{(1-\lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2(\lceil \sqrt{T} \rceil - \mathcal{W}_T)^2}{\delta}\right)}.$$

Multiplying by 2 yields

$$(IV) = 2 \sum_{t=1}^T \text{Drift}_t \leq 2 \lceil \sqrt{T} \rceil + \frac{2C'_{\text{drift}}}{(1-\lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2(\lceil \sqrt{T} \rceil - \mathcal{W}_T)^2}{\delta}\right)}.$$

Combine. Summing the four bounds for (I)–(IV) gives (191). The identity (192) follows directly from $1 - \lambda_t = (\log(e+t))^{-(1+\kappa)}$.

Finally, under (193):

- (I) = $O(\sqrt{T \log(1/\delta)}) = o(T)$.
- (II) = $O(T/(\log T)^{\kappa/2}) + O(T/(\log T)^{1+\kappa/2}) + O(\sqrt{T}) = o(T)$ since $\kappa > 0$.
- For (III), note that $L_T \rightarrow \infty$ and the second summand $2C_\eta L_T \cdot T/(\log(e+T))^{\kappa/4}$ is $o(T)$ because $\kappa > 0$. Moreover, the first summand in (III) satisfies

$$\frac{(V_T^{(\varepsilon)})^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} L_T}{T} = \left(\frac{V_T^{(\varepsilon)} L_T^3 (\log(e+T))^{3\kappa/4}}{T} \right)^{1/3} \rightarrow 0$$

under the first condition in (193).

- For (IV), we have $2\lceil \sqrt{T} \rceil = o(T)$. Using (192), the remaining part satisfies

$$\frac{1}{T} \cdot \frac{2C'_{\text{drift}}}{(1-\lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2(\lceil \sqrt{T} \rceil - \mathcal{W}_T)^2}{\delta}\right)} = \frac{2C'_{\text{drift}}}{\log\left(\frac{G\pi^2(\lceil \sqrt{T} \rceil - \mathcal{W}_T)^2}{\delta}\right)} \cdot \frac{(\log(e+T))^{2+2\kappa} V_T^{(\varepsilon)}}{T},$$

and the second condition in (193) makes the right-hand side vanish.

Hence every term is $o(T)$, proving (194). $\square$

Remark E.31 (Takeaway: a clean sufficient system for $R^{(\varepsilon)}(T, \delta) = o(T)$). Under the anytime schedules (80)–(82), the four contributions in (190) admit the following asymptotic scales:

$$(I) = O\left(\sqrt{T \log \frac{1}{\delta}}\right) = o(T), \quad (II) = O\left(\frac{T}{(\log T)^{\kappa/2}}\right) = o(T).$$

Moreover, since $L_T = \sqrt{\log(\frac{8GT}{\delta})} = \Theta(\sqrt{\log T})$ and $(\log(e+T))^{\kappa/4} = \Theta((\log T)^{\kappa/4})$, term (III) admits the intuitive scale

$$(III) = O\left((V_T^{(\varepsilon)})^{1/3} T^{2/3} (\log T)^{\frac{1}{2} + \frac{\kappa}{4}}\right) = O\left((V_T^{(\varepsilon)} T^2 (\log T)^{\frac{3}{2} + \frac{3\kappa}{4}})^{1/3}\right),$$

while the additional summand $O(T L_T / (\log T)^{\kappa/4})$ is $o(T)$.

Likewise, using $(1 - \lambda_T)^{-2} = (\log(e + T))^{2+2\kappa}$ and $\log(\frac{G\pi^2(\lceil\sqrt{T}\rceil - \mathcal{W}_T)^2}{\delta}) = \Theta(\log T)$, the drift term satisfies

$$(IV) = O(\sqrt{T}) + O\left(\frac{(\log T)^{2+2\kappa}}{\log(\frac{G\pi^2(\lceil\sqrt{T}\rceil - \mathcal{W}_T)^2}{\delta})} V_T^{(\varepsilon)}\right) = O(\sqrt{T}) + O\left(V_T^{(\varepsilon)} (\log T)^{1+2\kappa}\right).$$

Hence a readable sufficient condition for $R^{(\varepsilon)}(T, \delta) = o(T)$ is

$$V_T^{(\varepsilon)} = o\left(\frac{T}{L_T^3 (\log(e+T))^{3\kappa/4}}\right) \quad \text{and} \quad V_T^{(\varepsilon)} = o\left(\frac{T \log(\frac{G\pi^2(\lceil\sqrt{T}\rceil - \mathcal{W}_T)^2}{\delta})}{(\log(e+T))^{2+2\kappa}}\right),$$

which exactly makes (III)/ $T \rightarrow 0$ and the nontrivial part of (IV)/ $T \rightarrow 0$, i.e., (III) = $o(T)$ and (IV) = $o(T)$.

F. Recovering Oracle-rate Linear Convergence via Regret Controlling

This section formalizes how a high-probability dynamic regret guarantee for the bandit selector can be turned into an oracle-rate (linear) convergence statement for the coupled optimization dynamics. The key idea is to lower bound the cumulative effective progress $\sum_t \alpha_t$ (or its regularized version) by relating it to the realized bandit rewards plus a martingale correction term.

F.1. Setup and the target form

We first recall the supermartingale/Ville interface and identify the exact quantity that needs to be controlled: a *pathwise* lower bound on the cumulative progress parameter $\sum_t \alpha_t$. This provides the target form that subsequent subsections will connect to bandit regret.

Recall from the supermartingale/Ville argument (Theorem D.5) that if for some constant $c_U > 0$ and a predictable sequence $\{\alpha_t\}_{t \geq 1}$,

$$\mathbb{E}[\Delta_{t+1} \mid \mathcal{F}_t] \leq \exp(-c_U \alpha_t) \Delta_t, \quad \forall t \geq 1, \quad (195)$$

then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$, simultaneously for all $T \geq 2$,

$$\Delta_T \leq \frac{\Delta_0}{\delta} \exp\left(-c_U \sum_{t=1}^{T-1} \alpha_t\right). \quad (196)$$

Hence it suffices to obtain a *pathwise* lower bound on $\sum_{t=1}^{T-1} \alpha_t$.

In our bandit-coupled design, the bandit objective is aligned with $\sum_t \alpha_t$ through normalized rewards. Since we regularize the normalization by $\varepsilon > 0$ for stability, we work with ε -regularized quantities.

F.2. Regularized normalized rewards and their link to $\alpha_t^{(\varepsilon)}$

We next introduce the ε -regularized normalized rewards and define $\alpha_t^{(\varepsilon)}$ as a scaled conditional expectation under the sampling distribution. This subsection isolates the only place where regularization enters the convergence argument, and explains why working with $\sum_t \alpha_t^{(\varepsilon)}$ remains sufficient for the Ville bound.

Let $r_t^{(\varepsilon)}(g) \in [0, 1]$ denote the (possibly random) regularized normalized reward used by the selector, and let $\mathbb{P}_t(\cdot)$ be the bandit sampling distribution at time t . Define

$$\mu_t^{(\varepsilon)} := \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}], \quad \alpha_t^{(\varepsilon)} := G \mu_t^{(\varepsilon)}, \quad t \geq 1, \quad (197)$$

where $g_t \in [G]$ is the group selected at time t . Define the dynamic best arm and oracle reward:

$$g_t^* \in \arg \max_{g \in [G]} r_t^{(\varepsilon)}(g), \quad r_{t,\star}^{(\varepsilon)} := r_t^{(\varepsilon)}(g_t^*), \quad \alpha_{t,\text{orc}}^{(\varepsilon)} := G r_{t,\star}^{(\varepsilon)}, \quad (198)$$

for $t \geq 1$.

Remark F.1 (Using $\alpha_t^{(\varepsilon)}$ inside the supermartingale bound). Suppose the original analysis provides (196) with the unregularized $\{\alpha_t\}$, and moreover the regularization satisfies

$$\alpha_t^{(\varepsilon)} = \alpha_t \cdot \frac{W_t}{W_t + \varepsilon}, \quad W_t := \sum_{j=1}^G w_j(\theta_t),$$

which is the standard relationship induced by ε -regularized normalization. Then $\alpha_t^{(\varepsilon)} \leq \alpha_t$ for all t , and therefore

$$\exp\left(-c_U \sum_{t=1}^{T-1} \alpha_t\right) \leq \exp\left(-c_U \sum_{t=1}^{T-1} \alpha_t^{(\varepsilon)}\right), \quad (199)$$

so (196) remains valid (possibly weaker) if we replace $\sum \alpha_t$ by $\sum \alpha_t^{(\varepsilon)}$.

Combining (196) and Remark F.1, it is enough to lower bound $\sum_{t=1}^{T-1} \alpha_t^{(\varepsilon)}$ pathwise.

F.3. High-probability dynamic regret and a martingale bridge

We now connect the oracle cumulative progress $\sum_t \alpha_{t,\text{orc}}^{(\varepsilon)}$ to the algorithmic cumulative progress $\sum_t \alpha_t^{(\varepsilon)}$ through a dynamic regret term. The only remaining gap is that $\alpha_t^{(\varepsilon)}$ involves conditional expectations, while the regret is defined in terms of realized rewards; this is handled by a standard martingale concentration inequality stated below.

Define the *realized* (pathwise) dynamic regret random variable

$$\widehat{\text{Reg}}_T^{(\varepsilon)} := \sum_{t=1}^T (r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t)). \quad (200)$$

Assumption F.2 (High-probability dynamic regret bound). There exists a function $\mathcal{R}^{(\varepsilon)}(T, \delta)$ such that for all $T \geq 1$ and $\delta \in (0, 1)$,

$$\mathbb{P}\left(\widehat{\text{Reg}}_T^{(\varepsilon)} \leq \mathcal{R}^{(\varepsilon)}(T, \delta)\right) \geq 1 - \delta. \quad (201)$$

Next, to connect conditional expectations (which define $\alpha_t^{(\varepsilon)}$) to realized rewards, we use a standard bounded martingale difference concentration.

Lemma F.3 (Azuma–Hoeffding for bounded martingale differences (two-sided, standard constants)). *Let $X_t \in [0, 1]$ be $\mathcal{F}_t$ -measurable for $t = 1, \dots, T$, and let $\mu_t := \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$. Define the martingale difference $D_t := X_t - \mu_t$. Then $\{D_t\}_{t=1}^T$ is a martingale difference sequence and $|D_t| \leq 1$ almost surely. For any $\delta \in (0, 1)$ and any integer $T \geq 1$,*

$$\mathbb{P}\left(\left|\sum_{t=1}^T D_t\right| \geq \sqrt{2T \log \frac{2}{\delta}}\right) \leq \delta. \quad (202)$$

Equivalently, with probability at least $1 - \delta$,

$$\sum_{t=1}^T \mu_t \geq \sum_{t=1}^T X_t - \sqrt{2T \log \frac{2}{\delta}}. \quad (203)$$

Proof. Proof sketch. We apply Azuma–Hoeffding to the martingale difference sequence $D_t := X_t - \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$. The boundedness $X_t \in [0, 1]$ implies $|D_t| \leq 1$, yielding a sub-Gaussian tail for $\sum_{t=1}^T D_t$. A two-sided bound follows by applying the one-sided inequality to both $\sum D_t$ and $-\sum D_t$, and then taking a union bound. Finally, we rearrange $\sum D_t = \sum X_t - \sum \mu_t$ to obtain (203).

Since $\mathbb{E}[D_t \mid \mathcal{F}_{t-1}] = 0$ and $|D_t| \leq 1$ a.s., Azuma–Hoeffding gives

$$\mathbb{P}\left(\sum_{t=1}^T D_t \geq a\right) \leq \exp\left(-\frac{a^2}{2T}\right), \quad \mathbb{P}\left(\sum_{t=1}^T (-D_t) \geq a\right) \leq \exp\left(-\frac{a^2}{2T}\right).$$

By a union bound,

$$\mathbb{P}\left(\left|\sum_{t=1}^T D_t\right| \geq a\right) \leq 2 \exp\left(-\frac{a^2}{2T}\right).$$

Setting $a := \sqrt{2T \log(2/\delta)}$ yields (202). Rearranging $\sum_{t=1}^T D_t = \sum_{t=1}^T X_t - \sum_{t=1}^T \mu_t$ gives (203), which concludes the proof. $\square$

F.4. Bandit reduction: regularized normalized reward and regret-to-cumulative coverage

This subsection formalizes the reduction from the group-selection bandit to a cumulative *coverage* quantity. We first define group energies and the induced ε -regularized normalized reward, then introduce coverage coefficients and show how (pseudo-)regret controls the cumulative coverage in expectation.

Recall: group energies. For a parameter vector θ partitioned into G groups, define the nonnegative group energy

$$w_g(\theta) := \frac{\|\nabla_g \mathcal{L}(\theta)\|_2^2}{L_g} \geq 0, \quad W(\theta) := \sum_{j=1}^G w_j(\theta) \geq 0.$$

At time t , write $\theta_t := \theta(t)$, $w_t(g) := w_g(\theta_t)$, and $W_t := W(\theta_t)$.

Regularized normalized reward. Fix a regularization parameter $\varepsilon > 0$. Define the *regularized normalized reward* at time t as

$$r_t^{(\varepsilon)}(g) := \frac{w_t(g)}{W_t + \varepsilon}, \quad g \in [G], \quad (204)$$

and the instantaneous oracle reward

$$r_{t,\star}^{(\varepsilon)} := \max_{g \in [G]} r_t^{(\varepsilon)}(g). \quad (205)$$

Note that $0 \leq r_t^{(\varepsilon)}(g) \leq 1$ since $0 \leq w_t(g) \leq W_t \leq W_t + \varepsilon$.

Filtration and the online selector. Let $\mathcal{F}_{t-1}$ denote the sigma-field generated by all randomness up to time $t-1$ (including the optimization iterates), *before* sampling the group index g_t at time t . The online selector chooses a distribution $\mathbb{P}_t(\cdot)$ that is $\mathcal{F}_{t-1}$ -measurable, and then samples

$$g_t \sim \mathbb{P}_t(\cdot) \quad \text{conditional on } \mathcal{F}_{t-1}.$$

F.4.1. COVERAGE COEFFICIENTS AND THEIR ORACLE COUNTERPARTS

We introduce the coverage coefficients induced by the online selector and their oracle counterparts, and record a notational correspondence that will be repeatedly used in subsequent derivations.

Definition F.4 (Coverage coefficient induced by the online selector). Define the (regularized) coverage coefficient at time t by the conditional expectation

$$\alpha_t^{(\varepsilon)} := G \cdot \mathbb{E}\left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}\right] = G \sum_{g=1}^G \mathbb{P}_t(g) r_t^{(\varepsilon)}(g). \quad (206)$$

Define also the oracle coverage coefficient

$$\alpha_{t,\star}^{(\varepsilon)} := G \cdot r_{t,\star}^{(\varepsilon)} = G \max_{g \in [G]} r_t^{(\varepsilon)}(g). \quad (207)$$

Lemma F.5 (Exact correspondence between reward and coverage). *For every t ,*

$$\mathbb{E}\left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}\right] = \frac{1}{G} \alpha_t^{(\varepsilon)}, \quad r_{t,\star}^{(\varepsilon)} = \frac{1}{G} \alpha_{t,\star}^{(\varepsilon)}. \quad (208)$$

Proof. The first identity is exactly (206) rewritten:

$$\alpha_t^{(\varepsilon)} = G \cdot \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] \iff \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] = \frac{1}{G} \alpha_t^{(\varepsilon)}.$$

The second identity follows directly from (207):

$$\alpha_{t,\star}^{(\varepsilon)} = G r_{t,\star}^{(\varepsilon)} \iff r_{t,\star}^{(\varepsilon)} = \frac{1}{G} \alpha_{t,\star}^{(\varepsilon)}.$$

This proves (208). $\square$

F.4.2. REGRET DEFINITIONS AND A DETERMINISTIC IDENTITY

We next define the pathwise regret and the pseudo-regret, and then show a deterministic identity that links the expected cumulative coverage to the pseudo-regret.

Definition F.6 (Pathwise regret and pseudo-regret). Define the *pathwise* dynamic regret

$$\widehat{\text{Reg}}_T^{(\varepsilon)} := \sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t) \right), \quad (209)$$

and the (dynamic) *pseudo-regret*

$$\text{Regret}(T) := \sum_{t=1}^T \mathbb{E} \left[r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t) \right] = \sum_{t=1}^T \mathbb{E} \left[r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right]. \quad (210)$$

Theorem F.7 (Regret-to-cumulative-coverage identity (in expectation)). *For all $T \geq 1$,*

$$\sum_{t=1}^T \mathbb{E} \left[\alpha_t^{(\varepsilon)} \right] = \sum_{t=1}^T \mathbb{E} \left[\alpha_{t,\star}^{(\varepsilon)} \right] - G \cdot \text{Regret}(T). \quad (211)$$

Proof. Start from the definition of pseudo-regret in (210):

$$\text{Regret}(T) = \sum_{t=1}^T \mathbb{E} \left[r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right].$$

Apply Lemma F.5 (equivalently (208)) inside the expectation:

$$r_{t,\star}^{(\varepsilon)} = \frac{1}{G} \alpha_{t,\star}^{(\varepsilon)}, \quad \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] = \frac{1}{G} \alpha_t^{(\varepsilon)}.$$

Therefore,

$$\text{Regret}(T) = \sum_{t=1}^T \mathbb{E} \left[\frac{1}{G} \alpha_{t,\star}^{(\varepsilon)} - \frac{1}{G} \alpha_t^{(\varepsilon)} \right] = \frac{1}{G} \sum_{t=1}^T \mathbb{E} \left[\alpha_{t,\star}^{(\varepsilon)} \right] - \frac{1}{G} \sum_{t=1}^T \mathbb{E} \left[\alpha_t^{(\varepsilon)} \right].$$

Multiplying both sides by G and rearranging gives (211), which is exactly the claimed identity. $\square$

F.4.3. A HIGH-PROBABILITY *pathwise* LOWER BOUND ON $\sum_t \alpha_t^{(\varepsilon)}$

The expectation identity above is not sufficient for our later high-probability contraction analysis. We next derive a *pathwise* lower bound on $\sum_{t=1}^T \alpha_t^{(\varepsilon)}$ from a high-probability bound on $\widehat{\text{Reg}}_T^{(\varepsilon)}$.

Lemma F.8 (Azuma conversion: conditional regret vs pathwise regret). *Define the martingale difference sequence*

$$Z_t := r_t^{(\varepsilon)}(g_t) - \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right], \quad t = 1, \dots, T.$$

Then $\mathbb{E}[Z_t \mid \mathcal{F}_{t-1}] = 0$ and $|Z_t| \leq 1$ almost surely. Consequently, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\sum_{t=1}^T Z_t \leq \sqrt{2T \log \frac{1}{\delta}}. \quad (212)$$

Moreover, on the same event,

$$\sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right) \leq \widehat{\text{Reg}}_T^{(\varepsilon)} + \sqrt{2T \log \frac{1}{\delta}}. \quad (213)$$

Proof. Proof sketch. We introduce the deviation terms $Z_t = r_t^{(\varepsilon)}(g_t) - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}]$, which form bounded martingale differences because $r_t^{(\varepsilon)}(\cdot) \in [0, 1]$. Azuma–Hoeffding then provides a high-probability upper bound on $\sum_{t=1}^T Z_t$. Finally, we use the exact decomposition $r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] = (r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t)) - Z_t$ to relate the conditional-regret sum to the realized (pathwise) regret.

Step 1: martingale properties and boundedness. By construction,

$$\mathbb{E}[Z_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] = 0.$$

Also, since $0 \leq r_t^{(\varepsilon)}(g) \leq 1$ for all $g \in [G]$, we have $r_t^{(\varepsilon)}(g_t) \in [0, 1]$ and $\mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \in [0, 1]$, hence

$$|Z_t| = \left| r_t^{(\varepsilon)}(g_t) - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right| \leq 1.$$

Step 2: Azuma–Hoeffding. Let $M_T := \sum_{t=1}^T Z_t$. Since $\{Z_t\}$ is a martingale difference sequence with $|Z_t| \leq 1$, Azuma–Hoeffding yields, for any $a > 0$,

$$\mathbb{P}(M_T \geq a) \leq \exp\left(-\frac{a^2}{2 \sum_{t=1}^T 1^2}\right) = \exp\left(-\frac{a^2}{2T}\right).$$

Set $a := \sqrt{2T \log(1/\delta)}$ to obtain (212).

Step 3: relate conditional regret and pathwise regret. For every t ,

$$\begin{aligned} r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] &= (r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t)) + (r_t^{(\varepsilon)}(g_t) - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}]) \\ &= (r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t)) + Z_t. \end{aligned}$$

Summing over $t = 1, \dots, T$ gives the exact identity

$$\sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right) = \widehat{\text{Reg}}_T^{(\varepsilon)} + \sum_{t=1}^T Z_t.$$

On the event where (212) holds, we have $\sum_{t=1}^T Z_t \leq \sqrt{2T \log(1/\delta)}$, and plugging this into the above identity yields (213), which concludes the proof. $\square$

Theorem F.9 (High-probability cumulative coverage lower bound (true coverage)). *Assume that for some $\delta \in (0, 1)$ and the regret interface $\mathcal{R}(T, \delta) = R^{(\varepsilon)}(T, \delta) \geq 0$ from Definition E.25, the online selector satisfies the high-probability pathwise regret bound*

$$\mathbb{P}\left(\widehat{\text{Reg}}_T^{(\varepsilon)} \leq \mathcal{R}(T, \delta/2)\right) \geq 1 - \delta/2. \quad (214)$$

Then with probability at least $1 - \delta$,

$$\sum_{t=1}^T \alpha_t^{(\varepsilon)} \geq \sum_{t=1}^T \alpha_{t,\star}^{(\varepsilon)} - G \mathcal{R}(T, \delta/2) - G \sqrt{2T \log \frac{2}{\delta}}. \quad (215)$$

Proof. Proof sketch. We start from the definition $\alpha_t^{(\varepsilon)} = G \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}]$ and rewrite the cumulative coverage as the oracle cumulative coverage minus a cumulative conditional-regret term. We then express this conditional-regret term as the realized regret $\widehat{\text{Reg}}_T^{(\varepsilon)}$ plus a martingale deviation $\sum_{t=1}^T Z_t$. Finally, we combine the assumed high-probability bound on $\widehat{\text{Reg}}_T^{(\varepsilon)}$ with an Azuma–Hoeffding bound on $\sum_{t=1}^T Z_t$ and apply a union bound.

Step 1 (rewrite coverage using conditional expected reward). By Definition F.4, for each $t \in \{1, \dots, T\}$,

$$\alpha_t^{(\varepsilon)} = G \cdot \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right], \quad \alpha_{t,\star}^{(\varepsilon)} = G r_{t,\star}^{(\varepsilon)}. \quad (216)$$

Summing (216) over $t = 1, \dots, T$ gives

$$\begin{aligned} \sum_{t=1}^T \alpha_t^{(\varepsilon)} &= G \sum_{t=1}^T \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] \\ &= G \sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - \left(r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right) \right) \\ &= G \sum_{t=1}^T r_{t,\star}^{(\varepsilon)} - G \sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right) \\ &= \sum_{t=1}^T \alpha_{t,\star}^{(\varepsilon)} - G \sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right). \end{aligned} \quad (217)$$

Step 2 (convert the conditional-regret sum into pathwise regret plus a martingale deviation). Recall the pathwise dynamic regret

$$\widehat{\text{Reg}}_T^{(\varepsilon)} := \sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t) \right),$$

and define the martingale difference sequence

$$Z_t := r_t^{(\varepsilon)}(g_t) - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}], \quad t = 1, \dots, T.$$

Then for every t ,

$$r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] = (r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t)) + (r_t^{(\varepsilon)}(g_t) - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}]) = (r_{t,\star}^{(\varepsilon)} - r_t^{(\varepsilon)}(g_t)) + Z_t.$$

Summing over $t = 1, \dots, T$ yields the exact identity

$$\sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right) = \widehat{\text{Reg}}_T^{(\varepsilon)} + \sum_{t=1}^T Z_t. \quad (218)$$

Step 3 (apply the assumed regret event and Azuma–Hoeffding, then union bound). By assumption (214),

$$\mathbb{P} \left(\widehat{\text{Reg}}_T^{(\varepsilon)} \leq \mathcal{R}(T, \delta/2) \right) \geq 1 - \delta/2.$$

Moreover, applying Lemma F.8 with confidence parameter $\delta/2$ gives

$$\mathbb{P} \left(\sum_{t=1}^T Z_t \leq \sqrt{2T \log \frac{2}{\delta}} \right) \geq 1 - \delta/2.$$

Therefore, by a union bound, with probability at least $1 - \delta$ both events hold simultaneously, hence by (218),

$$\sum_{t=1}^T \left(r_{t,\star}^{(\varepsilon)} - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right) \leq \mathcal{R}(T, \delta/2) + \sqrt{2T \log \frac{2}{\delta}}.$$

Plugging this inequality into (217) gives exactly (215), which concludes the proof. $\square$

F.4.4. INCORPORATING BIASED FEEDBACK $\hat{r}_t^{(\varepsilon)}$ VIA A ζ_t TERM

In practice, the bandit update may use an observable feedback $\hat{r}_t^{(\varepsilon)}(g_t)$ that is *biased* relative to the target reward $r_t^{(\varepsilon)}(g_t)$. We now state a minimal and fully checkable way to propagate this bias into the cumulative coverage guarantee.

Assumption F.10 (Conditional mean mismatch controlled by ζ_t). There exists a deterministic sequence $\{\zeta_t\}_{t=1}^T$ with $\zeta_t \geq 0$ such that almost surely,

$$\left| \mathbb{E} \left[\hat{r}_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] - \mathbb{E} \left[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right] \right| \leq \zeta_t \quad \text{for all } t = 1, \dots, T. \quad (219)$$

Remark F.11 (On whether ζ_t must be deterministic). Assumption F.10 is stated with a deterministic upper bound sequence $\{\zeta_t\}$ only for presentation clarity. All subsequent arguments use (219) in a *pathwise* way, hence the same pipeline remains valid if ζ_t is merely $\mathcal{F}_{t-1}$ -measurable (predictable), or if one proves that (219) holds on a high-probability event with a deterministic envelope. Verifiable sufficient conditions are provided in Sec. G, in particular Sec. G.3.

Definition F.12 (Practical coverage coefficient induced by the observable feedback). Define

$$\hat{\alpha}_t^{(\varepsilon)} := G \cdot \mathbb{E} \left[\hat{r}_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1} \right]. \quad (220)$$

Lemma F.13 (Bias propagation to cumulative coverage). Under Assumption F.10, for every sample path,

$$\sum_{t=1}^T \hat{\alpha}_t^{(\varepsilon)} \geq \sum_{t=1}^T \alpha_t^{(\varepsilon)} - G \sum_{t=1}^T \zeta_t. \quad (221)$$

Proof. Fix any $t \in \{1, \dots, T\}$. By Definitions F.4 and F.12,

$$\hat{\alpha}_t^{(\varepsilon)} - \alpha_t^{(\varepsilon)} = G \left(\mathbb{E}[\hat{r}_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right).$$

Assumption F.10 implies

$$\mathbb{E}[\hat{r}_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \geq -\zeta_t,$$

hence

$$\hat{\alpha}_t^{(\varepsilon)} - \alpha_t^{(\varepsilon)} \geq -G\zeta_t \quad \Longleftrightarrow \quad \hat{\alpha}_t^{(\varepsilon)} \geq \alpha_t^{(\varepsilon)} - G\zeta_t.$$

Summing the last inequality over $t = 1, \dots, T$ yields (221), which concludes the proof. $\square$

Corollary F.14 (High-probability cumulative coverage lower bound (practical coverage)). Under the assumptions of Theorem F.9 and Assumption F.10, with probability at least $1 - \delta$,

$$\sum_{t=1}^T \hat{\alpha}_t^{(\varepsilon)} \geq \sum_{t=1}^T \alpha_{t,\star}^{(\varepsilon)} - G \mathcal{R}(T, \delta/2) - G \sqrt{2T \log \frac{2}{\delta}} - G \sum_{t=1}^T \zeta_t. \quad (222)$$

Proof. By Lemma F.13 (mismatch control), for every sample path,

$$\sum_{t=1}^T \hat{\alpha}_t^{(\varepsilon)} \geq \sum_{t=1}^T \alpha_t^{(\varepsilon)} - G \sum_{t=1}^T \zeta_t. \quad (223)$$

On the event of Theorem F.9, we have

$$\sum_{t=1}^T \alpha_t^{(\varepsilon)} \geq \sum_{t=1}^T \alpha_{t,\star}^{(\varepsilon)} - G \mathcal{R}(T, \delta/2) - G \sqrt{2T \log \frac{2}{\delta}}.$$

Combining this inequality with (223) yields (222), which concludes the proof. $\square$

Remark F.15 (Why this is the right place for ζ_t). The only role of ζ_t in (222) is to account for the gap between the *conditional mean* of the observable feedback $\hat{r}_t^{(\varepsilon)}(g_t)$ and that of the target reward $r_t^{(\varepsilon)}(g_t)$. Once (219) is verified for the specific estimator (e.g., EMA-based normalization), the subsequent regret-to-coverage-to-rate pipeline remains unchanged, with an extra additive term $G \sum_{t=1}^T \zeta_t$.

F.5. Main theorem: high-probability oracle-rate exponential gap bound

This subsection combines the two high-probability ingredients established earlier: (i) the Ville/supermartingale bound that turns a lower bound on cumulative coverage into an exponential gap decay, and (ii) the regret-to-coverage lower bound (with the bias correction via ζ_t). Putting them together yields a single oracle-rate exponential convergence statement with an explicit exponent.

Theorem F.16 (High-probability oracle-rate exponential convergence (no extra constant factor)). *Assume the one-step supermartingale contraction condition of Theorem D.5 holds with constant $c_U > 0$, namely (195). Fix any horizon $T \geq 2$ and confidence level $\delta \in (0, 1)$. Suppose (i) Assumption F.2 holds with regret function $\mathcal{R}^{(\varepsilon)}(T, \cdot)$, and (ii) Assumption F.10 holds with sequence $\{\zeta_t\}_{t=1}^{T-1}$. Then, with probability at least $1 - \delta$,*

$$\Delta_T \leq \frac{2\Delta_0}{\delta} \exp \left(-c_U \sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} + c_U G \mathcal{R}^{(\varepsilon)} \left(T, \frac{\delta}{4} \right) + c_U G \sqrt{2T \log \frac{4}{\delta}} + c_U G \sum_{t=0}^{T-1} \zeta_t \right). \quad (224)$$

Proof. Proof sketch. We intersect (a) the Ville/supermartingale event that upper bounds Δ_T in terms of $\sum_{t=0}^{T-1} \alpha_t^{(\varepsilon)}$, and (b) the high-probability cumulative-coverage lower bound that lower bounds $\sum_{t=0}^{T-1} \alpha_t^{(\varepsilon)}$ by the oracle coverage minus regret and bias terms. Substituting (b) into (a) yields the stated exponential bound.

Step 1: Ville/supermartingale event at confidence $\delta/2$. By Theorem D.5, with probability at least $1 - \delta/2$,

$$\Delta_T \leq \frac{\Delta_0}{\delta/2} \exp \left(-c_U \sum_{t=0}^{T-1} \alpha_t^{(\varepsilon)} \right) = \frac{2\Delta_0}{\delta} \exp \left(-c_U \sum_{t=0}^{T-1} \alpha_t^{(\varepsilon)} \right). \quad (225)$$

Denote this event by E_{sm} .

Step 2: practical coverage lower bound at confidence $\delta/2$. Apply Corollary F.14 with confidence parameter $\delta/2$. Then, with probability at least $1 - \delta/2$,

$$\sum_{t=0}^{T-1} \alpha_t^{(\varepsilon)} \geq \sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} - G \mathcal{R}^{(\varepsilon)} \left(T, \frac{\delta}{4} \right) - G \sqrt{2T \log \frac{4}{\delta}} - G \sum_{t=0}^{T-1} \zeta_t. \quad (226)$$

Denote this event by E_{cov} .

Step 3: intersect and combine. By a union bound,

$$\mathbb{P}(E_{\text{sm}} \cap E_{\text{cov}}) \geq 1 - \delta/2 - \delta/2 = 1 - \delta.$$

On $E_{\text{sm}} \cap E_{\text{cov}}$, substitute (226) into (225):

$$\begin{aligned} \Delta_T &\leq \frac{2\Delta_0}{\delta} \exp \left(-c_U \sum_{t=0}^{T-1} \alpha_t^{(\varepsilon)} \right) \\ &\leq \frac{2\Delta_0}{\delta} \exp \left(-c_U \sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} + c_U G \mathcal{R}^{(\varepsilon)} \left(T, \frac{\delta}{4} \right) + c_U G \sqrt{2T \log \frac{4}{\delta}} + c_U G \sum_{t=0}^{T-1} \zeta_t \right), \end{aligned}$$

which is exactly (224) and concludes the proof. $\square$

F.6. Corollaries: oracle linear rate is preserved under sublinear bandit errors

This subsection extracts an interpretable consequence of Theorem F.16. If the oracle cumulative coverage grows linearly while the bandit regret and the cumulative bias grow sublinearly, then the achieved exponential rate matches the oracle rate up to a vanishing perturbation.

Corollary F.17 (Oracle linear rate constant is preserved under sublinear errors). *Assume there exists a constant $\alpha > 0$ such that for all $T \geq 2$,*

$$\sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} \geq \alpha \cdot T. \quad (227)$$

Fix $\delta \in (0, 1)$. Suppose moreover that

$$\mathcal{R}^{(\varepsilon)}(T, \delta) = o(T), \quad \sum_{t=0}^{T-1} \zeta_t = o(T) \quad \text{as } T \rightarrow \infty. \quad (228)$$

Then Theorem F.16 implies that, with probability at least $1 - \delta$,

$$\Delta_T \leq \frac{2\Delta_0}{\delta} \exp\left(- (c_U \alpha - o(1))T\right), \quad \text{as } T \rightarrow \infty, \quad (229)$$

i.e., Δ_T decays geometrically with the same leading exponential constant $c_U \alpha$ as the oracle, up to a vanishing perturbation $o(1)$ in the rate.

Proof. Start from Theorem F.16:

$$\Delta_T \leq \frac{2\Delta_0}{\delta} \exp\left(-c_U \sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} + c_U G \mathcal{R}^{(\varepsilon)}\left(T, \frac{\delta}{4}\right) + c_U G \sqrt{2T \log \frac{4}{\delta}} + c_U G \sum_{t=0}^{T-1} \zeta_t\right).$$

By (227),

$$-c_U \sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} \leq -c_U \cdot \alpha \cdot T.$$

Divide the exponent by T :

$$\begin{aligned} \frac{1}{T} \left[-c_U \sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} + c_U G \mathcal{R}^{(\varepsilon)}\left(T, \frac{\delta}{4}\right) + c_U G \sqrt{2T \log \frac{4}{\delta}} + c_U G \sum_{t=0}^{T-1} \zeta_t \right] \\ \leq -c_U \alpha + c_U G \frac{\mathcal{R}^{(\varepsilon)}(T, \delta/4)}{T} + c_U G \sqrt{\frac{2 \log(4/\delta)}{T}} + c_U G \frac{1}{T} \sum_{t=0}^{T-1} \zeta_t. \end{aligned}$$

Under (228), the last three terms converge to 0 as $T \rightarrow \infty$. Hence the exponent equals $-(c_U \alpha - o(1))T$, proving (229) and concluding the proof. $\square$

F.7. Implementation to achieve the preserved oracle linear rate

This subsection instantiates the regret interface in Theorem F.16 under the anytime schedules (80)–(82). The goal is to obtain a fully explicit high-probability oracle-gap bound whose exponent separates the oracle linear term from sublinear perturbations.

Lemma F.18 (A rigorous $o(T)$ condition in the presence of polylog factors). *Let $\{V_T\}_{T \geq 1}$ be any nonnegative sequence and let $\{L_T\}_{T \geq 1}$ be any positive sequence. Fix any exponent $\beta \geq 0$. If*

$$\frac{V_T}{T} L_T^3 (\log(e+T))^{3\beta} \xrightarrow{T \rightarrow \infty} 0, \quad (230)$$

then

$$(V_T)^{1/3} T^{2/3} L_T (\log(e+T))^\beta = o(T). \quad (231)$$

Proof. Consider the ratio to T :

$$\frac{(V_T)^{1/3} T^{2/3} L_T (\log(e+T))^\beta}{T} = \left(\frac{V_T}{T}\right)^{1/3} L_T (\log(e+T))^\beta = \left(\frac{V_T}{T} L_T^3 (\log(e+T))^{3\beta}\right)^{1/3}.$$

By (230), the quantity inside the parentheses converges to 0, hence the right-hand side converges to 0 as well. This proves (231) and concludes the proof. $\square$

Corollary F.19 (Instantiating the bandit term in Theorem F.16 under the anytime schedule). *Assume the conditions required by Theorem E.26 and Proposition E.28 hold, and adopt the anytime schedules (80)–(82). Let*

$$L_T(\delta) := \sqrt{\log\left(\frac{8GT}{\delta}\right)}.$$

Then there exist explicit constants $C_{\text{UCB},1}, C_{\text{UCB},2}, C_{\text{UCB},3} > 0$ (independent of T) such that, with probability at least $1 - \delta$,

$$\begin{aligned} R^{(\varepsilon)}(T, \delta) \leq & \underbrace{C_{\text{UCB},1} \sqrt{T \log \frac{2}{\delta}}}_{\text{(I) statistical}} + \underbrace{\frac{C_{\text{UCB},2} T}{(\log(e+T))^\kappa} + \frac{C_{\text{UCB},2} T}{(\log(e+T))^{1+\kappa}}}_{\text{(II) exploration}} + C_{\text{UCB},2} \sqrt{T} \\ & + \underbrace{C_{\text{UCB},3} L_T(\delta) \left[(V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right]}_{\text{(III) mismatch/confidence}} + \underbrace{\frac{8G C_{\text{drift}}}{(1-\lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)}}_{\text{(IV) drift}} \end{aligned} \quad (232)$$

Here $m_T := \lceil \sqrt{T} \rceil - \mathcal{W}_T$ is as defined in Corollary E.19.

Consequently, substituting (232) into Theorem F.16 yields the explicit oracle-gap bound: on the same event and on the supermartingale/Ville event used in Theorem F.16, with probability at least $1 - \delta$,

$$\Delta_T \leq \frac{2\Delta_0}{\delta} \exp \left(-c_U \sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} + c_U G R^{(\varepsilon)}\left(T, \frac{\delta}{4}\right) + c_U G \sqrt{2T \log \frac{4}{\delta}} + c_U G \sum_{t=0}^{T-1} \zeta_t \right), \quad (233)$$

where $R^{(\varepsilon)}(T, \delta/4)$ is further upper bounded by (232) with δ replaced by $\delta/4$.

Moreover, assume $\sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} = \Theta(T)$ and $\sum_{t=0}^{T-1} \zeta_t = o(T)$. If in addition

$$\frac{V_T^{(\varepsilon)}}{T} L_T(\delta)^3 (\log(e+T))^{3\kappa/4} \xrightarrow{T \rightarrow \infty} 0 \quad \text{and} \quad \frac{V_T^{(\varepsilon)}}{T} \cdot \frac{(\log(e+T))^{2+2\kappa}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)} \xrightarrow{T \rightarrow \infty} 0, \quad (234)$$

then every additive perturbation term in the exponent of (233) is $o(T)$, so the oracle exponential rate constant is preserved (cf. Corollary F.17).

Proof. Proof sketch. We upper bound the regret interface $R^{(\varepsilon)}(T, \delta)$ by combining the high-probability decomposition from Theorem E.26 with the confidence accumulation bound in Proposition E.28 and the drift control in Corollary E.19. We then plug the resulting explicit bound into Theorem F.16. Finally, we verify that each additive term in the exponent is $o(T)$ under (234).

Step 1 (instantiate $R^{(\varepsilon)}(T, \delta)$). The statistical and exploration contributions are those already separated in Corollary E.30. For the mismatch/confidence contribution, Proposition E.28 yields the two-term bound in (232)-(III), including both the $T^{2/3}$ term with factor $(\log(e+T))^{\kappa/4}$ and the additive $T(\log(e+T))^{-\kappa/4}$ term. The drift term (IV) follows from Corollary E.19 and retains the logarithmic denominator $\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)$, which is necessary for the inequality direction.

Step 2 (plug into Theorem F.16). Theorem F.16 applies to any valid high-probability regret interface. Substituting the bound (232) (with δ replaced by $\delta/2$) into (224) yields (233). We use c_U for the Ville constant to avoid conflict with the exploration-schedule parameter c .

Step 3 ($o(T)$ verification under (234)). Under (234), each term in (232) divided by T vanishes: (I) and (II) are standard sublinear terms for $\kappa > 0$, (III) follows from Lemma F.18 with $\beta = \kappa/4$ and the explicit additive $T(\log(e+T))^{-\kappa/4}$ term, and (IV) follows from the second limit. Together with $\sum_{t < T} \zeta_t = o(T)$, the exponent is $-\Theta(T) + o(T)$, so the oracle rate constant is preserved. This concludes the proof. $\square$

Remark F.20 (No constant-factor pollution of the linear rate). All terms other than $\sum_{t < T} \alpha_{t,\text{orc}}^{(\varepsilon)}$ appear additively in the exponent. Hence, as long as each perturbation term is $o(T)$, the leading linear term dominates and the oracle exponential rate constant is preserved.

Corollary F.21 (Readable asymptotic rate implied by Corollary F.19). *Assume the oracle coverage is linear: there exists $\alpha > 0$ such that for all $T \geq 1$,*

$$\sum_{t=0}^{T-1} \alpha_{t,\text{orc}}^{(\varepsilon)} \geq \alpha T.$$

Fix any $\delta \in (0, 1)$ and adopt the anytime schedules (80)–(82). Assume also $\sum_{t=0}^{T-1} \zeta_t = o(T)$. If, in addition, $V_T^{(\varepsilon)}$ satisfies the two-term sufficient system (234), then, with probability at least $1 - \delta$,

$$\Delta_T \leq \frac{2\Delta_0}{\delta} \exp\left(- (c_U \alpha - o(1))T\right), \quad (T \rightarrow \infty), \quad (235)$$

so the leading exponential constant $c_U \alpha$ matches the oracle and is unaffected.

Proof. Start from (233) and divide the exponent by T . Using $\sum_{t < T} \alpha_{t,\text{orc}}^{(\varepsilon)} \geq \alpha T$ gives the leading term $-c_U \alpha$. All remaining contributions are $o(1)$ after dividing by T under (234) and $\sum_{t < T} \zeta_t = o(T)$, hence the exponent equals $-(c_U \alpha - o(1))T$, which is exactly (235) and concludes the proof. $\square$

G. Theoretical Analysis on the Mismatch Sequence $\{\zeta_t\}$ and Proving $\sum_{t < T} \zeta_t = o(T)$

This section makes the mismatch term $\{\zeta_t\}$ in Sec. F fully verifiable for the concrete feedback used by the bandit selector. The core idea is to start from a pointwise control of the observable reward proxy and then translate it into a conditional-mean mismatch bound, which is exactly the form required by Assumption F.10. Finally, we identify sufficient conditions under which $\sum_{t < T} \zeta_t = o(T)$.

G.1. Notation comparison: $\eta_t(\cdot)$ in Sec. E vs. ζ_t in Sec. F

We begin by clarifying the roles of the two error terms that appear in different parts of the analysis.

Mismatch sequence in Sec. F. Sec. F introduces a one-step *conditional-mean mismatch* sequence $\{\zeta_t\}_{t \geq 1}$: for the actually used reward signal $X_t \in [0, 1]$ and the ideal (regularized) bandit reward $r_t^{(\varepsilon)}(g_t) \in [0, 1]$,

$$\left| \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right| \leq \zeta_t, \quad (236)$$

which is exactly Assumption F.10 (see (219) in Sec. F).

Confidence radius in Sec. E. In contrast, the symbol $\eta_t(\cdot)$ is defined in Sec. E as a *statistical confidence radius* that controls plug-in estimation errors of the form $|\hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)|$ (e.g., Lemma E.11 and subsequent results).

Throughout this section, we keep the two notions distinct:

$$\eta_t(g) \quad (\text{statistical confidence radius; Sec. E}), \quad \zeta_t \quad (\text{conditional-mean mismatch; Sec. F}).$$

G.2. A general lemma: pointwise control $\Rightarrow$ conditional-mean control

The next lemma is a basic but useful tool: a pointwise bound immediately implies a bound in conditional expectation, which is the format required by Assumption F.10.

Lemma G.1 (Pointwise deviation implies conditional-mean deviation). *Fix $t \geq 1$. Let $X_t \in [0, 1]$ be $\mathcal{F}_t$ -measurable and let $Y_t \in [0, 1]$ be any random variable. Suppose that on some event $\mathcal{E}$ we have the pointwise inequality*

$$|X_t - Y_t| \leq \varepsilon_t, \quad (237)$$

where $\varepsilon_t \geq 0$ is deterministic (or more generally $\mathcal{F}_{t-1}$ -measurable). Then on the same event $\mathcal{E}$,

$$\left| \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] - \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \right| \leq \varepsilon_t. \quad (238)$$

In particular, Assumption F.10 holds with $\zeta_t := \varepsilon_t$ whenever $Y_t = r_t^{(\varepsilon)}(g_t)$.

Proof. We start from the difference of conditional expectations:

$$\left| \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] - \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \right| = \left| \mathbb{E}[X_t - Y_t \mid \mathcal{F}_{t-1}] \right| \leq \mathbb{E}[|X_t - Y_t| \mid \mathcal{F}_{t-1}]. \quad (239)$$

On the event $\mathcal{E}$ we have $|X_t - Y_t| \leq \varepsilon_t$ pointwise by (237), hence

$$\mathbb{E}[|X_t - Y_t| \mid \mathcal{F}_{t-1}] \leq \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}].$$

If ε_t is deterministic, or merely $\mathcal{F}_{t-1}$ -measurable, then $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = \varepsilon_t$. This yields (238), which concludes the proof. $\square$

G.3. Instantiating ζ_t

This subsection provides a concrete, checkable construction of ζ_t by relating the observable feedback used in the bandit update to the ideal regularized normalized reward.

From pointwise control to the mismatch term ζ_t . Fix any time $t \geq 1$ and condition on $\mathcal{F}_{t-1}$. Recall that the algorithm uses the observable feedback $\hat{r}_t^{(\varepsilon)}(g_t)$, whereas the ideal bandit analysis is phrased in terms of the regularized normalized reward $r_t^{(\varepsilon)}(g_t)$.

Step 1: a pathwise bound for the conditional mismatch. Define the conditional-mean mismatch at time t :

$$\Delta_t^{\text{cm}} := \mathbb{E}[\hat{r}_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}].$$

By Jensen's inequality applied to the conditional expectation,

$$|\Delta_t^{\text{cm}}| \leq \mathbb{E}[|\hat{r}_t^{(\varepsilon)}(g_t) - r_t^{(\varepsilon)}(g_t)| \mid \mathcal{F}_{t-1}]. \quad (240)$$

Step 2: plug a pointwise estimator-error bound. Assume that on a high-probability event (denoted $\mathcal{E}_H(\delta)$) one has the following pointwise control for the realized g_t :

$$|\hat{r}_t^{(\varepsilon)}(g_t) - r_t^{(\varepsilon)}(g_t)| \leq \eta_t(g_t) + \text{Drift}_t, \quad \forall t \leq T. \quad (241)$$

Taking $\mathcal{F}_{t-1}$ -conditional expectations on both sides of (241) yields

$$\mathbb{E}[|\hat{r}_t^{(\varepsilon)}(g_t) - r_t^{(\varepsilon)}(g_t)| \mid \mathcal{F}_{t-1}] \leq \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \text{Drift}_t, \quad (242)$$

where we used that Drift_t is $\mathcal{F}_{t-1}$ -measurable by construction. The bounds (241)–(242) will be established later (see Proposition G.3).

Combining (240) and (242) gives the conditional mismatch bound

$$|\Delta_t| \leq \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \text{Drift}_t.$$

Step 3: define ζ_t on the same high-probability event. On $\mathcal{E}_H(\delta)$, define

$$\zeta_t := \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \text{Drift}_t = \sum_{g=1}^G \mathbb{P}_t(g) \eta_t(g) + \text{Drift}_t, \quad (243)$$

where $\mathbb{P}_t(\cdot)$ is the bandit sampling distribution at time t (as in Sec. F). Then the mismatch condition (219) in Assumption F.10 holds pathwise on $\mathcal{E}_H(\delta)$:

$$\left| \mathbb{E}[\hat{r}_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] - \mathbb{E}[r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right| \leq \zeta_t.$$

Proposition G.2 (Concrete choice of ζ_t implying Assumption F.10). *On the event $\mathcal{E}_H(\delta)$, Assumption F.10 holds with the explicit choice*

$$\zeta_t := \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \text{Drift}_t. \quad (244)$$

Proposition G.3 (Pointwise feedback error bound for the regularized normalized reward). *Fix $\delta \in (0, 1)$ and consider any horizon $T \geq 1$. On the master event $\mathcal{E}_H(\delta)$ (Lemma G.4), for every $t \in \{1, \dots, T\}$ and every group $g \in [G]$, we have*

$$|\hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq \eta_t(g) + \text{Drift}_t, \quad (245)$$

where $\eta_t(g)$ is the confidence-radius term (cf. (30)) and

$$\text{Drift}_t := \|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_\infty.$$

Consequently, in particular for the played group g_t ,

$$|\hat{r}_t^{(\varepsilon)}(g_t) - r_t^{(\varepsilon)}(g_t)| \leq \eta_t(g_t) + \text{Drift}_t.$$

Moreover, taking $\mathcal{F}_{t-1}$ -conditional expectations yields the bound

$$\left| \mathbb{E} [\hat{r}_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] - \mathbb{E} [r_t^{(\varepsilon)}(g_t) \mid \mathcal{F}_{t-1}] \right| \leq \mathbb{E} [\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \text{Drift}_t. \quad (246)$$

Proof. Proof sketch. On the master event $\mathcal{E}_H(\delta)$, the discounted-UCB construction guarantees a uniform deviation bound between the observable plug-in reward $\hat{r}_t^{(\varepsilon)}(g)$ and its smoothed/noise-free target $\bar{r}_t^{(\varepsilon)}(g)$, with radius $\eta_t(g)$ (as encoded by (30)). Separately, the definition of Drift_t controls the deviation between the smoothed target $\bar{r}_t^{(\varepsilon)}(g)$ and the true reward $r_t^{(\varepsilon)}(g)$. Combining these two deviations by the triangle inequality yields (245). Finally, Jensen's inequality and conditional expectation yield (246).

Fix any $t \in \{1, \dots, T\}$ and any $g \in [G]$.

Step 1: confidence part (observable vs smoothed). By the definition of the master event $\mathcal{E}_H(\delta)$ and the reward radius construction (cf. (30)), on $\mathcal{E}_H(\delta)$ we have

$$|\hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g)| \leq \eta_t(g). \quad (247)$$

Step 2: drift part (smoothed vs true). By definition of $\text{Drift}_t = \|\bar{r}_t^{(\varepsilon)} - r_t^{(\varepsilon)}\|_\infty$, we have for every $g \in [G]$,

$$|\bar{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq \text{Drift}_t. \quad (248)$$

Step 3: triangle inequality (pointwise bound). Combining (247) and (248) gives, on $\mathcal{E}_H(\delta)$,

$$|\hat{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq |\hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g)| + |\bar{r}_t^{(\varepsilon)}(g) - r_t^{(\varepsilon)}(g)| \leq \eta_t(g) + \text{Drift}_t,$$

which is exactly (245). The specialization to $g = g_t$ is immediate.

Step 4: conditional-mean version. Define $D_t := \hat{r}_t^{(\varepsilon)}(g_t) - r_t^{(\varepsilon)}(g_t)$. Then Jensen's inequality yields

$$|\mathbb{E}[D_t \mid \mathcal{F}_{t-1}]| \leq \mathbb{E}[|D_t| \mid \mathcal{F}_{t-1}].$$

On $\mathcal{E}_H(\delta)$, (245) gives $|D_t| \leq \eta_t(g_t) + \text{Drift}_t$, hence

$$\mathbb{E}[|D_t| \mid \mathcal{F}_{t-1}] \leq \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \text{Drift}_t,$$

where we used that Drift_t is $\mathcal{F}_{t-1}$ -measurable by construction. This is exactly (246), which concludes the proof. $\square$

G.4. A fully explicit high-probability upper bound on $\sum_{t < T} \zeta_t$

This subsection derives an explicit high-probability upper bound on the cumulative mismatch $\sum_{t=1}^T \zeta_t$ by combining: (i) the concrete instantiation of ζ_t from Sec. G.3, and (ii) the high-probability controls already established for the confidence radii and the drift terms in Sec. E. The resulting bound is stated in a form that directly supports verifying $\sum_{t < T} \zeta_t = o(T)$.

Lemma G.4 (A master high-probability event for the regret interface). *Fix $\delta \in (0, 1)$ and a horizon $T \geq 1$. Let $\mathcal{E}_H(\delta)$ denote the intersection of the high-probability events used in Sec. E to establish the regret interface, including:*

- *the event on which the confidence-radius bounds (used to define $\eta_t(g)$ via (30)) hold simultaneously for all $t \leq T$ and all $g \in [G]$;*
- *the event on which the anytime drift control holds (Corollary E.19, in particular the safe form (104));*
- *the event on which the cumulative confidence bound along the realized greedy sequence holds (e.g., (138));*
- *any additional concentration events that are explicitly union-bounded in the proofs leading to Theorem E.26 and Corollary E.30.*

Then $\mathbb{P}(\mathcal{E}_H(\delta)) \geq 1 - \delta$. Moreover, on $\mathcal{E}_H(\delta)$ the regret instantiation in Corollary E.30 (and hence (190)–(191)) holds.

Recall Proposition G.2: on $\mathcal{E}_H(\delta)$, for all $t \leq T$,

$$\zeta_t = \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \text{Drift}_t.$$

Summing over $t = 1, \dots, T$ yields

$$\sum_{t=1}^T \zeta_t = \sum_{t=1}^T \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] + \sum_{t=1}^T \text{Drift}_t. \quad (249)$$

Deterministic control of the drift part. On $\mathcal{E}_H(\delta)$, the safe drift bound (104) in Corollary E.19 gives

$$\sum_{t=1}^T \text{Drift}_t \leq t_0(T) + \frac{C'_{\text{drift}}}{(1 - \lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta}\right)}. \quad (250)$$

For notational convenience, we may denote $m_T := t_0(T) - \mathcal{W}_T$ (no new label is introduced), so the logarithmic term is $\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)$.

High-probability control of the statistical part. By the definition of the bandit sampling distribution $\mathbb{P}_t(\cdot)$ (see Sec. F),

$$\mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] = \sum_{g=1}^G \mathbb{P}_t(g) \eta_t(g). \quad (251)$$

To relate this predictable quantity to the confidence-sum bounds available in Sec. E, we use the standard mixture structure of $\mathbb{P}_t(\cdot)$ (greedy-with-exploration): writing

$$\mathbb{P}_t(g) = (1 - \gamma_t) \mathbf{1}\{g = \hat{g}_t\} + \frac{\gamma_t}{G},$$

we obtain the deterministic (pathwise) inequality

$$\mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] \leq \eta_t(\hat{g}_t) + \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g). \quad (252)$$

Therefore,

$$\sum_{t=1}^T \mathbb{E}[\eta_t(g_t) \mid \mathcal{F}_{t-1}] \leq \sum_{t=1}^T \eta_t(\hat{g}_t) + \sum_{t=1}^T \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g). \quad (253)$$

On $\mathcal{E}_H(\delta)$, the first term is controlled by Proposition E.28 (equivalently (138)). The second term will be controlled using the same uniform confidence-radius event (the first bullet in Lemma G.4) together with the explicit exploration schedule; we formalize the resulting $o(T)$ verification in Lemma G.5 below.

Putting the two parts together. Combining (249), (250), and (253) yields: on $\mathcal{E}_H(\delta)$,

$$\sum_{t=1}^T \zeta_t \leq \sum_{t=1}^T \eta_t(\hat{g}_t) + \sum_{t=1}^T \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g) + t_0(T) + \frac{C'_{\text{drift}}}{(1 - \lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2(t_0(T) - \mathcal{W}_T)^2}{\delta}\right)}. \quad (254)$$

When is $\sum_{t=1}^T \zeta_t = o(T)$? Divide (254) by T . It suffices that each term on the right-hand side is $o(T)$. The first and the drift terms are already in the same form as in Sec. E. The remaining exploration-weighted term $\sum_{t \leq T} \frac{\gamma_t}{G} \sum_{g \leq G} \eta_t(g)$ will be shown to be $o(T)$ under the same two-term sufficient system used later for the regret perturbations; we formalize this in Lemma G.5 below.

G.5. A rigorous sufficient condition for $\sum_{t \leq T} \zeta_t = o(T)$

This subsection provides a checkable sufficient condition guaranteeing that the cumulative mismatch $\sum_{t=1}^T \zeta_t$ is sublinear on the master event $\mathcal{E}_H(\delta)$. Starting from the explicit decomposition (254), the proof must simultaneously control: (i) the realized-path confidence sum $\sum_{t \leq T} \eta_t(\hat{g}_t)$; (ii) the exploration-weighted confidence term $\sum_{t \leq T} \frac{\gamma_t}{G} \sum_{g \leq G} \eta_t(g)$; (iii) the schedule term $t_0(T)$; and (iv) the drift term with denominator $\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)$. The two statistical terms (i)–(ii) share the same underlying difficulty: the appearance of the poly-logarithmic factor $L_T(\delta)$ in the one-step confidence radii, meaning that $V_T^{(\varepsilon)} = o(T)$ alone does *not* automatically imply $(V_T^{(\varepsilon)})^{1/3} T^{2/3} L_T(\delta) (\log(e + T))^{\kappa/4} = o(T)$. In addition, for (ii), one cannot assume a crude bound such as $\eta_t(g) \leq 1$; instead, one must use the same uniform one-step envelope (and the early-time capped radii) already established in the proof of the cumulative bound for $\sum_{t \leq T} \eta_t(\hat{g}_t)$. We therefore state a two-limit sufficient system that matches the one used later in Section F to ensure that all exponent perturbations are $o(T)$.

Lemma G.5 (A sufficient condition ensuring $\sum_{t \leq T} \zeta_t = o(T)$). *Fix $\delta \in (0, 1)$ and assume that the explicit high-probability bound (254) holds on the event $\mathcal{E}_H(\delta)$. Assume moreover that the two limits*

$$\frac{V_T^{(\varepsilon)}}{T} L_T(\delta)^3 (\log(e + T))^{3\kappa/4} \xrightarrow{T \rightarrow \infty} 0 \quad \text{and} \quad \frac{V_T^{(\varepsilon)}}{T} \cdot \frac{(\log(e + T))^{2+2\kappa}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)} \xrightarrow{T \rightarrow \infty} 0 \quad (255)$$

hold, where $m_T = t_0(T) - \mathcal{W}_T$ is as in (107). Then, on $\mathcal{E}_H(\delta)$,

$$\sum_{t=1}^T \zeta_t = o(T).$$

Proof. Start from (254). On $\mathcal{E}_H(\delta)$,

$$\sum_{t=1}^T \zeta_t \leq \underbrace{\sum_{t=1}^T \eta_t(\hat{g}_t)}_{(A)} + \underbrace{\sum_{t=1}^T \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g)}_{(B)} + \underbrace{t_0(T)}_{(C)} + \underbrace{\frac{C'_{\text{drift}}}{(1 - \lambda_T)^2} \cdot \frac{V_T^{(\varepsilon)}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)}}_{(D)}.$$

We show that each term is $o(T)$.

Step 1: Term (A) is $o(T)$. By Proposition E.28, on $\mathcal{E}_H(\delta)$ we have the explicit bound (138), i.e.,

$$\sum_{t=1}^T \eta_t(\hat{g}_t) \leq C_\eta L_T \left((V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e + T))^{\kappa/4} + \frac{T}{(\log(e + T))^{\kappa/4}} \right).$$

Dividing by T , the second term is $O((\log(e + T))^{-\kappa/4}) = o(1)$. For the first term, use the first limit in (255) (equivalently Lemma F.18 with $\beta = \kappa/4$) to conclude it is $o(T)$. Hence (A) = $o(T)$.

Step 2: Term (B) is $o(T)$ (no assumption like $\eta_t(g) \leq 1$). Let

$$S_{\text{expl}}(T) := \sum_{t=1}^T \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g).$$

We split the sum into an early segment $t \leq 2H$ and a tail segment $t \geq 2H + 1$, where H is the same window parameter used in the proof of Proposition E.28.

Step 2.1 (Early segment $t \leq 2H$: boundedness via capped radii). In Step 4 of the proof of Proposition E.28, the *capped* energy radius is introduced: for every t and g ,

$$\beta_t^{\text{cap}}(g) := 2\sqrt{2} w_{\max} \sqrt{\frac{\log\left(\frac{4GT}{\delta}\right)}{N_t(g) \vee 1}}, \quad (\text{cf. (163)}).$$

Since $N_t(g) \vee 1 \geq 1$, we have the pointwise bound

$$\beta_t^{\text{cap}}(g) \leq 2\sqrt{2} w_{\max} \sqrt{\log\left(\frac{4GT}{\delta}\right)} = 2\sqrt{2} w_{\max} L_T(\delta),$$

and therefore also

$$\sum_{j=1}^G \beta_t^{\text{cap}}(j) \leq 2\sqrt{2} w_{\max} G L_T(\delta).$$

Applying the reward-radius definition (30) with $\beta_t(\cdot)$ replaced by $\beta_t^{\text{cap}}(\cdot)$ yields, for every $t \leq 2H$ and every $g \in [G]$,

$$\eta_t(g) \leq \frac{\beta_t^{\text{cap}}(g)}{\varepsilon} + \frac{w_{\max}}{\varepsilon^2} \sum_{j=1}^G \beta_t^{\text{cap}}(j) \leq \underbrace{2\sqrt{2} w_{\max} \left(\frac{1}{\varepsilon} + \frac{w_{\max} G}{\varepsilon^2} \right)}_{=: C_{\text{cap}}} L_T(\delta).$$

Using $\gamma_t \leq \frac{1}{2}$ (see (137)), we obtain

$$\sum_{t=1}^{2H} \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g) \leq \sum_{t=1}^{2H} \frac{1}{2} \max_{g \in [G]} \eta_t(g) \leq H \cdot C_{\text{cap}} L_T(\delta).$$

Since $H \leq T$ and $L_T(\delta) = O(\sqrt{\log T})$, we have $H L_T(\delta) = o(T)$ (in particular in the worst case $H = O(T^{2/3})$ as in (26), so $H L_T(\delta) = O(T^{2/3} \sqrt{\log T}) = o(T)$). Thus the early segment contribution is $o(T)$.

Step 2.2 (Tail segment $t \geq 2H + 1$: reduce to the same tail sum as Proposition E.28). For $t \geq 2H + 1$, the proof of Proposition E.28 establishes the uniform bounds (160) and (161):

$$\beta_t(g) \leq \frac{L_T}{\sqrt{\Lambda_{t,H}}} \sqrt{\frac{G}{H \underline{\gamma}_T}}, \quad \sum_{j=1}^G \beta_t(j) \leq \frac{G L_T}{\sqrt{\Lambda_{t,H}}} \sqrt{\frac{G}{H \underline{\gamma}_T}},$$

uniformly for all $g \in [G]$. Substituting these two bounds into the definition (30) yields the uniform one-step envelope (identical to (162)): for all $t \geq 2H + 1$ and all $g \in [G]$,

$$\eta_t(g) \leq C_1 \frac{L_T(\delta)}{\sqrt{\Lambda_{t,H}}} \sqrt{\frac{G}{H \underline{\gamma}_T}}.$$

Consequently, using again $\gamma_t \leq \frac{1}{2}$,

$$\begin{aligned} \sum_{t=2H+1}^T \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g) &\leq \sum_{t=2H+1}^T \frac{1}{2} \max_{g \in [G]} \eta_t(g) \\ &\leq \frac{1}{2} \sum_{t=2H+1}^T C_1 \frac{L_T(\delta)}{\sqrt{\Lambda_{t,H}}} \sqrt{\frac{G}{H \underline{\gamma}_T}}. \end{aligned}$$

The remaining tail sum is exactly the same object controlled in Proposition E.28: by (176),

$$\sum_{t=2H+1}^T \frac{1}{\sqrt{\Lambda_{t,H}}} \leq C_{\Lambda} T \left(\frac{1}{\sqrt{H}} + \frac{V_T^{(\varepsilon)}}{H} \right)^{1/2}.$$

Therefore,

$$\sum_{t=2H+1}^T \frac{\gamma_t}{G} \sum_{g=1}^G \eta_t(g) \leq \frac{1}{2} C_1 C_\Lambda L_T(\delta) T \sqrt{\frac{G}{H \gamma_T}} \left(\frac{1}{\sqrt{H}} + \frac{V_T^{(\varepsilon)}}{H} \right)^{1/2}.$$

The tuning step for H carried out in Proposition E.28 (see (179)–(138)) shows that the right-hand side is bounded by

$$C L_T \left((V_T^{(\varepsilon)} \vee 1)^{1/3} T^{2/3} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right)$$

for an absolute constant $C > 0$. By the same argument as in Step 1, this is $o(T)$ under the first limit in (255). Hence the tail segment is $o(T)$, and thus $S_{\text{expl}}(T) = o(T)$.

Step 3: Terms (C) and (D) are $o(T)$. The schedule uses $t_0(T) = \lceil \sqrt{T} \rceil$, hence $t_0(T) = o(T)$. For the drift term (D), divide by T and use the second limit in (255) together with $(1 - \lambda_T)^{-2} = (\log(e+T))^{2+2\kappa}$ under the anytime schedule.

Combining Steps 1–3 yields $\sum_{t=1}^T \zeta_t = o(T)$ on $\mathcal{E}_H(\delta)$. $\square$

Remark G.6 (How this is used in Sec. F). The oracle-rate preservation argument in Sec. F requires $\sum_{t < T} \zeta_t = o(T)$ so that all perturbations in the exponent are $o(T)$. Lemma G.5 provides a checkable sufficient condition that is compatible with the polylog factor and also handles the exploration-weighted confidence term appearing in (254).

Remark G.7 (How to interpret and satisfy the conditions on $V_T^{(\varepsilon)}$). Recall that $V_T^{(\varepsilon)}$ (see Sec. E) is a variation budget controlling the time drift of the regularized reward functions $r_t^{(\varepsilon)}(\cdot)$. Intuitively, $V_T^{(\varepsilon)}$ is small when the underlying group-level energies evolve gradually over time.

A common sufficient growth model. A standard gradual-drift model assumes a polynomial envelope

$$V_T^{(\varepsilon)} = O(T^\rho) \quad \text{for some } \rho \in [0, 1). \quad (256)$$

Why (256) implies (255) (for fixed δ). Fix $\delta \in (0, 1)$. Assume (256). Recall $L_T(\delta) = \sqrt{\log(\frac{8GT}{\delta})}$.

For the first limit in (255),

$$\frac{V_T^{(\varepsilon)}}{T} L_T(\delta)^3 (\log(e+T))^{3\kappa/4} = O\left(T^{\rho-1} (\log(e+T))^{3/2+3\kappa/4}\right) \xrightarrow{T \rightarrow \infty} 0,$$

since $\rho - 1 < 0$ and any polylog factor is $o(T^a)$ for every $a > 0$.

For the second limit in (255), we use the anytime drift construction: Corollary E.19 shows that

$$\log\left(\frac{G\pi^2 m_T^2}{\delta}\right) = O(\log T) + O(1),$$

hence in particular there exist constants $c_0 > 0$ and T_0 such that $\log\left(\frac{G\pi^2 m_T^2}{\delta}\right) \geq c_0 \log(e+T)$ for all $T \geq T_0$. Therefore,

$$\frac{V_T^{(\varepsilon)}}{T} \cdot \frac{(\log(e+T))^{2+2\kappa}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)} = O\left(T^{\rho-1} (\log(e+T))^{1+2\kappa}\right) \xrightarrow{T \rightarrow \infty} 0.$$

Thus (256) is an easy-to-check sufficient model ensuring $\sum_{t \leq T} \zeta_t = o(T)$.

Corollary G.8 (A simple polynomial-growth sufficient condition). Fix $\delta \in (0, 1)$ and suppose the anytime schedule (80) is used. Assume that (254) holds on $\mathcal{E}_H(\delta)$ and that

$$V_T^{(\varepsilon)} = O(T^\rho) \quad \text{for some } \rho \in [0, 1).$$

Then the sufficient system (255) in Lemma G.5 holds, and hence on $\mathcal{E}_H(\delta)$,

$$\sum_{t=1}^T \zeta_t = o(T).$$

More explicitly, for all sufficiently large T on $\mathcal{E}_H(\delta)$,

$$\sum_{t=1}^T \zeta_t \leq C_\eta L_T(\delta) \left(T^{\frac{2+\rho}{3}} (\log(e+T))^{\kappa/4} + \frac{T}{(\log(e+T))^{\kappa/4}} \right) + \frac{4G C_{\text{drift}}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)} \cdot T^\rho \cdot (\log(e+T))^{2+2\kappa},$$

and each term is $o(T)$ since $\frac{2+\rho}{3} < 1$ and $T^{\rho-1}$ dominates polylogarithmic factors.

Remark G.9 (Another sufficient growth model: near-linear budget with a polylog gap). The polynomial envelope (256) is sufficient but not necessary. A weaker model allows $V_T^{(\varepsilon)}$ to be nearly linear up to a polylogarithmic divisor.

Fix $\delta \in (0, 1)$ and treat G as fixed. Since $L_T(\delta) = \sqrt{\log(\frac{8GT}{\delta})}$, there exist constants $C_L \geq 1$ and $p_L \geq 0$ such that for all $T \geq 2$,

$$L_T(\delta) \leq C_L (\log(e+T))^{p_L}. \quad (257)$$

Assume there exist constants $C_V > 0$ and $m > 0$ such that for all $T \geq 2$,

$$V_T^{(\varepsilon)} \leq C_V \frac{T}{(\log(e+T))^m}. \quad (258)$$

If

$$m > 3p_L + \frac{3\kappa}{4} \quad \text{and} \quad m > 1 + 2\kappa, \quad (259)$$

then the two limits in (255) hold. Indeed, using (257)–(258),

$$\frac{V_T^{(\varepsilon)}}{T} L_T(\delta)^3 (\log(e+T))^{3\kappa/4} \leq C_V C_L^3 (\log(e+T))^{3p_L+3\kappa/4-m} \xrightarrow{T \rightarrow \infty} 0$$

whenever $m > 3p_L + 3\kappa/4$.

For the second limit, (258) gives $V_T^{(\varepsilon)}/T \leq C_V (\log(e+T))^{-m}$, hence

$$\frac{V_T^{(\varepsilon)}}{T} \cdot \frac{(\log(e+T))^{2+2\kappa}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)} \leq C_V \cdot \frac{(\log(e+T))^{2+2\kappa-m}}{\log\left(\frac{G\pi^2 m_T^2}{\delta}\right)}.$$

Under the anytime drift construction, Corollary E.19 yields $\log\left(\frac{G\pi^2 m_T^2}{\delta}\right) = \log T + O(1)$, so the denominator is $\Omega(\log(e+T))$. Therefore the right-hand side converges to 0 whenever $m > 1 + 2\kappa$. Thus (258) with (259) is a convenient near-linear sufficient model.

Corollary G.10 (A near-linear polylog-divided sufficient condition). *Fix $\delta \in (0, 1)$ and suppose the anytime schedule (80) is used. Assume that (254) holds on $\mathcal{E}_H(\delta)$, and that there exist constants $C_L \geq 1$, $p_L \geq 0$ satisfying (257). If (258) holds with an exponent m obeying (259), then the sufficient conditions (255) in Lemma G.5 hold, and hence*

$$\sum_{t=1}^T \zeta_t = o(T).$$

H. The derivation of the confidence radius in our algorithm

This section derives the explicit confidence radii used by our discounted estimator and by the induced regularized normalized reward. Concretely, we first obtain a high-probability deviation bound for the discounted energy estimator $\hat{w}_t(b)$ around its discounted conditional mean $\bar{w}_t(b)$, which yields the per-arm radius $\beta_t(b)$ after a union bound over all (t, b) up to horizon T . We then propagate this bound through the projection $x \mapsto x^+$ and the normalization map $w \mapsto w/(\sum_j w_j + \varepsilon)$ to obtain a uniform reward-radius bound $|\hat{r}_t^{(\varepsilon)}(b) - \bar{r}_t^{(\varepsilon)}(b)| \leq \eta_t(b)$. These radii are exactly the quantities invoked later when establishing the high-probability regret interface and the mismatch control.

Fix a target confidence level $\delta \in (0, 1)$ and a horizon $T \geq 1$. For each pair (t, g) with $t \in [T]$ and $g \in [G]$, we will apply a concentration inequality with failure probability

$$\delta' := \frac{\delta}{2GT}.$$

This choice ensures that a union bound over all $t \in [T]$ and $g \in [G]$ (at most GT pairs) yields a simultaneous “good event” with probability at least $1 - \delta/2$.

Step 1: bounded martingale differences for the discounted estimator. Recall the discounted count and sum statistics (26) and the corresponding estimator (27): for each $t \geq 2$ and $g \in [G]$,

$$N_t(g) = \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}, \quad S_t(g) = \sum_{s=1}^{t-1} \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\} \tilde{w}_s(g),$$

and $\hat{w}_t(g) = S_t(g)/N_t(g)$ whenever $N_t(g) > 0$. For fixed (t, g) , define the predictable weights

$$a_{t,s}(g) := \left(\prod_{j=s+1}^{t-1} \lambda_j \right) \mathbf{1}\{g_s = g\}, \quad s = 1, \dots, t-1.$$

Since each $\lambda_j \in (0, 1)$, we have $0 \leq a_{t,s}(g) \leq 1$, and $N_t(g) = \sum_{s=1}^{t-1} a_{t,s}(g)$ and $S_t(g) = \sum_{s=1}^{t-1} a_{t,s}(g) \tilde{w}_s(g)$.

Define the conditional mean at time s (given the past and the chosen group) as

$$m_s(g) := \mathbb{E}[\tilde{w}_s(g) \mid \mathcal{F}_{s-1}, g_s = g],$$

and the centered noise

$$\xi_s(g) := \tilde{w}_s(g) - m_s(g).$$

Then $\mathbb{E}[\xi_s(g) \mid \mathcal{F}_{s-1}, g_s = g] = 0$ by construction, and hence the sequence $\{a_{t,s}(g)\xi_s(g)\}_{s=1}^{t-1}$ is a martingale difference sequence with respect to $(\mathcal{F}_s)$.

Moreover, by Assumption E.2 we have $|\tilde{w}_s(g)| \leq w_{\max}$ almost surely, and therefore

$$|m_s(g)| \leq \mathbb{E}[|\tilde{w}_s(g)| \mid \mathcal{F}_{s-1}, g_s = g] \leq w_{\max} \quad \text{a.s.}$$

Consequently,

$$|\xi_s(g)| \leq |\tilde{w}_s(g)| + |m_s(g)| \leq 2w_{\max} \quad \text{a.s.}, \quad \text{and hence} \quad |a_{t,s}(g)\xi_s(g)| \leq 2w_{\max} a_{t,s}(g) \quad \text{a.s.}$$

Step 2: Azuma–Hoeffding for the discounted average. For fixed (t, g) , define the martingale

$$M_{t,g} := \sum_{s=1}^{t-1} a_{t,s}(g)\xi_s(g) = \sum_{s=1}^{t-1} a_{t,s}(g) (\tilde{w}_s(g) - m_s(g)).$$

Also define the corresponding discounted mean quantities

$$\bar{S}_t(g) := \sum_{s=1}^{t-1} a_{t,s}(g)m_s(g), \quad \bar{w}_t(g) := \frac{\bar{S}_t(g)}{N_t(g)} \quad (N_t(g) > 0).$$

Then, whenever $N_t(g) > 0$,

$$\hat{w}_t(g) - \bar{w}_t(g) = \frac{S_t(g) - \bar{S}_t(g)}{N_t(g)} = \frac{M_{t,g}}{N_t(g)}.$$

By the two-sided Azuma–Hoeffding inequality and the bound $|a_{t,s}(g)\xi_s(g)| \leq 2w_{\max}a_{t,s}(g)$, for any $x > 0$,

$$\Pr(|M_{t,g}| \geq x) \leq 2 \exp \left(-\frac{x^2}{8w_{\max}^2 \sum_{s=1}^{t-1} a_{t,s}(g)^2} \right).$$

Since $a_{t,s}(g) \in [0, 1]$, we have $a_{t,s}(g)^2 \leq a_{t,s}(g)$ and thus $\sum_{s=1}^{t-1} a_{t,s}(g)^2 \leq N_t(g)$. Plugging this in yields

$$\Pr(|M_{t,g}| \geq x) \leq 2 \exp \left(-\frac{x^2}{8w_{\max}^2 N_t(g)} \right).$$

Setting $x := \beta N_t(g)$ gives, for $N_t(g) > 0$,

$$\Pr(|\hat{w}_t(g) - \bar{w}_t(g)| \geq \beta) \leq 2 \exp\left(-\frac{\beta^2 N_t(g)}{8w_{\max}^2}\right).$$

Therefore, choosing β such that $2 \exp\left(-\frac{\beta^2 N_t(g)}{8w_{\max}^2}\right) \leq \delta'$ is sufficient; equivalently,

$$\beta \geq 2\sqrt{2} w_{\max} \sqrt{\frac{\log(2/\delta')}{N_t(g)}} = 2\sqrt{2} w_{\max} \sqrt{\frac{\log\left(\frac{4GT}{\delta}\right)}{N_t(g)}}.$$

This is exactly the confidence radius $\beta_t(g)$ defined in (29) for $N_t(g) > 0$. For uniform statements that also cover the case $N_t(g) = 0$, Sec. E uses the capped version with $N_t(g) \vee 1$ (see (163)).

A union bound over all $t \in [T]$ and $g \in [G]$ shows that with probability at least $1 - GT\delta' = 1 - \delta/2$, the inequality $|\hat{w}_t(g) - \bar{w}_t(g)| \leq \beta_t(g)$ holds simultaneously for all pairs (t, g) with $N_t(g) > 0$ (and similarly the capped bound holds for all (t, g)).

Step 3: from $\beta_t(\cdot)$ to the reward radius $\eta_t(\cdot)$. Recall the projected estimator $\hat{w}_t^+(g) := \max\{\hat{w}_t(g), 0\}$ (see (28)) and the plug-in regularized normalized reward

$$\hat{r}_t^{(\varepsilon)}(g) := \frac{\hat{w}_t^+(g)}{\widehat{W}_t + \varepsilon}, \quad \widehat{W}_t := \sum_{j=1}^G \hat{w}_t^+(j).$$

Also define the corresponding discounted population quantities

$$\bar{W}_t := \sum_{j=1}^G \bar{w}_t(j), \quad \bar{r}_t^{(\varepsilon)}(g) := \frac{\bar{w}_t(g)}{\bar{W}_t + \varepsilon}.$$

On the union-bounded good event from Step 2, we have for every $t \leq T$ and every $g \in [G]$ with $N_t(g) > 0$, $|\hat{w}_t(g) - \bar{w}_t(g)| \leq \beta_t(g)$. Since $\bar{w}_t(g) \geq 0$ and $x \mapsto x^+$ is 1-Lipschitz,

$$|\hat{w}_t^+(g) - \bar{w}_t(g)| \leq \beta_t(g).$$

Summing over g gives

$$|\widehat{W}_t - \bar{W}_t| \leq \sum_{j=1}^G \beta_t(j).$$

By direct algebra,

$$|\hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g)| \leq \frac{|\hat{w}_t^+(g) - \bar{w}_t(g)|}{\widehat{W}_t + \varepsilon} + \frac{\bar{w}_t(g) |\widehat{W}_t - \bar{W}_t|}{(\widehat{W}_t + \varepsilon)(\bar{W}_t + \varepsilon)}.$$

Since $\widehat{W}_t, \bar{W}_t \geq 0$, we have $\widehat{W}_t + \varepsilon \geq \varepsilon$ and $\bar{W}_t + \varepsilon \geq \varepsilon$, hence

$$\frac{1}{\widehat{W}_t + \varepsilon} \leq \frac{1}{\varepsilon}, \quad \frac{1}{(\widehat{W}_t + \varepsilon)(\bar{W}_t + \varepsilon)} \leq \frac{1}{\varepsilon^2}.$$

Also $\bar{w}_t(g) \leq w_{\max}$ by Assumption E.2. Combining these bounds yields

$$|\hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g)| \leq \frac{\beta_t(g)}{\varepsilon} + \frac{w_{\max}}{\varepsilon^2} \sum_{j=1}^G \beta_t(j),$$

which is exactly the induced reward confidence radius $\eta_t(g)$ defined in (30). In particular, on the above union-bounded good event (probability at least $1 - \delta/2$), $|\hat{r}_t^{(\varepsilon)}(g) - \bar{r}_t^{(\varepsilon)}(g)| \leq \eta_t(g)$ holds uniformly for all (t, g) .

I. Proofs of All Theoretical Results in the Main Text

In this section, we provide the proofs for all theorems in the main text. Since the proofs are organized in the above sections and may be difficult to find, we provide a cross-reference for each of them:

- Theorem 4.7: Theorem D.5 in Appendix.
- Corollary 4.8: Corollary D.6 in Appendix.
- Theorem 4.10: Theorem D.10 in Appendix.
- Theorem 5.2: Theorem E.26 in Appendix.
- Theorem 5.5: Theorem F.9 in Appendix.
- Theorem 5.6: Theorem F.16 in Appendix.
- Corollary 5.7: Corollary F.17 in Appendix.